

Analog To Digital Converter (ADC) Module

This chapter describes the analog to digital converter (ADC) interface module.

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22.1 Overview

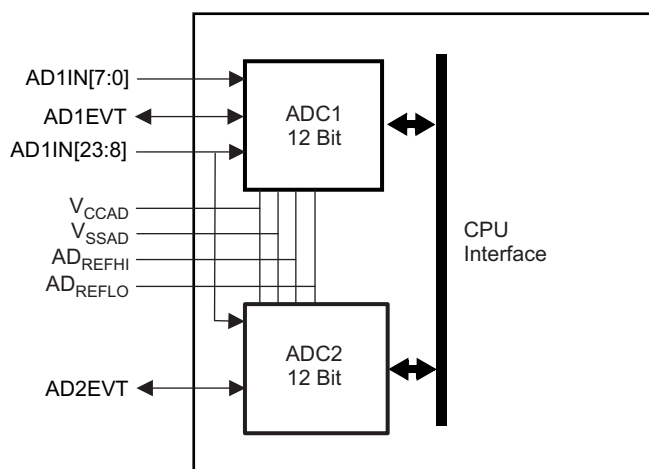
This microcontroller implements up to two instances of the ADC module. The main features of the ADC module are:

- Selectable 10-bit or 12-bit resolution, 10-bit mode is the default
- Successive-approximation-register (SAR) architecture
- Three conversion groups – Group1, Group2 and Event Group
- All three conversion groups can be configured to be hardware-triggered; group1 and group2 can also be triggered by software
- Conversion results are stored in a 64-word memory (SRAM)
 - These 64 words are divided between the three conversion groups and are configurable by software
 - Accesses to the conversion result RAM are protected by parity
- Flexible options for generating DMA requests for transferring conversion results
- Selectable channel conversion order:
 - Sequential conversions in ascending order of channel number, OR
 - User-defined channel conversion order with the Enhanced Channel Selection Mode
- Single or continuous conversion modes
- Embedded self-test logic for input channel failure detection (open / short to power / short to ground)
- Embedded calibration logic for offset error correction
- Enhanced Power-down mode
- External event pin (ADxEVT) to trigger conversions
 - ADxEVT is also programmable as general-purpose I/O
- Eight hardware events to trigger conversions

The two instances of the 12-bit ADC modules on the microcontroller share 16 analog input channels. The connections are shown in [Figure 22-1](#).

- ADC1 supports 24 channels.
- ADC2 supports 16 channels, all of which are shared with ADC1.
- When using both ADC1 and ADC2 on a shared channel, the sample windows must be identical such that the sample windows completely match each other or non-overlapping with a minimum of 2 ADC cycles buffer between the end of one ADC's sample window and the start of the other ADC's sample window.
- The reference voltages, as well as operating supply and ground, are shared between the two ADC cores.

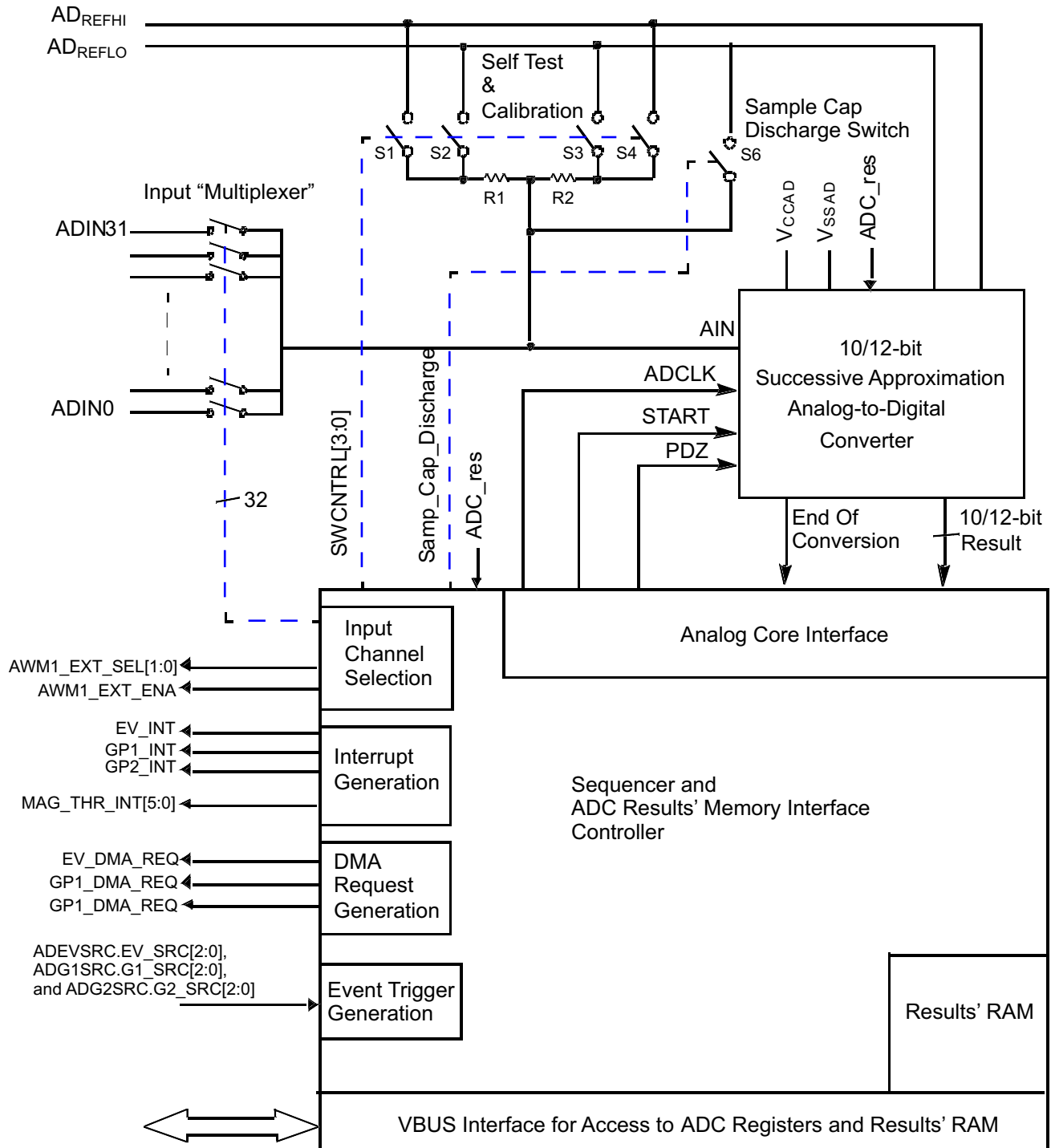
Figure 22-1. Channel Assignments of Two ADC Cores



22.2 Introduction

This section presents a brief functional description of the analog-to-digital converter (ADC) module. [Figure 22-2](#) illustrates the components of the ADC module.

Figure 22-2. ADC Block Diagram



22.2.1 Input Multiplexor

The input multiplexor (MUX) connects the selected input channel to the AIN input of the ADC core. The ADC1 module supports up to 24 inputs as shown in [Figure 22-2](#). The ADC2 module supports up to 16 inputs. The sequencer selects the channel to be converted. Enabling the enhanced channel selection mode also allows one or more of the analog input channels to be connected to the output of an external analog switch or multiplexor.

22.2.2 Self-Test and Calibration Cell

The ADC includes specific hardware that allows a software algorithm to detect open/short on an ADC analog input. It also allows the application program to calibrate the ADC. Also see [Section 22.8.1](#) and [Section 22.8.2](#).

22.2.3 Analog-to-Digital Converter Core

The ADC core is a combination voltage scaling, charge redistribution Successive Approximation Register (SAR) based analog-to-digital converter. The core can be configured for operation in 10-bit resolution (default) or 12-bit resolution. This is controlled by the sequencer logic. This selection applies to all conversions performed by the ADC module. It is not possible to convert some channels with a 12-bit resolution and some with a 10-bit resolution.

A single conversion from an analog input to a digital conversion result occurs in two distinct periods:

- Sampling Period:
 - The sequencer generates a START signal to the ADC core to signal the start of the sampling period.
 - The analog input signal is sampled directly on to the switched capacitor array during this period, providing an inherent sample-and-hold function.
 - The sampling period ends one full ADCLK after the falling edge of the START signal.
 - The sequencer can control the sampling period duration by configuring the conversion group's sample time control register (ADEVSAMP, ADG1SAMP, ADG2SAMP). This register controls the time for which the START signal stays high.
- Conversion Period:
 - The conversion period starts one full ADCLK after the falling edge of START.
 - One bit of the conversion result is output on each rising edge of ADCLK in the conversion period, starting with the most-significant bit first.
 - The conversion period is 12 ADCLK cycles in case of a 12-bit ADC, and is 10 ADCLK cycles in case of a 10-bit ADC.
 - The ADC core generates an End-Of-Conversion (EOC) signal to the sequencer at the end of the conversion period. At this time the complete 12-, or 10-bit conversion result is available.
 - The sequencer captures the ADC core conversion result output as soon as EOC is driven High.

The analog conversion range is determined by the reference voltages: AD_{REFHI} and AD_{REFLO} . AD_{REFHI} is the top reference voltage and is the maximum analog voltage that can be converted. An analog input voltage equal to AD_{REFHI} or higher results in an output code of 0x3FF for 10-bit resolution and 0xFFF for 12-bit resolution. AD_{REFLO} is the bottom reference voltage and is the minimum analog voltage that can be converted. Applying an input voltage equal to AD_{REFLO} or lower results in an output code of 0x000. Both AD_{REFHI} and AD_{REFLO} must be chosen not to exceed the analog power supplies: V_{CCAD} and V_{SSAD} , respectively. Input voltages between AD_{REFHI} and AD_{REFLO} produce a conversion result given by [Equation 31](#) for 10-bit resolution and by [Equation 32](#) for 12-bit resolution.

$$DigitalResult = \frac{1024 \times (InputVoltage - AD_{REFLO})}{AD_{REFHI} - AD_{REFLO}} - 0.5 \quad (31)$$

$$DigitalResult = \frac{4096 \times (InputVoltage - AD_{REFLO})}{(AD_{REFHI} - AD_{REFLO})} - 0.5 \quad (32)$$

22.2.4 Sequencer

The sequencer coordinates the operations of the ADC, including the input multiplexor, the ADC core, and the result memory. In addition, the logic of the sequencer sets the status register flags when the conversion is ongoing, stopped, or finished.

All the features of the sequencer are discussed in detail in the following sections of this document.

22.2.5 Conversion Groups

Several applications require groups of channels to be converted using a single trigger source for example. There could also be some groups of channels identified which require a specific setting of the acquisition time. The ADC module supports three conversion groups for this purpose – Group1, Group2 and the Event Group.

Any of the available analog input channels can be assigned to any of the conversion groups. This also allows a particular channel to be repeatedly sampled by selecting it in multiple groups. There is an inherent priority scheme used when multiple conversion groups are triggered at once. The Event Group is the highest-priority, followed by the Group1 and then the Group2.

The Event Group is always hardware event-triggered. Group1 and Group2 are software-triggered by default and can be configured to be hardware-, or event-triggered as well. The triggering of conversions in each group is discussed in [Section 22.3.6](#).

Each conversion group has a separate set of control registers to:

- Select the input channels to be converted
- Configure the mode of conversion: single conversion sequence or continuous conversions
- Configure the input channel sampling time
- Configure the interrupt and/or DMA request generation conditions

22.3 Basic Features and Usage of the ADC

This section describes the usage of the basic features of the ADC module.

22.3.1 How to Select Between 12-bit and 10-bit Resolutions

The 10_12_BIT field of the ADC Operating Mode Control Register (ADOPMODECR) configures the ADC to be in 10-bit or 12-bit resolution mode.

- If 10_12_BIT = 0, the module is in 10-bit resolution mode. This is the default mode of operation.
- If 10_12_BIT = 1, the module is in 12-bit resolution mode.

22.3.2 How to Set Up the ADCLK Speed

The ADC sequencer generates the clock for the ADC core, ADCLK. The ADC core uses the ADCLK signal for its timing. The ADCLK is generated by dividing down the input clock to the ADC module, which is the VBUSP interface clock, VCLK. A 5-bit field (PS) in the ADC Clock Control Register (ADCLOCKCR) is used to divide down the VCLK by 1 up to 32. The ADCLK valid frequency range is specified in the device datasheet.

$$f_{\text{ADCLK}} = f_{\text{VCLK}} / (\text{PS} + 1)$$

The maximum frequency for ADCLK is specified in the device datasheet.

22.3.3 How to Set Up the Input Channel Acquisition Time

The signal acquisition time for each group is separately configurable using the ADG1SAMP[11:0], ADG2SAMP[11:0], and ADEVSAMP[11:0] registers.

The acquisition time is specified in terms of ADCLK cycles and ranges from a minimum of 2 ADCLK cycles to a maximum of 4098 ADCLK cycles.

For example, Group1 acquisition time, $t_{\text{ACQG1}} = \text{G1SAMP}[11:0] + 2$, in ADCLK cycles.

The minimum acquisition time is specified in the device datasheet. This time also depends on the impedance of the circuit connected to the analog input channel being converted. See the *ADC Source Impedance for Hercules™ ARM® Safety MCUs Application Report* ([SPNA118](#)).

22.3.4 How to Select an Input Channel for Conversion

The ADC module needs to be enabled first before selecting an input channel for conversion. The ADC module can be enabled by setting the ADC EN bit in the ADC Operating Mode Control Register (ADOPMODECR). Multiple input channels can be selected for conversion in each group. Only one input channel is converted at a time. The channels to be converted are configured in one or more of the three conversion groups' channel selection registers. Channels to be converted in Group1 are configured in the Group1 Channel-Select Register (ADG1SEL), those to be converted in Group2 are configured in the Group2 Channel-Select Register (ADG2SEL), and those to be converted in the Event Group are configured in the Event Group Channel-Select Register (ADEVSEL).

The description in this section only refers to the case when the enhanced channel selection mode is not enabled. Input channel selection in the enhanced channel selection mode is defined in [Section 22.4](#).

22.3.5 How to Select Between Single Conversion Sequence or Continuous Conversions

Each group has its own mode control register. The MODE field of these control registers allow the application to select between a single conversion sequence or continuous conversion mode.

NOTE: Selecting continuous conversion mode for all three groups

All three conversion groups cannot be configured to be in a continuous conversion mode. If the application configures the group mode control registers to enable continuous conversion mode for all three groups, then the Group2 will be automatically be configured to be in a single conversion sequence mode.

With conversions ongoing in continuous conversion mode, if the MODE field of a group is cleared, then that group switches to the single conversion sequence mode. Conversions for this group will stop once all channels selected for that group have been converted.

22.3.6 How to Start a Conversion

The conversion groups Group1 and Group2 are software-triggered by default. A conversion in these groups can be started just by writing the desired channels to the respective Channel-Select Registers. For example, in order to convert channels 0, 1, 2, and 3 in Group1 and channels 8, 9, 10, and 11 in Group2, the application just has to write 0x0000000F to ADG1SEL and 0x00000F00 to ADG2SEL. The ADC module will start by servicing the group that was triggered first, Group1 in this example.

The conversions for all groups are performed in ascending order of the channel number. For the Group1 the conversions will be performed in the order: channel 0 first, followed by channel 1, then channel 2, and then channel 3. The Group2 conversions will be performed in the order: channels 8, 9, 10, and 11.

The Event Group is only hardware-triggered. There are up to eight hardware event trigger sources defined for the ADC module. Check the device datasheet for a complete listing of these eight hardware trigger options.

The trigger source to be used needs to be configured in the ADEVSR register. Similar registers also exist for the Group1 and Group2 as these can also be configured to be event-triggered.

The polarity of the event trigger is also configurable, with a falling edge being the default.

An Event Group conversion starts when at least one channel is selected for conversion in this group, and when the defined event trigger occurs.

If any conversion group is configured to be in a continuous conversion mode, then it needs to only be triggered once. All the channels selected for conversion in that group will be converted repeatedly.

22.3.7 How to Know When the Group Conversion is Completed

Each conversion group has a status flag to indicate when its conversion has ended. See ADEVSR, ADG1SR, and ADG2SR. This bit is set when a conversion sequence for a group ends. This bit does is always set if a group is configured for continuous conversions.

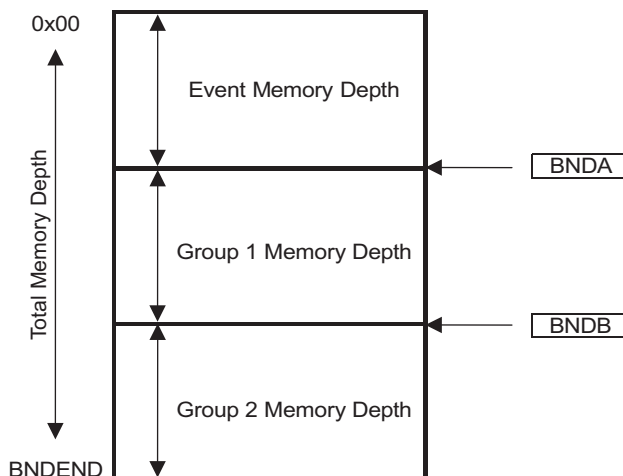
22.3.8 How Results are Stored in the Results' Memory

The ADC stores the conversion results in three separate memory regions in the ADC Results' RAM, one region for each group. Each memory region is a stack of buffers, with each buffer capable of holding one conversion result. The number of buffers allocated for each group is programmed by configuring the ADC module registers ADBNDCR and ADBNDEND.

ADBNDCR contains two 9-bit pointers BNDA and BNDB. BNDA, BNDB, and BNDEND are used to partition the total memory available into three memory regions as shown in [Figure 22-3](#). Both BNDA and BNDB are pointers referenced from the start of the results' memory. BNDA specifies the number of buffers allocated for the Event Group conversion results in units of two buffers; BNDB specifies the number of buffers allocated for the Event Group plus Group1 in units of two buffers. Refer to for more details on configuring the ADC results' memory.

ADBNDEND contains a 3-bit field called BNDEND that configures the total memory available. The ADC module can support up to 1024 buffers. The device supports a maximum of 64 buffers for both the ADC modules.

Figure 22-3. FIFO Implementation



- Number of buffers for Event Group = $2 \times \text{BNDA}$
- Number of buffers for Group1 = $2 \times (\text{BNDB} - \text{BNDA})$
- Number of buffers for Group2 = Total number of buffers – $2 \times \text{BNDB}$

22.3.9 How to Read the Results from the Results' Memory

The CPU can read the conversion results in one of two ways:

1. By using the conversion results memory as a FIFO queue
2. By accessing the conversion results memory directly

22.3.9.1 Reading Conversion Results from a FIFO

The conversion results for each group can be accessed via a range of addresses provided to facilitate the use of the ARM Cortex-R4 CPU's Load-Multiple (LDM) instruction. A single read performed using the LDR instruction can also be used to read out a single conversion result. The results are read out from the group's memory region as a FIFO queue by reading from any location inside this address range. The conversion result that got stored first gets read first. A result that is read from the memory in this method is removed from the memory. For example, a read from any address in the range ADEVBUFFER (offset 90h to AFh) pulls out one conversion result from the Event Group memory.

Figure 22-4. Format of Conversion Result Read from FIFO, 12-bit ADC

Offset Address Register		31 15	30 14	29 13	28 12	27 11	26 10	25 9	24 8	23 7	22 6	21 5	20 4	19 3	18 2	17 1	16 0
0x90 to 0xAF ADEVBUFFER	EV EMPTY	Reserved											EV_CHID				
	Reserved					EV_DR											
0xB0 to 0xCF ADG1BUFFER	G1 EMPTY	Reserved											G1_CHID				
	Reserved					G1_DR											
0xD0 to 0xEF ADG2BUFFER	G2 EMPTY	Reserved											G2_CHID				
	Reserved					G2_DR											

Figure 22-5. Format of Conversion Result Read from FIFO, 10-bit ADC

Offset Address Register		31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x90 to 0xAF ADEVBUFFER	Reserved																
	EV_EMPTY	EV_CHID						EV_DR									
0xB0 to 0xCF ADG1BUFFER	Reserved																
	G1_EMPTY	G1_CHID						G1_DR									
0xD0 to 0xEF ADG2BUFFER	Reserved																
	G2_EMPTY	G2_CHID						G2_DR									

Option to read channel id along with conversion result:

The application has an option to read the channel id along with the conversion result. This is controlled by the CHID field of the group's mode control register. If the option to read the channel id is not selected, the channel id field of the conversion result reads as zeros.

Protection against reading from empty FIFO:

There is also a hardware mechanism to protect the application from reading past the number of new conversion results held in the FIFO. Once all available conversion results have been read out of the FIFO by the application, a subsequent read from the FIFO causes the mechanism to indicate that the FIFO is empty by setting the EMPTY field.

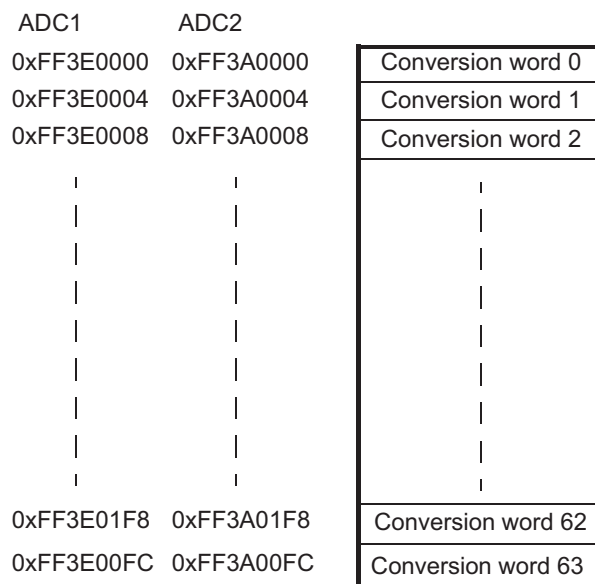
Debug / Emulation Support:

For debug purposes, each conversion group also provides an address that the application can read from for extracting the group's conversion results. However, no status flags for a conversion group are affected by reading from these emulation buffer addresses. For example, reading from ADEVEMUBUFFER (offset 0xF0) returns the next result in the Event Group buffer but does not actually remove that result from the buffer or change the amount of data held in the buffer.

22.3.9.2 Reading Conversion Results Directly from the Conversion Results' Memory

The conversion result memory is part of the device's memory map. The base address for the ADC1 result memory is 0xFF3E0000, while that for the ADC2 result memory is 0xFF3A0000.

Figure 22-6. ADC Memory Mapping



The application can identify the address ranges for each of the three memory regions for the three conversion groups after performing the segmentation as described in [Section 22.3.8](#). It is up to the application to read the desired results from the three conversion groups. The formats of the conversion results when reading from RAM directly are shown in [Figure 22-7](#) and [Figure 22-8](#).

Figure 22-7. Format of Conversion Result Directly Read from ADC RAM, 12-bit ADC

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
ADC RAM address	Reserved															channel id [4]	
	channel id [3-0]					12-bit conversion result											

Figure 22-8. Format of Conversion Result Directly Read from ADC RAM, 10-bit ADC

	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
ADC RAM address	Reserved															channel id [4-0]	
	Rsrd		channel id [4-0]					10-bit conversion result									

Note that there is no EMPTY field to protect the application from reading data that has been previously read.

Each group does have a separate register which holds the address in the group's result memory where the ADC will write the next conversion result. These are the ADEV RAMWRADDR, ADG1RAMWRADDR, and ADG2RAMWRADDR registers. The application can use this information to calculate how many valid conversion results are available to be read.

Benefit of reading conversion results directly from ADC RAM:

The application does not have to read out conversion results sequentially as in the case of reading from a FIFO. As a result, the application can selectively read the conversion results for any particular input channel of interest without having to read other channels' conversion results.

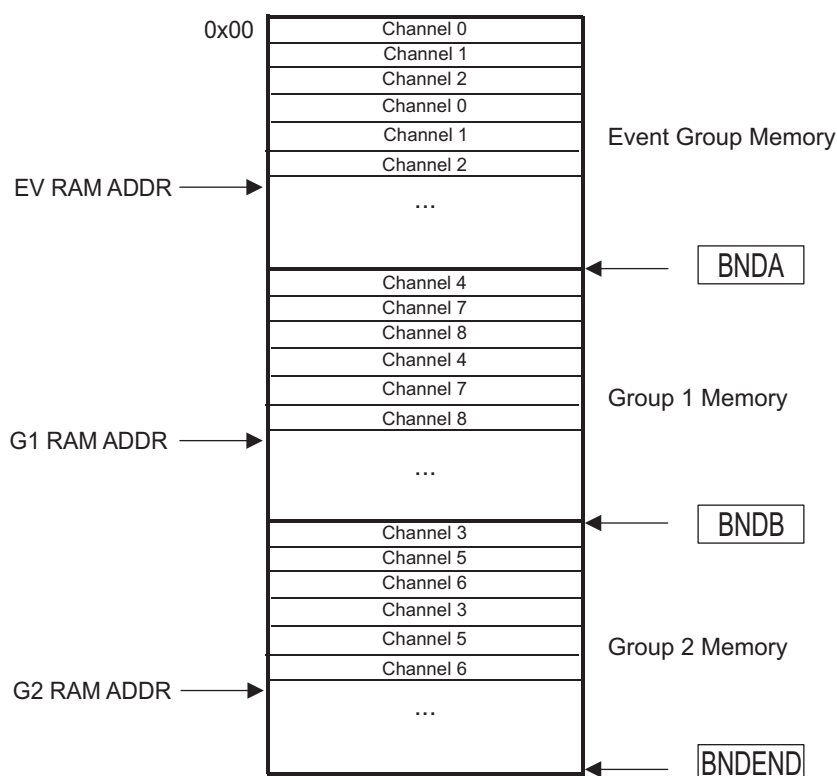
22.3.9.3 Example

Suppose that channels 0, 1, and 2 are selected for conversion in the Event Group, channels 4, 7, and 8 are selected for conversion in group 1, and channels 3, 5, and 6 are selected for conversion in group 2. The conversion results will get stored in the three memory regions as shown in Figure 22-9.

Suppose that the CPU wants to read out the results for the Event Group from a FIFO queue. The CPU needs to read from any address in the range ADEVBUFFER (offset 0x90 to 0xAF) multiple times, or do a "load multiple" from this range of addresses. This will cause the ADC to return the results for channel 0, then channel 1, then channel 2, then channel 0, and so on for each read access to this address range.

Now suppose that the application wants to read out the results for the group 1 from the RAM directly. The conversion results for the group 1 are accessible starting from address ADC RAM Base Address + BNDA. Also, it is known that the first result at this address is for the input channel 4, the next one is for input channel 7, and so on. So the application can selectively read the conversion results for only one channel if so desired.

Figure 22-9. Conversion Results Storage



22.3.10 How to Stop a Conversion

A group's conversion can be stopped by clearing the group's channel select register.

22.3.11 Example Sequence for Basic Configuration of ADC Module

The following sequence is necessary to configure the ADC to convert channels 0, 2, 4, and 8 in single-conversion mode using Group1:

1. Write 0 to the Reset Control Register (ADRSTCR) to release the module from the reset state
2. Write 1 to the ADC_EN bit of the Operating Mode Control Register (ADOPMODECR) to enable the ADC state machine
3. Configure the ADCLK frequency by programming the desired divider into the Clock Control Register (ADCLOCKCR)
4. Configure the acquisition time for the group that is to be used. For example, configure the Group1 Sampling Time Control Register (ADG1SAMP) to set the acquisition time for Group1.
5. Select the channels that need to be converted in Group1 by writing to the Group1 Channel Select Register (ADG1SEL). In this example, a value of 0x115 needs to be written to ADG1SEL in order to select channels 0, 2, 4, and 8 for conversion in Group1.
 - The ADC sequencer will start the Group1 conversions as soon as the write to the ADG1SEL register is completed.
6. Wait for the G1_END bit to be set in the Group1 Conversion Status Register (ADG1SR). This bit gets set when all the channels selected for conversion in Group1 are converted and the results are stored in the Group1 memory.
7. Read the conversion results by reading from the Group1 FIFO access location (ADG1BUFFER) or by reading directly from the Group1 results' memory.

22.4 Advanced Conversion Group Configuration Options

Figure 22-10 shows the operating mode control registers and the status registers for each of the three conversion groups. The register addresses shown are offsets from the base address. The ADC1 register frame base address is FFF7 C000h and the ADC2 register frame base address is FFF7 C200h.

Figure 22-10. ADC Groups' Operating Mode Control and Status Registers

Offset Address Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x010 ADEVMODECR	Reserved															No Reset On ChnSel
	Reserved						EV_ DATA_FMT	Reserved	EV_ CHID	OVR_ EV_ RAM_ IGN	Rsvd	EV_ 8BIT	EV_ MODE	FRZ_ EV		
0x014 ADG1MODECR	Reserved															No Reset On ChnSel
	Reserved						G1_ DATA_FMT	Reserved	G1_ CHID	OVR_ G1_ RAM_ IGN	Rsvd	G1_ 8BIT	G1_ MODE	FRZ_ G1		
0x018 ADG2MODECR	Reserved															No Reset On ChnSel
	Reserved						G2_ DATA_FMT	Reserved	G2_ CHID	OVR_ G2_ RAM_ IGN	Rsvd	G2_ 8BIT	G2_ MODE	FRZ_ G2		
0x06C ADEVSR	Reserved															
	Reserved											EV_ MEM_ EMPTY	EV_ BUSY	EV_ STOP	EV_ END	
0x070 ADG1SR	Reserved															
	Reserved											G1_ MEM_ EMPTY	G1_ BUSY	G1_ STOP	G1_ END	
0x074 ADG2SR	Reserved															
	Reserved											G2_ MEM_ EMPTY	G2_ BUSY	G2_ STOP	G2_ END	
0x19C ADEVCURRCOUNT	Reserved															
	Reserved											EV_CURRENT_COUNT				
0x1A0 ADEVMAXCOUNT	Reserved															
	Reserved											EV_MAX_COUNT				
0x1A4 ADG1CURRCOUNT	Reserved															
	Reserved											G1_CURRENT_COUNT				
0x1A8 ADG1MAXCOUNT	Reserved															
	Reserved											G1_MAX_COUNT				
0x1AC ADG2CURRCOUNT	Reserved															
	Reserved											G2_CURRENT_COUNT				
0x1B0 ADG2MAXCOUNT	Reserved															
	Reserved											G2_MAX_COUNT				

22.4.1 Group Trigger Options

The Group1 and Group2 operating mode control registers have an extra control bit: HW_TRIG. This bit configures the group to be hardware event-triggered instead of software-triggered, which is the default.

When a group is configured to be event-triggered, the group conversion starts when at least one channel is selected for conversion in this group, and when the defined event trigger occurs. The event trigger source is defined for each group in the ADEVSRC, ADG1SRC, and the ADG2SRC registers. The actual connections used as the event trigger sources are defined in the device datasheet for both the ADC modules.

22.4.2 Analog Input Channel Selection Mode Options

The ADC modules on this device support two different modes for selecting the analog input channel to be converted. These are the sequential channel selection mode (default), and the enhanced channel selection mode. These two modes are now described:

22.4.2.1 Sequential Channel Selection Mode

This is the default mode and allows the ADC module to be used in a backwards compatible mode to the ADC module on other Hercules ARM Safety MCUs from Texas Instruments. As discussed in [Section 22.3.4](#), an analog input channel can be selected for conversion in one or more conversion groups by setting the bit corresponding to that channel number in the group's channel select register.

22.4.2.2 Enhanced Channel Selection Mode

There are some important concepts related to the enhanced channel selection mode. These are defined first:

- Look-Up Table

This is a 32-word deep memory-mapped region used to define the analog input channel number to be converted. The LUTs for the three groups are stacked together so that the entire LUT occupies 96 words. Each word is aligned on a 32-bit boundary. The LUTs for ADC1 start at 0xFF3E2000 and the LUTs for ADC2 start at 0xFF3A2000.

- Conversion Group Sub-Sequence

A group sub-sequence is defined as the conversion for a set of channels that is converted on each conversion trigger. The number of channels selected for conversion in a group sub-sequence is defined by the number of bits that are set in the group's channel select register. For example, setting bits 0, 1, 29, and 31 in ADG1SEL means that each Group1 conversion sub-sequence consists of 4 conversions.

- LUT Index

A "CURRENT_COUNT" register for each group is maintained as an index into that group's LUT. This register increments each time a channel conversion is completed. Therefore, as its name suggests, a read from this register returns the number of conversions completed since the last write to the group's channel select register. The CURRENT_COUNT register resets to all zeros under any of the following conditions:

1. The ADC peripheral is reset via a global peripheral reset
2. The ADC peripheral is reset via the ADC Reset Control Register,
3. The CURRENT_COUNT becomes equal to the MAX_COUNT defined for that conversion group
4. The application writes zeros to the CURRENT_COUNT register
5. The conversion group's result RAM is reset

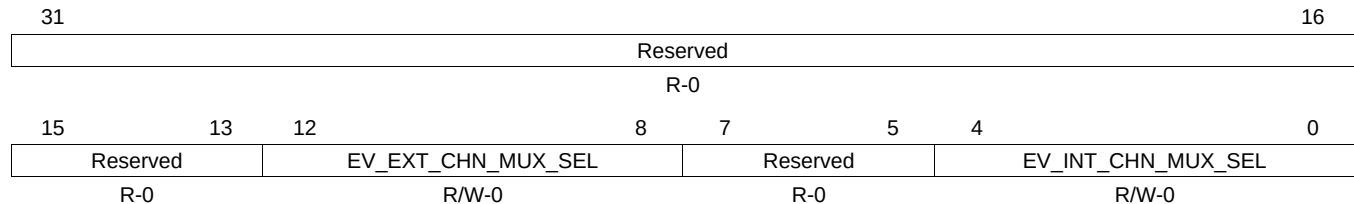
- Maximum Number of Conversions

A "MAX_COUNT" register for each conversion group stores the maximum number of conversions to be performed before the index into a group's LUT is reset to zero. This register can be programmed to a value between 0 and 31. It is recommended to program the MAX_COUNT register with a value that is one less than a multiple of the number of channels in that group's conversion sub-sequence (number of bits that are set in the group's channel select register).

22.4.2.2.1 Look-Up Table Details

As described earlier, each conversion group has a look-up table (LUT) that is used when the enhanced channel selection mode is enabled. This look-up table starts at an offset of 8kB from the base of the ADC results RAM. The LUT holds 32 entries for each of the three conversion groups. The first 32 entries are for the event group, the next 32 entries are for Group1 and the last 32 entries are for Group2. [Figure 22-11](#) shows an example LUT entry for the Event group.

Figure 22-11. Example Look-Up Table Entry



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 22-1. ADC Look-Up Table Field Descriptions

Bit	Field	Value	Description
31–13	Reserved	0	Reads return zeros, writes have no effect.
12–8	EV_EXT_CHN_MUX_SEL		This field defines the external analog mux select that is output from the ADC module when the Event group CURRENT_COUNT register points to this LUT entry, and when the Event group conversion is triggered with the enhanced channel selection mode enabled.
7–5	Reserved	0	Reads return zeros, writes have no effect.
4–0	EV_INT_CHN_MUX_SEL		This field defines the internal analog mux select that is output from the ADC module when the Event group CURRENT_COUNT register points to this LUT entry, and when the Event group conversion is triggered with the enhanced channel selection mode enabled. This can be a value between 0 and 31, which corresponds to the internal analog input channel number between 0 and 31. Note that this device only supports 24 input channels for ADC1 and 16 input channels for ADC2. If the application configures an unavailable channel number in the EV_INT_CHN_MUX_SEL field, the ADC will still perform the conversion and the result will be indeterminate.

Consider the example conversion Group1 configuration shown in [Figure 22-12](#). Only bits 0 and 31 of ADG1SEL are set. Assume that all other bits in this register are zeros.

In case of the default sequential channel selection mode, the write to the ADG1SEL register would cause the Group1 conversions to start with channel 0 followed by channel 31. The conversions would then stop or repeat in this order depending on whether Group1 is in single or continuous conversion mode.

Figure 10 is a block diagram illustrating the internal structure of the ADGxSEL register and its connection to the ADC. The diagram shows the following components and connections:

- On-chip Input "Multiplexer":** This block receives inputs from the external 8:1 multiplexers and the internal channel select. It outputs to the ADC Sample/Hold Circuit.
- Internal Channel Select, 32 bits:** This block receives the 32-bit internal channel select and outputs to the On-chip Input Multiplexer.
- 1-bit Enable or nEnable for ext. channel mux:** This block receives the 1-bit enable signal and outputs to the On-chip Input Multiplexer.
- 5-bit Select for ext. channel mux:** This block receives the 5-bit select signal and outputs to the On-chip Input Multiplexer.
- Channel Identifiers:** This block contains a table mapping external and internal channel identifiers to the LUT index.
- LUT index, 0 to 31:** This block receives the LUT index and outputs to the On-chip Input Multiplexer.
- Current Count:** This block receives the current count and outputs to the On-chip Input Multiplexer.
- Reset when Current Count = Max Count:** This block receives the current count and outputs to the On-chip Input Multiplexer.
- Max Count:** This block receives the maximum count and outputs to the On-chip Input Multiplexer.

The diagram also shows the internal structure of the register, including the On-chip Input Multiplexer, Internal Channel Select, 1-bit Enable or nEnable for ext. channel mux, 5-bit Select for ext. channel mux, and the LUT (Look-Up Table) index. The LUT index is used to select the internal channel select and the 5-bit select for the external channel mux.

Now suppose that the application has enabled the enhanced channel selection mode for Group1 with the G1_MAX_COUNT register configured to be 3. Also suppose that the application has programmed the Group1 LUT as shown in [Figure 22-12](#). Now suppose that the application triggers Group1 conversions by writing 0x80000001 to ADG1SEL, that is, bits 0 and 31 are set and all others are zeros. The ADC conversions will proceed in the following sequence:

- **Input Channel Selection**

The initial value of G1_CURRENT_COUNT is 0, which is used as the index into the Group1 LUT. The row 0 of Group1's LUT has values of 1 for the G1_EXT_CHN_MUX_SEL and a value of 1 for the G1_INT_CHN_MUX_SEL. The 5-bit external channel id of 0b00001 is driven out on the AWM1_EXT_SEL terminals. This selects channel 1 for all the connected external analog multiplexors, as shown in [Figure 22-12](#). The ADC module also outputs an enable signal to the external analog multiplexors via the AWM1_EXT_ENA terminal.

Now consider the fact that the internal channel id is also configured to be 1 in row 0 of the Group1 LUT. This causes the switch for ADC's internal channel 1 (ADIN1) to be closed. All other internal ADC input switches (ADIN0, ADIN2, ADIN3, ..., ADIN31) will be open. Note that the ADIN1 input channel is actually connected to the output of an 8:1 analog multiplexor.

In effect, the ADC will convert channel 1 of the 8:1 analog multiplexor connected to the ADIN1 terminal of the microcontroller.

- **After Completion of Conversion**

Once the first conversion is completed, the CURRENT_COUNT value of 0 is stored in the "channel id" field of the conversion result RAM of Group1 along with the actual conversion result from the ADC core. Then the G1_CURRENT_COUNT value of 0 is compared against the G1_MAX_COUNT value of 3. The values do not match, so that G1_CURRENT_COUNT is incremented from 0 to 1.

- **Next Channel Selection**

There are two bits set in the ADG1SEL register, so that the ADC module now uses the G1_CURRENT_COUNT value of 1 to index the Group1 LUT. As shown in [Figure 22-12](#), this row in Group1 LUT contains 4 as the G1_EXT_CHN_MUX_SEL and 2 as the G1_INT_CHN_MUX_SEL.

ADC input channel ADIN2 is not connected to any external analog multiplexor and is connected directly to the analog signal to be converted. Note that the ADC module still drives the AWM1_EXT_ENA and the AWM1_EXT_SEL (value of 4, that is, 0b00100) to all the external analog multiplexors connected to the microcontroller.

- **End of Conversion Sub-Sequence**

Once the conversion of the internal channel ADIN2 is completed, the G1_CURRENT_COUNT of 1 is stored in the "channel id" field of the Group1 result RAM along with the actual conversion result. This value of 1 is compared against the G1_MAX_COUNT value of 3. The values do not match, so that G1_CURRENT_COUNT is incremented from 1 to 2.

There are no more conversions required in this sub-sequence as only two bits are set in ADG1SEL.

- **Continuation on Next Group1 Trigger**

When the ADC Group1 is triggered again or if Group1 is in continuous conversion mode, the G1_CURRENT_COUNT of 2 is again used to index the Group1 LUT. Following the same reasoning as before, this will cause the channel 1 of the 8:1 analog multiplexor connected to ADIN1 to be converted.

Once this conversion is done, the G1_CURRENT_COUNT value of 2 is stored in the "channel id" field of the result RAM along with the conversion result. This still does not match the G1_MAX_COUNT of 3, so that G1_CURRENT_COUNT is now incremented from 2 to 3.

This index value of 3 is used to again convert channel ADIN2, following the same reasoning as before.

When this conversion is completed, the G1_CURRENT_COUNT of 3 is stored as the "channel id" field of the result RAM along with the conversion result.

Also, now this G1_CURRENT_COUNT value of 3 matches the G1_MAX_COUNT. This resets the G1_CURRENT_COUNT to zero.

The above sequence proceeds as described whenever Group1 is next triggered, or if Group1 is configured to be in a continuous conversion mode.

22.4.3 Single or Continuous Conversion Modes

The EV_MODE, G1_MODE, and G2_MODE bits are used to select between either single or continuous conversion mode for each of the three groups.

22.4.3.1 Single Conversion Mode

A conversion group configured to be in single-conversion mode gets serviced only once by the ADC for each group trigger. The trigger can be a software trigger as in the case of Group1 and Group2 by default, or it could be a hardware event trigger as in the case of the Event Group or Group1 or Group2.

The entire conversion sequence, from the acceptance of the group conversion request to the end of the last channel's conversion, is flagged for each group by the corresponding BUSY bit in that group's status register. After single-conversion mode is started, the BUSY bit is read as 1 until the conversion of the last channel is complete. The END bit for the group is set once all the channels in that group are converted.

For example, say channels 0, 2, 4, and 6 are selected for conversion in Group1 in single-conversion mode. When the Group1 gets serviced, the ADC will start conversion for channel 0, then channel 2, then channel 4, and then channel 6. It will then stop servicing the Group1, set the G1_END status bit, and look to service the Event Group or the Group2, if required.

22.4.3.2 Continuous Conversion Mode

A conversion group configured to be in continuous-conversion mode gets serviced by the ADC continuously. The group still needs to be triggered appropriately for the first conversion to start. The conversions are performed continuously thereafter.

The entire conversion sequence, from the acceptance of the group conversion request to the end of the last channel's conversion, is flagged for each group by the corresponding BUSY bit in that group's status register. After continuous-conversion mode is started, the BUSY bit is read as 1 as long as the continuous-conversion mode for this group is selected.

As an example, say the channels 0, 2, 4, and 6 are selected for conversion in Group1, now in continuous-conversion mode. When the Group1 gets serviced, the ADC will complete conversions for channels 0, 2, 4 and 6, and then look to service the Event Group or the Group2. Once it is done servicing the Event Group or the Group2, it will return to service the Group1 again. The Group1 does not need to be triggered again for the repeated conversion.

NOTE: Configuring all conversion groups in continuous conversion mode

All the three groups cannot operate in continuous-conversion mode at the same time. If the application program configures all three groups to be in continuous-conversion mode, the Group2 is automatically reset to single-conversion mode, and the G2_MODE bit in the ADG2MODECR register is cleared to reflect the single-conversion mode of Group2.

22.4.4 Conversion Group Freeze Capability

The ADC module has an inherent priority order between the three conversion groups. This group priority determines the order of conversion in case multiple groups are triggered. The priority of conversions between the three groups in descending order is:

1. Event Group
2. Group1
3. Group2

Examples of conversion group priority:

- If an Event Group conversion is ongoing in single conversion sequence mode and Group2 and Group1 conversions are requested, then the ADC will finish conversion of channels selected in Event Group, then switch over to converting channels selected in Group1, and then convert channels selected in Group2.
- If Group1 conversions are ongoing in continuous conversion mode and Group2 conversion is requested, then the ADC will complete converting the current channel for Group1 and switch over to converting channels selected in Group2. The new conversion request for Group2 has a higher priority than the pending continuous conversion request for Group1.

The conversion group freeze capability allows the application to override this default priority between the conversion groups. Enabling the freeze capability allows the ADC to freeze a higher-priority conversion group's conversions whenever there is a request for conversion in another (lower-priority) group.

For example, setting the FRZ_EV bit in the ADEVMODECR register will allow the ADC to freeze ongoing Event Group conversions whenever there is a pending request, or a new request for a Group1 or Group2 conversion. The conversions for the Event Group will be frozen as long as the Group1 or Group2 conversions are active. Once the Group1 or Group2 conversions are completed, the Event Group conversions start from where they were frozen.

While a group's conversions are frozen, the group's STOP status bit is set. This bit is cleared once the group's conversions are restarted.

22.4.5 Conversion Group Memory Overrun Option

An overrun condition occurs when the ADC module tries to store more conversion results to a group's results' memory which is already full. In this case, the ADC allows two options.

If the OVR_RAM_IGN bit in the group's operating mode control register (ADEVMODECR, ADG1MODECR, ADG2MODECR) is set, then the ADC module ignores the contents of the group's results' memory and wraps around to overwrite the memory with the results of new conversions.

If the OVR_RAM_IGN bit is not set, then the application program has to read out the group's results' memory upon an overrun condition; only then can the ADC continue to write new results to the memory.

22.4.6 Response on Writing Non-Zero Value to Conversion Group's Channel Select Register

If the application writes a non-zero value to a group's channel select register while that group's conversions are already being serviced, then that group's conversions will be restarted with the new configuration programmed in the channel select registers.

The following rules apply in terms of the effect on the ADC conversion sequence:

- If the new conversion request comes from the same group as the ongoing conversion, then the ongoing conversion will be stopped in whichever stage it is in, and the new sequence of conversions will be started.
- If the new conversion request comes from a separate group, then the ongoing channel's conversion will be completed before starting the new sequence of conversions.

The following rules apply in terms of the effect on the group's results memory:

- If a group conversion is ongoing or is frozen, writing a non-zero value to the group's channel select register will also reset its results FIFO. This does not clear the contents of the results FIFO; only the ADC module is allowed to overwrite the FIFO's contents with new conversion results starting from the first location.
- If the group conversion is completed (<GRP>_END flag is set), or the group is not being used, then writing a non-zero value to the group's channel select register will either be reset or not depending on the value of the No_Reset_on_ChSel bit for that group (ADEVMODECR, ADG1MODECR, ADG2MODECR).
 - If the No_Reset_on_ChSel bit is 0, then the group's FIFO will be reset.
 - If the No_Reset_on_ChSel bit is 1, then the group's FIFO will not be reset.

22.4.7 Conversion Result Size on Reading: 8-bit, 10-bit, or 12-bit

Some applications do not need the full 12-bit resolution of the ADC modules on the device and can work with 8-bit or 10-bit conversion results.

22.4.7.1 ADC Configured in 12-bit Resolution

The mode control register for each conversion group contains a field called DATA_FMT, which defines the format of the conversion result read out of the result RAM, when accessed as a FIFO.

The DATA_FMT field is encoded as follows:

- If DATA_FMT = 00, the complete 12-bit conversion result is read out of the FIFO.
- If DATA_FMT = 01, the 12-bit conversion result is right-shifted by 2 and the resulting 10-bit result is read out of the FIFO.
- If DATA_FMT = 10, the 12-bit conversion result is right-shifted by 4 and the resulting 8-bit result is read out of the FIFO.

This control field is not effective when the application chooses to access the conversion result memory directly. In that case, the application can choose to mask off the number of bits as required.

22.4.7.2 ADC Configured in 10-bit Resolution

The DATA_FMT field is not effective in this mode and the application has the choice to read either the full 10-bit conversion result or an 8-bit conversion result. This is controlled by the 8BIT field of the group's operating mode control register.

- If 8BIT = 0, the complete 10-bit conversion result is read out of the FIFO.
- If 8BIT = 1, the 10-bit conversion result is right-shifted by 2 and the resulting 8-bit result is read out of the FIFO.

22.4.8 Option to Read Group Channel Id Along With Conversion Result

The ADC module allows the application program to also read out the analog input channel number along with its conversion result. This capability is enabled by setting the CHID bit in the group's operating mode control register.

- If CHID = 0, the bits [14-10] are forced to 00000 when the conversion results are read out from the group's results' FIFO.
- If CHID = 1, the bits [14-10] in the group's results' memory contain the input channel number to which the conversion result belongs.

NOTE: Actual Storage of Channel ID

Regardless of whether the CHID bit is set or not, the channel number is **always stored** in the memory along with the conversion result. The CHID bit only affects whether the channel number is available with the conversion result **when the group's memory is read**. Therefore, the CHID bit for a group can be changed dynamically without affecting that group's ongoing conversions.

22.5 ADC Module Basic Interrupts

This section describes the basic interrupts generated by the ADC module.

22.5.1 Group Conversion End Interrupt

The ADC module sets the group's conversion end flag (EV_END, G1_END, or G2_END) in that group's interrupt flag register (ADEVINTFLG, ADG1INTFLG, ADG2INTFLG) when all the channels selected for conversion in that group are converted. This causes a group conversion end interrupt to be generated if this interrupt is enabled by setting the group's END_INT_EN control bit (EV_END_INT_EN, G1_END_INT_EN, or G2_END_INT_EN).

This interrupt can be easily used for conversion groups configured to be in the single-conversion mode. The application program can read out the conversion results, change the group's configuration if necessary, and restart the conversions by triggering the group from within the interrupt service routine.

For groups configured to be in continuous conversion mode, this interrupt condition is not practical as the conversions are always in progress. In this case, the Group Memory Threshold Interrupt is more practical as the application can allow a programmable number of conversion results to accumulate before interrupting the CPU.

22.5.2 Group Memory Threshold Interrupt

The ADC module has the ability to generate an interrupt for a fixed number of conversions for each group. A group memory threshold register determines how many conversion results must be in a group's memory region before the CPU is interrupted. This feature can be used to significantly reduce the CPU load when using interrupts for reading the conversion results.

The group's threshold register needs to be configured before the group conversions are triggered. This threshold register value behaves like a down-counter, which decrements each time the ADC writes a conversion result to this group's memory. This counter is incremented each time the application program reads a conversion result from the results' memory by accessing the FIFO queue. Simultaneous read (by application program) and write (by ADC module) operations from the group's results' memory leave the threshold counter unchanged.

The threshold counter can decrement past 0 and become negative. It always increments back to its original value when the memory region is emptied. To determine how many samples are in the memory region at a given moment, the threshold counter can be subtracted from the originally configured threshold count.

Whenever the threshold counter transitions from +1 to 0, it sets the group's threshold interrupt flag, and the CPU is interrupted if the group's threshold interrupt is enabled. The CPU is expected to clear the interrupt flag after reading the conversion results from the memory.

The interrupt flag is not set when the threshold counter stays at 0 or transitions from -1 to 0.

22.5.3 Group Memory Overrun Interrupt

An interrupt can be generated for each group if the number of ADC conversions for that group exceed the number of buffers allocated for that conversion group. The application program can choose to read out all the conversion results using the CPU or the DMA. Alternatively, the application program can set the group's OVR_RAM_IGN bit and allow the ADC module to overwrite the group's results' memory contents with new conversion results.

22.6 ADC Module DMA Requests

This section describes the capabilities of the ADC module to take advantage of the Platform DMA controller module. The ADC module can generate a DMA request under two conditions:

22.6.1 DMA Request for Each Conversion Result Written to the Results' Memory

In this mode, the ADC module will generate the first DMA request as soon as a conversion result gets written to the group's results' memory. Subsequent writes to the results' memory will cause DMA requests to be generated. This mode allows a smaller amount of ADC results' memory to suffice for an application.

This DMA request generation is enabled by setting the group's DMA_EN bit in the group's DMA control register. The BLK_XFER bit in this register must be left cleared (default) if a DMA request is desired to be generated for new results getting written to the results' memory.

22.6.2 DMA Request for a Fixed Number of Conversion Results

This mode is enabled by setting both the group's DMA_EN and the group's BLK_XFER bits in the group's DMA control registers.

In this mode, a DMA request will be generated for a specified number of conversion results being available in the group's results' memory. The number of conversion results desired are configured using the group's BLOCKS field in the control registers.

For example, if the BLOCK count is configured for 10, then ADC module will generate a DMA request at the end of 10th conversion. DMA controller should complete reading out 10 data before next set of 10 conversions complete.

NOTE: Usage of Block DMA transfers with Threshold Interrupts

It is not recommended to enable the block DMA transfers for a group at the same time as the group threshold interrupt. The group's BLOCKS field is essentially the same as the group's THRESHOLD field in the group's interrupt control register described in [Section 22.5.2](#).

22.7 ADC Magnitude Threshold Interrupts

The ADC allows up to three magnitude threshold interrupts to be generated. The comparison parameters are programmed via the Magnitude Compare Interrupt Control Register (ADMAGINTxCR).

22.7.1 Magnitude Threshold Interrupt Configuration

The following fields are configurable for each of the three available magnitude threshold interrupts:

1. **CHN_THR_COMP**: Specifies whether to compare two channels' conversion results, or to compare a channel's conversion result to a programmable threshold value. A value of 0 will select the programmable threshold to be compared, and a value of 1 will select the conversion result of the channel identified by the **COMP_CHID** field to be compared.
2. **MAG_CHID**: Specifies the channel number from 0 to 23 whose conversion result needs to be monitored.
3. **COMP_CHID**: Specifies the channel number from 0 to 23 whose last conversion result is used for the comparison with the conversion result of the channel being monitored.
4. **MAG_THR**: Specifies the value for comparison with the conversion result of the channel identified by the **MAG_CHID** field.
5. **CMP_GE_LT**: Specifies whether the conversion result of the channel identified by **MAG_CHID** is compared to be "greater than or equal to", or "less than" the reference value. The reference value can be the conversion result of another channel identified by the **COMP_CHID** field, or it could be a threshold value specified in the **MAG_THR** field. A value of 0 in the **CMP_GE_LT** field indicates a "less than" comparison and a value of 1 indicates a "greater than or equal to" comparison.

22.7.2 Magnitude Threshold Interrupt Comparison Mask Configuration

There is also a separate comparison mask register (ADMAGINTxMASK) for each of the three magnitude threshold interrupts. This register is used to specify the bits that are masked off for the sake of the comparison. For example, the lower 4 bits of the conversion result can be masked off by writing 0xf to the interrupt comparison mask register, allowing a gross comparison to be made. By default, the full 10/12-bit conversion results are compared.

22.7.3 Magnitude Threshold Interrupt Enable / Disable Control

Each of the three magnitude interrupts also have separate interrupt enable set (ADMAGINTENASET) and clear (ADMAGINTENACLR) registers. These are used to respectively enable and disable that particular magnitude threshold interrupt from being generated. To enable a magnitude threshold interrupt, write a 1 to the corresponding bit of the interrupt enable set register. Conversely, to disable a magnitude threshold interrupt, write a 1 to the corresponding bit of the interrupt enable clear register.

22.7.4 Magnitude Threshold Interrupt Flags

There is a separate Magnitude Interrupt Flag register (ADMAGINTFLG) that holds the flags for these three interrupts. This flag gets set whenever the comparison condition for the corresponding interrupt is met. A magnitude threshold interrupt is generated if the corresponding flag is set inside the flag register, and the interrupt generation is enabled. This flag can be cleared by writing a 1 to the flag or by reading from the interrupt offset register in case of this interrupt being the current highest-priority pending interrupt.

22.7.5 Magnitude Threshold Interrupt Offset Register

It is possible to have multiple magnitude threshold interrupts pending at the same time. The magnitude threshold interrupt offset register (ADMAGINTOFF) holds the index of the currently pending highest priority magnitude threshold interrupt. The magnitude threshold interrupt 1 has the highest priority while the magnitude threshold interrupt 3 has the lowest priority. This is a read-only register and returns zeros if none of the magnitude threshold interrupts are pending. Writes to this register have no effect.

A read from this register updates the register to the next highest-priority pending magnitude threshold interrupt. This read also clears the corresponding flag from the magnitude threshold interrupt flag register. However, a read from the magnitude threshold interrupt offset register in emulation mode does not affect the interrupt flag register or the interrupt offset register.

22.8 ADC Special Modes

The ADC module supports some special modes for diagnostics and power saving purposes.

22.8.1 ADC Error Calibration Mode

The application program can activate a calibration sequence any time self-test mode is disabled (`SELF_TEST = 0`). This calibration sequence includes the conversion of an embedded calibration reference voltage followed by the calculation of an offset error correction value.

NOTE: Disable Self-Test Mode Before Calibration

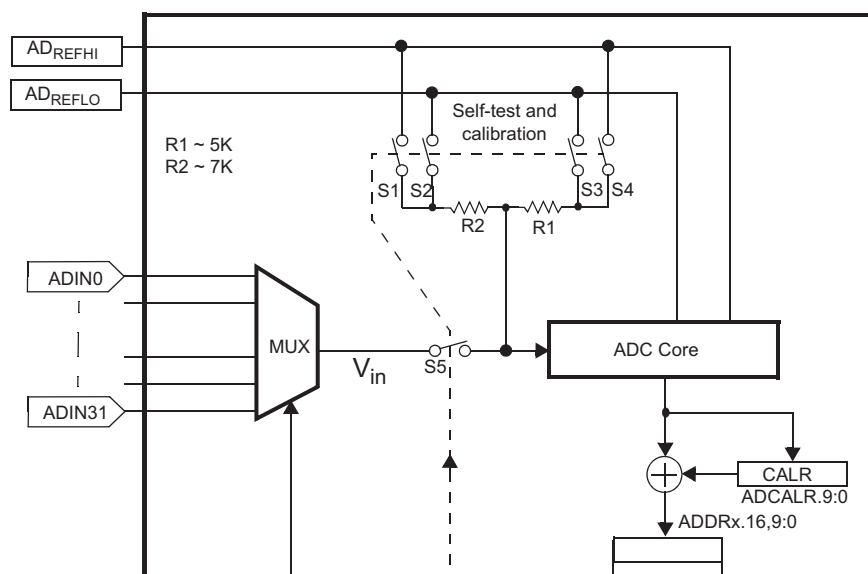
To avoid errors during the calibration operation, self-test mode must *not* be enabled during a calibration sequence. In addition, to ensure accurate results, calibrate the ADC in an environment with minimum noise.

Calibration mode is enabled by setting the `CAL_EN` bit (`ADCALCR.0`). The application needs to ensure that no conversion group is being serviced when the calibration mode is enabled.

The input multiplexor gets disabled and only the reference voltage is connected to the ADC core input. Switch `S5` of [Figure 22-13](#) is opened. In addition, the digital result issued from a conversion is output from the ADC core to the calibration and offset error correction register, `ADCALR`. The ADC results' memory is not affected by the calibration conversion.

When calibration mode is disabled, the ADC can be configured for normal conversions.

Figure 22-13. Self-Test and Calibration Logic



22.8.1.1 Calibration Conversion

The calibration conversion also needs to meet the minimum sampling time specification for the ADC. This value is typically 1 μ s. The Event Group sample time register (`ADEVSAMP`) is used to specify the number of `ADCLK` cycles for the calibration conversion.

The `BRIDGE EN` and `HILO` bits (`ADCALCR.9:8`) control the voltage to the calibration reference device shown in [Figure 22-15](#). The positions of the switches in calibration mode are listed in [Table 22-2](#).

Table 22-2. Calibration Reference Voltages⁽¹⁾

CAL_EN	BRIDGE_EN	HILO	S1	S2	S3	S4	S5	Reference Voltage
1	0	0	1	0	1	0	0	$(AD_{REFHI} \times R1 + AD_{REFLO} \times R2) / (R1 + R2)$
1	0	1	0	1	0	1	0	$(AD_{REFLO} \times R1 + AD_{REFHI} \times R2) / (R1 + R2)$
1	1	0	0	1	1	0	0	AD_{REFLO}
1	1	1	1	0	0	1	0	AD_{REFHI}
0	X	X	0	0	0	0	1	V_{in}

⁽¹⁾ The state of the switches in this table assumes that self-test mode is not enabled.

When CAL_ST (ADCALCR.16) is set, a calibration conversion is started. The voltage source selected via the bits BRIDGE_EN and HILO is converted once (single conversion mode) and the digital result is returned to the calibration and correction register, ADCALR, where it can be read by the CPU. The CAL_ST bit acts as a flag and must be polled by the CPU. It is held set during the conversion process and automatically clears to indicate the end of the reference voltage conversion.

NOTE: No Interrupt for end of calibration

The ADC does not generate an interrupt to signal the end of the calibration conversion. The application must poll the CAL_ST bit to determine the end of the calibration conversion.

After the CAL_ST bit is set by the application program, it can only be reset by the end of the ongoing conversion generated by the ADC core. If the calibration conversion is interrupted (CAL_EN bit is cleared), the CAL_ST bit is held at 1 until a new calibration conversion has been set and completed. Setting the CAL_ST bit while calibration is disabled (CAL_EN = 0) has no effect; however, in this situation, setting CAL_EN immediately starts a calibration conversion. When the calibration conversion is interrupted by an ADC enable (ADC_EN = 0, CAL_EN = 1, and CAL_ST = 1), a new conversion is automatically restarted as soon as the ADC enable bit is released (ADC_EN = 1).

22.8.1.2 Calibration and Offset Error Correction Sequences

The number of measurements and the source to measure for an ADC calibration are application dependent. The CAL_ST bit must be set for each calibration source to be measured. While calibration mode is enabled, any available calibration sources can be converted according to the BRIDGE_EN and HILO bits (see Table 22-2). The digital results of the calibration measurements should be read from ADCALR by the application after each reference conversion so that a correction value can be computed and written back into ADCALR.

When the application has the necessary calibration data, it should compute the offset error correction value and load it into the calibration and correction register, ADCALR. After the CAL_EN bit is cleared, normal conversion mode restarts, continuing from where it was frozen, but with the addition of self-correction data.

In normal mode, the self-correction system adds the correction value stored in ADCALR to each digital result before it is written to the respective group's FIFO.

The basic calibration routine is as follows:

1. Enable calibration via CAL_EN (ADCALCR.0).
2. Select the voltage source via BRIDGE_EN and HILO (ADCALCR.9:8).
3. Start the conversion with CAL_ST (ADCALCR.16).
4. Wait for CAL_ST to go to 0.
5. Get the results from ADCALR and save to memory.
6. Loop to step 2 until the calibration conversion data is collected for the desired reference voltages.
7. Compute the error correction value using calibration data saved in memory.
8. Load the ADCALR register with the 2s complement of the computed error correction value.
9. Disable calibration mode.

At this point, the ADC can be configured for normal operation, and it corrects each digital result with the error correction value loaded in ADCALR.

NOTE: Prevent ADC Calibration Data From Being Overwritten

In calibration mode, the conversion result is written to ADCALR which overwrites any previous calibration data; therefore, the ADCALR register must be read before a new conversion is started.

For no correction, a value of 0x0000 must be written to ADCALR. In noncalibration mode, the ADCALR register can be read and written. Any value written to ADCALR in normal mode (CAL_EN = 0) is added to each digital result from the ADC core.

22.8.1.3 Mid-Point Calibration

Because of its connections to the ADC's reference voltage (VrefHi, VrefLo), the precision of the calibration reference is voltage independent. On the other hand, the accuracy of the switched bridge resistor (R1 and R2) relies on the manufacturing process deviation. Consequently, the mid-point voltage's accuracy can be affected due to the imperfections in the two resistors (expected mismatch error is around 1.5%).

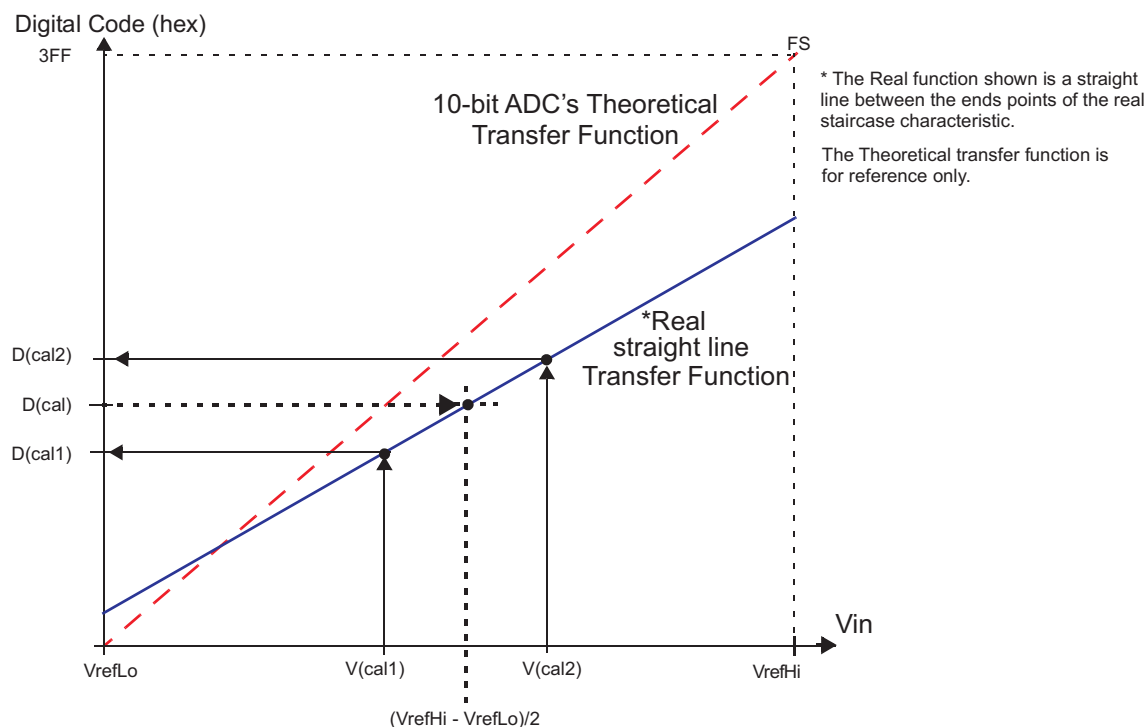
The switched reference voltage device has been specially designed to support a differential measurement of its mid-point voltage. This ensures the accuracy of the mid-point reference, and hence the efficiency of the calibration.

The differential mid-point calibration is software controlled; the algorithm (voltage source measurements and associated calculation) is inserted within the calibration software module included in the application program.

The basic differential mid-point calibration flow is illustrated here after:

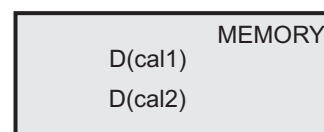
1. The application program connects the voltage VrefHi to R1 and VrefLo to R2, (BRIDGE_EN = 0, HILO = 0), launches a conversion of the input voltage V(cal1), and stores the digital result D(cal1) into the memory.
2. Then the application program switches the voltage VrefHi to R2 and VrefLo to R1 (BRIDGE_EN = 0, HILO = 1), converts this new input voltage V(cal2) and again stores the issued digital result D(cal2) into the memory.
3. The actual value of the real middle point is obtained by computing the average of these two results. $[D(cal1)+D(cal2)] / 2$; [Figure 22-14](#) summarizes the mid-point calibration flow.

Figure 22-14. Mid-point Value Calculation

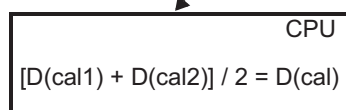


$$V(\text{cal1}) = [V_{\text{REFHI}} \cdot R1 + V_{\text{REFLO}} \cdot R2] / (R1 + R2)$$

$$V(\text{cal2}) = [V_{\text{REFLO}} \cdot R1 + V_{\text{REFHI}} \cdot R2] / (R1 + R2)$$



$$[V(\text{cal1}) + V(\text{cal2})] / 2 = (V_{\text{REFHI}} - V_{\text{REFLO}}) / 2$$

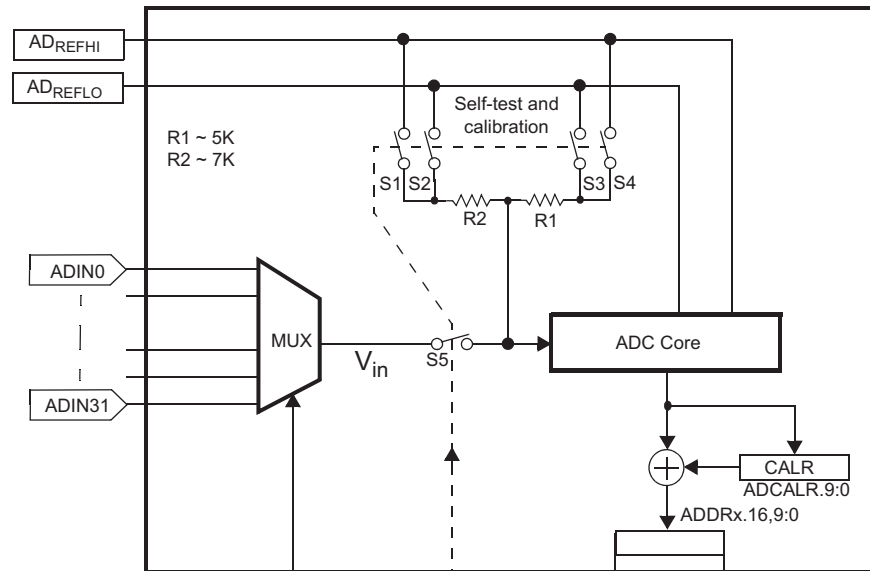


22.8.2 ADC Self-Test Mode

The ADC module supports a self-test mode which can be used to detect an open or a short on the ADC input channels. Self-test mode is enabled by setting the SELF_TEST bit (ADCALCR.24). Any conversion type (continuous or single conversion, freeze enabled or non-freeze enabled, interrupts enabled or disabled) can be performed in this mode.

In normal mode, setting the self-test mode while a conversion sequence is in process can corrupt the current channel conversion results. However, the next channel in the sequence is converted correctly during the additional self-test cycle. The logic associated with both self-test and calibration is shown in Figure 22-15.

Figure 22-15. Self-Test and Calibration Logic



In self-test mode, a test voltage defined by the HILO bit (ADCALCR.8) is provided to the ADC core input through a resistor (see Table 22-3). To change the test source, this bit can be toggled before any single conversion mode request. Changing this bit while a conversion is in progress can corrupt the results if the source switches during the acquisition period.

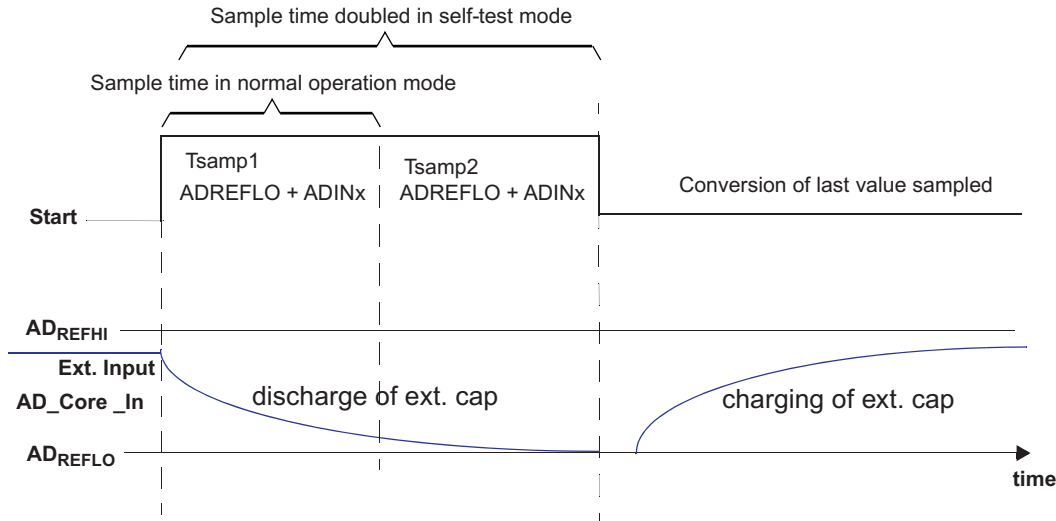
Note that the switch S5 shown in Figure 22-15 is only for the purpose of explaining the self-test sequence. There is no physical switch.

Table 22-3. Self-Test Reference Voltages⁽¹⁾

SELF_TEST	HILO	S1	S2	S3	S4	S5	Reference Voltage
1	0	0	1	1	0	1	AD_REFLO via R1 R2 connected to V _{in}
1	1	1	0	0	1	1	AD_REFHI via R1 R2 connected to V _{in}
0	X	0	0	0	0	1	V _{in}

⁽¹⁾ Switches refer to Figure 22-15.

Conversions in self-test mode are started just as they are in the normal operating mode (see Section 22.3.6). The conversion starts according to the configuration set in the three mode control registers (ADEVMODECR, ADG1MODECR, ADG2MODECR) and the sampling time control registers (ADEVSAMP, ADG1SAMP, ADG2SAMP). The acquisition time for each conversion in self-test mode is extended to twice the normal configured acquisition time. The selected reference voltage and the input voltage from the ADINx input channel are both connected to the ADC internal sampling capacitor throughout this extended acquisition period. Figure 22-16 shows the self-test mode timing when the ADREFLO is chosen as the reference voltage for the self-test mode conversion. It also assumes an external capacitor connected to the ADC input channel.

Figure 22-16. Timing for Self-Test Mode


22.8.2.1 Use of Self-Test Mode to Determine Open/Short on ADC Input Channels

The following sequence needs to be used to deduce the ADC pin status:

- Convert the channel with self test enabled and with the reference voltage as V_{reflo} . Store the conversion result, say V_d .
- Convert the channel with self test enabled and with the reference voltage as V_{refhi} . Store the conversion result, say V_u .
- Convert the channel with self test disabled. Store the conversion result, say V_n .

The results can be interpreted using the following table.

Table 22-4. Determination of ADC Input Channel Condition

Normal Conversion Result, V_n	Self-test Conversion Result, V_u	Self-test Conversion Result, V_d	Pin Condition
V_n	$V_n < V_u < AD_{REFHI}$	$AD_{REFLO} < V_d < V_n$	Good
AD_{REFHI}	AD_{REFHI}	approx. AD_{REFHI}	Shorted to AD_{REFHI}
AD_{REFLO}	approx. AD_{REFLO}	AD_{REFLO}	Shorted to AD_{REFLO}
Unknown	AD_{REFHI}	AD_{REFLO}	Open

22.8.3 ADC Power-Down Mode

This is an inactive mode in which the clocks to the ADC module are stopped leaving the module in a static state. The clock to the ADC core ($ADCLK$) is stopped whenever there are no ongoing conversions. This is the clock-gating implementation requirement. Also, the ADC module places the ADC core into the power down mode such that there is minimal current drawn from the ADC operating and reference supplies.

22.8.3.1 Powering Down Just The ADC Core

The ADC core can be individually powered down without stopping the clocks to the ADC module. This can be done by setting the **POWERDOWN** bit of the ADC Operating Mode Control Register ($ADOPMODECR.3$). Whenever a conversion is required the **POWERDOWN** bit must be cleared, and a minimum time $t_{d(PU-ADV)}$, (see the specific device data sheet for actual value) has to be allowed before starting a new conversion. This wait must be implemented in the application software.

22.8.3.2 Enhanced Power-Down Mode

A bit in the ADC operating mode control register, IDLE_PWRDN (ADOPMODECR.4) enables the enhanced power-down mode of the ADC.

Once this bit is set, the ADC module will power down the ADC core whenever there are no more ongoing or pending ADC conversions. The ADC core will be powered down regardless of the state of the POWERDOWN bit (ADOPMODECR.3).

The ADC module releases the ADC core from power down mode as soon as a new conversion is requested. The ADC logic state machine then has to wait for at least $t_{d(PU-ADV)}$ (see the device data sheet for actual value) before starting a new conversion. The IDLE_PWRDN bit will remain set at all times. The logic state machine can use this bit to determine that it needs to wait for a programmable number of VCLK cycles before it allows the input channel to be sampled. This time is configured by the ADC Power Up Delay Control register (ADPWRUPDLYCTRL).

If IDLE_PWRDN is not set, the ADC module does not wait for any additional delay before sampling the input channel and the application software has to take account of this required delay.

22.8.3.3 Managing Clocks to the ADC Module

The clock to the ADC module can be turned off via the appropriate Peripheral Central Resource (PCR) controller PSPWRDNSET register (check the specific device datasheet to identify the register and the bit to be set). If a conversion is ongoing when this bit is set, the ADC module will wait until the current conversion completes before allowing the ADC module clock to be stopped.

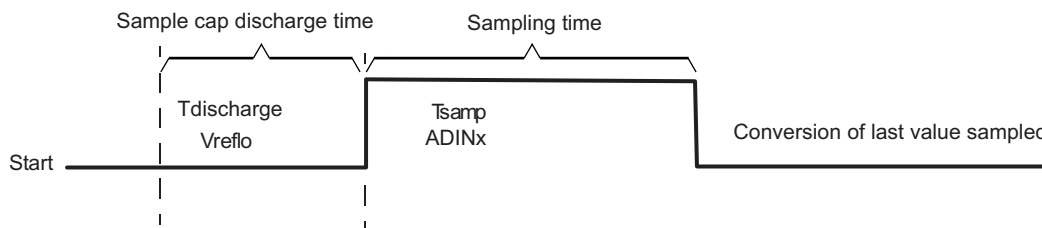
22.8.4 ADC Sample Capacitor Discharge Mode

This mode allows the charge on the ADC core's internal sampling capacitor to be discharged before starting the sampling phase of the next channel.

The ADC Sample Cap Discharge Mode is enabled by setting the SAMP_DIS_EN bit of the group's ADSAMPDISEN register. A discharge period for the sampling capacitor is added before the sampling period for each channel as shown in Figure 22-17. The duration of this discharge period is configurable via the corresponding group's SAMP_DIS_CYC field in the ADSAMPDISEN register. The discharge time is specified in terms of number of ADCLK cycles.

During the sample capacitor discharge period, the V_{REFLO} reference voltage is connected to the input voltage terminal of the ADC core. This allows any charge collected on the sampling capacitor from the previous conversion to be discharged to ground. The V_{REFLO} reference voltage is usually connected to ground.

Figure 22-17. Timing for Sample Capacitor Discharge Mode



22.9 ADC Results' RAM Special Features

The following sections describe some of the special features supported by the ADC module to enhance the results' RAM testability and integrity.

22.9.1 ADC Results' RAM Auto-Initialization

The ADC module allows the application to auto-initialize the ADC results' RAM to all zeros. The application must ensure that the ADC module is not in any of the conversion modes before triggering off the auto-initialization process.

The auto-initialization sequence is as follows:

1. Enable the global hardware memory initialization key by programming a value of 1010 to the bits [3-0] of the MINITGCR register of the System module.
2. Set the control bit for the ADC results' RAM in the MSINENA System module register. The bit 8 of the MSINENA register is used to control the initialization of the ADC1 results' RAM, while bit 14 controls the initialization of the ADC2 results' RAM. This starts the initialization process. The BUF_INIT_ACTIVE flag in the ADC module ADBNDEND register will get set to reflect that the initialization is ongoing.
3. When the memory initialization is completed, the corresponding status bit in the MINISTAT register will be set. Also, the BUF_INIT_ACTIVE flag will get cleared.

22.9.2 ADC Results' RAM Test Mode

In the defined conversion modes of the ADC, the application can only read from the ADC results' RAM. Only the ADC module is allowed to write to the results' RAM. A special test mode is defined to allow the application to also write into the ADC results' RAM - this mode is the ADC Results' RAM Test Mode. Only 32-bit reads and writes are allowed to the ADC results' RAM in this test mode.

NOTE: Contention on access to ADC Results' RAM

The ADC module cannot handle a contention between the application write to the results' RAM and the ADC writing a conversion result to the results' RAM. The application must ensure that the ADC is not likely to write a new conversion result to the results' RAM when the ADC Results' RAM Test Mode is enabled.

The ADC Results' RAM Test Mode is enabled by setting the RAM_TEST_EN bit in the ADOPMODECR.

22.9.3 ADC Results' RAM Parity

The below shows the ADC Results' RAM parity control registers.

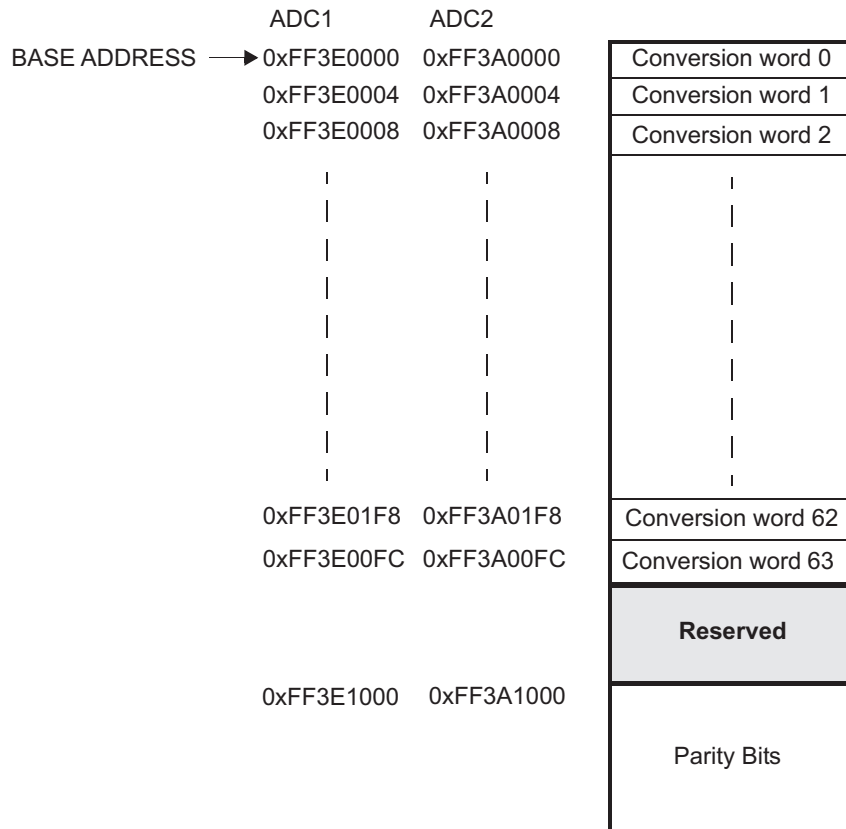
Parity checking is implemented using parity on a per-half word basis for the ADC RAM. That is, there is one parity bit for 16 bits of the ADC RAM. The polarity of the ADC RAM parity is controlled by the DEVCR1 register in the system module (address = 0xFFFFFDC). The parity checking is enabled by the ADPARCR register. After reset, the parity checking is disabled and must be enabled if parity protection is required.

During a read access, the parity is calculated based on the data read from the ADC RAM and compared with the good parity value stored in the parity bits. If any word fails the parity check then the ADC generates an error signal hooked up to the Error Signaling Module (ESM). The ADC RAM address which generated the parity error is captured for host system debugging, and is frozen from being updated until it is read by the application.

Testing the Parity Checking Mechanism:

To test the parity checking mechanism itself, the parity RAM is made writable by the CPU in a special test mode. This is done by a control bit called TEST in the AD PAR CR register. Once this bit is set, the parity bits are mapped to an address starting at an address offset of 4KB from the base address of the ADC RAM. See [Figure 22-18](#). The CPU can now manually insert parity errors. Note that the ADC RAM only supports 32-bit accesses.

Figure 22-18. ADC Memory Map in Parity Test Mode



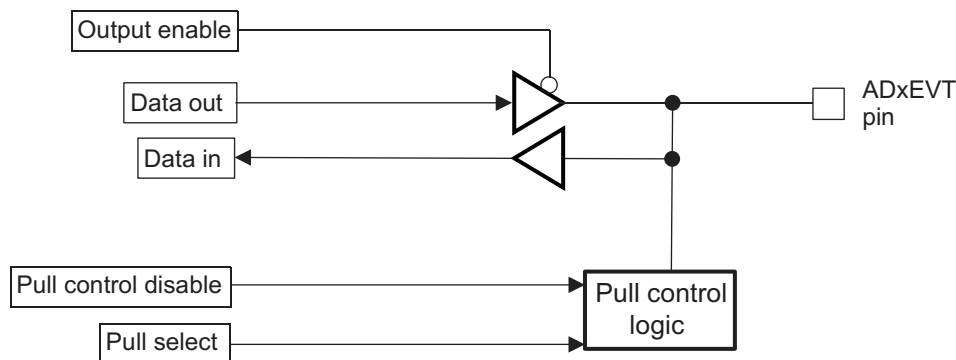
22.10 ADEVT Pin General Purpose I/O Functionality

The AD1EVT pin for ADC1 and AD2EVT pin for ADC2 can be configured as general-purpose I/O signals. The following sections describe the different ways in which the application can configure the ADxEVT pins.

22.10.1 GPIO Functionality

Figure 22-19 illustrates the GPIO functionality of the ADxEVT pin.

Figure 22-19. GPIO Functionality of ADxEVT



Once the device power-on reset is released, the ADC module controls the state of the ADxEVT pin.

- **Pull control:** The pull control can either be enabled or disabled by default (while system reset is active and after it is released). The actual default state of the pull control is specified in the device datasheet. The application can enable pull control by clearing the PDIS (pull control disable) bit in the ADEVTPDIS register. In this case, if the PSEL (pull select) bit in the ADEVTPSEL register is set, the pin will have a pull-up. If the PSEL bit is cleared, the pin will have a pull-down. If the PDIS bit is set in the control register, there is no pull-up or pull-down on the pin.

NOTE: Pull Behavior when ADxEVT is configured as output

If the ADxEVT pin is configured as output, then the pulls are disabled automatically. If the pin is configured as input, the pulls are enabled or disabled depending on bit PDIS in the pull disable register ADEVTPDIS.

- **Output buffer:** The ADxEVT pin can be driven as an output pin if the ADEVTDIR bit is set in the pin direction control register.
- **Open-Drain Feature:** The open drain output capability is enabled via the ADEVTPDR control register. The ADxEVT pin must be also configured to be an output pin for this mode.
 - The output buffer is enabled if a low signal is being driven on to the pin.
 - The output buffer is disabled if a high signal is being driven on to the pin.

22.10.2 Summary

The behavior of the output buffer, and the pull control is summarized in [Table 22-5](#). The input buffer for the ADxEVT pins are enabled once the device power-on reset is released.

Table 22-5. Output Buffer and Pull Control Behavior for ADxEVT as GPIO Pins

System Reset Active?	Pin Direction (DIR) ⁽¹⁾⁽²⁾	Pull Disable (PDIS) ⁽¹⁾⁽³⁾	Pull Select (PSEL) ⁽¹⁾⁽⁴⁾	Pull Control	Output Buffer
Yes	X	X	X	Enabled	Disabled
No	0	0	0	Pull down	Disabled
No	0	0	1	Pull up	Disabled
No	0	1	0	Disabled	Disabled
No	0	1	1	Disabled	Disabled
No	1	X	X	Disabled	Enabled

⁽¹⁾ X = Don't care

⁽²⁾ DIR = 0 for input, 1 for output

⁽³⁾ PULDIS = 0 for enabling pull control, 1 for disabling pull control

⁽⁴⁾ PULSEL = 0 for pull-down functionality, 1 for pull-up functionality

22.11 ADC_CONTROL Registers

Table 22-6 lists the memory-mapped registers for the ADC_CONTROL. All register offset addresses not listed in Table 22-6 should be considered as reserved locations and the register contents should not be modified.

All registers in the ADC module are 32-bit, word-aligned; 8-bit, 16-bit and 32-bit accesses are allowed. The application must ensure that the reserved bits are always written as 0 to ensure software compatibility to future revisions of the module. The base address of ADC1 registers is FFF7 C000h.

Table 22-6. ADC_CONTROL Registers

Offset	Acronym	Register Name	Section
0h	ADRSTCR	ADC Reset Control Register	Section 22.11.1
4h	ADOPMODECR	ADC Operating Mode Control Register	Section 22.11.2
8h	ADCLOCKCR	ADC Clock Control Register	Section 22.11.3
Ch	ADCALCR	ADC Calibration Mode Control Register	Section 22.11.4
10h	ADEVMODECR	ADC Event Group Operating Mode Control Register	Section 22.11.5
14h	ADG1MODECR	ADC Group1 Operating Mode Control Register	Section 22.11.6
18h	ADG2MODECR	ADC Group2 Operating Mode Control Register	Section 22.11.7
1Ch	ADEVSR	ADC Trigger Source Select Register	Section 22.11.8
20h	ADG1SRC	ADC Group1 Trigger Source Select Register	Section 22.11.9
24h	ADG2SRC	ADC Group2 Trigger Source Select Register	Section 22.11.10
28h	ADEVINTENA	ADC Event Interrupt Enable Control Register	Section 22.11.11
2Ch	ADG1INTENA	ADC Group1 Interrupt Enable Control Register	Section 22.11.12
30h	ADG2INTENA	ADC Group2 Interrupt Enable Control Register	Section 22.11.13
34h	ADEVINTFLG	ADC Event Group Interrupt Flag Register	Section 22.11.14
38h	ADG1INTFLG	ADC Group1 Interrupt Flag Register	Section 22.11.15
3Ch	ADG2INTFLG	ADC Group2 Interrupt Flag Register	Section 22.11.16
40h	ADEVTHRINTCR	ADC Event Group Threshold Interrupt Control Register	Section 22.11.17
44h	ADG1THRINTCR	ADC Group1 Threshold Interrupt Control Register	Section 22.11.18
48h	ADG2THRINTCR	ADC Group2 Threshold Interrupt Control Register	Section 22.11.19
4Ch	ADEVDMACR	ADC Event Group DMA Control Register	Section 22.11.20
50h	ADG1DMACR	ADC Group1 DMA Control Register	Section 22.11.21
54h	ADG2DMACR	ADC Group2 DMA Control Register	Section 22.11.22
58h	ADBNDRCR	ADC Results Memory Configuration Register	Section 22.11.23
5Ch	ADBNDEND	ADC Results Memory Size Configuration Register	Section 22.11.24
60h	ADEVSAMP	ADC Event Group Sampling Time Configuration Register	Section 22.11.25
64h	ADG1SAMP	ADC Group1 Sampling Time Configuration Register()	Section 22.11.26
68h	ADG2SAMP	ADC Group2 Sampling Time Configuration Register	Section 22.11.27
6Ch	ADEVSR	ADC Event Group Status Register	Section 22.11.28
70h	ADG1SR	ADC Group1 Status Register	Section 22.11.29
74h	ADG2SR	ADC Group2 Status Register	Section 22.11.30
78h	ADEVSEL	ADC Event Group Channel Select Register	Section 22.11.31
7Ch	ADG1SEL	ADC Group1 Channel Select Register	Section 22.11.32
80h	ADG2SEL	ADC Group2 Channel Select Register	Section 22.11.33
84h	ADCALR	ADC Calibration and Error Offset Correction Register	Section 22.11.34
88h	ADSMSTATE	ADC State Machine Status Register	Section 22.11.35
8Ch	ADLASTCONV	ADC Channel Last Conversion Value Register	Section 22.11.36
90h to ACh	ADEVBUFFER_0 to ADEVBUFFER_7	ADC Event Group Results FIFO Register	Section 22.11.37
B0h to CCh	ADG1BUFFER_0 to ADG1BUFFER_7	ADC Group1 Results FIFO Register	Section 22.11.38
D0h to ECh	ADG2BUFFER_0 to ADG2BUFFER_7	ADC Group2 Results FIFO Register	Section 22.11.39
F0h	ADEVEMUBUFFER	ADC Event Group Results Emulation FIFO Register	Section 22.11.40

Table 22-6. ADC_CONTROL Registers (continued)

Offset	Acronym	Register Name	Section
F4h	ADG1EMUBUFFER	ADC Group1 Results Emulation FIFO Register	Section 22.11.41
F8h	ADG2EMUBUFFER	ADC Group2 Results Emulation FIFO Register	Section 22.11.42
FCh	ADEVTDIR	ADC ADEVT Pin Direction Control Register	Section 22.11.43
100h	ADEVTOUT	ADC ADEVT Pin Output Value Control Register	Section 22.11.44
104h	ADEVTIN	ADC ADEVT Pin Input Value Register	Section 22.11.45
108h	ADEVTSET	ADC ADEVT Pin Set Register	Section 22.11.46
10Ch	ADEVTCLR	ADC ADEVT Pin Clear Register	Section 22.11.47
110h	ADEVTTPDR	ADC ADEVT Pin Open Drain Enable Register	Section 22.11.48
114h	ADEVTTPDIS	ADC ADEVT Pin Pull Control Disable Register	Section 22.11.49
118h	ADEVTTPSEL	ADC ADEVT Pin Pull Control Select Register	Section 22.11.50
11Ch	ADEVSAMPDISEN	ADC Event Group Sample Cap Discharge Control Register	Section 22.11.51
120h	ADG1SAMPDISEN	ADC Group1 Sample Cap Discharge Control Register	Section 22.11.52
124h	ADG2SAMPDISEN	ADC Group2 Sample Cap Discharge Control Register	Section 22.11.53
128h	ADMAGINT1CR	ADC Magnitude Compare Interrupt 1 Control Register	Section 22.11.54
12Ch	ADMAGINT1MASK	ADC Magnitude Compare Mask 1 Register	Section 22.11.55
130h	ADMAGINT2CR	ADC Magnitude Compare Interrupt 2 Control Register	Section 22.11.56
134h	ADMAGINT2MASK	ADC Magnitude Compare Mask 2 Register	Section 22.11.57
138h	ADMAGINT3CR	ADC Magnitude Compare Interrupt 3 Control Register	Section 22.11.58
13Ch	ADMAGINT3MASK	ADC Magnitude Compare Mask 3 Register	Section 22.11.59
158h	ADMAGINTENASET	ADC Magnitude Compare Interrupt Enable Set Register	Section 22.11.60
15Ch	ADMAGINTENACLR	ADC Magnitude Compare Interrupt Enable Clear Register	Section 22.11.61
160h	ADMAGINTFLG	ADC Magnitude Compare Interrupt Flag Register	Section 22.11.62
164h	ADMAGINTOFF	ADC Magnitude Compare Interrupt Offset Register	Section 22.11.63
168h	ADEVFIFORESETCR	ADC Event Group FIFO Reset Control Register	Section 22.11.64
16Ch	ADG1FIFORESETCR	ADC Group1 FIFO Reset Control Register	Section 22.11.65
170h	ADG2FIFORESETCR	ADC Group2 FIFO Reset Control Register	Section 22.11.66
174h	ADEVWRAMWRADDR	ADC Event Group RAM Write Address Register	Section 22.11.67
178h	ADG1RAMWRADDR	ADC Group1 RAM Write Address Register	Section 22.11.68
17Ch	ADG2RAMWRADDR	ADC Group2 RAM Write Address Register	Section 22.11.69
180h	ADPARCR	ADC Parity Control Register	Section 22.11.70
184h	ADPARADDR	ADC Parity Error Address Register	Section 22.11.71
188h	ADPWRUPDLYCTRL	ADC Power-Up Delay Control Register	Section 22.11.72
190h	ADEVCHNSELMODECTRL	ADC Event Group Channel Selection Mode Control Register	Section 22.11.73
194h	ADG1CHNSELMODECTRL	ADC Group1 Channel Selection Mode Control Register	Section 22.11.74
198h	ADG2CHNSELMODECTRL	ADC Group2 Channel Selection Mode Control Register	Section 22.11.75
19Ch	ADEVCURRCOUNT	ADC Event Group Current Count Register	Section 22.11.76
1A0h	ADEVMAXCOUNT	ADC Event Group Max Count Register	Section 22.11.77
1A4h	ADG1CURRCOUNT	ADC Group1 Current Count Register	Section 22.11.78
1A8h	ADG1MAXCOUNT	ADC Group1 Max Count Register	Section 22.11.79
1ACh	ADG2CURRCOUNT	ADC Group2 Current Count Register	Section 22.11.80
1B0h	ADG2MAXCOUNT	ADC Group2 Max Count Register	Section 22.11.81

Complex bit access types are encoded to fit into small table cells. [Table 22-7](#) shows the codes that are used for access types in this section.

Table 22-7. ADC_CONTROL Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
RC	C R	to Clear Read
Write Type		
W	W	Write
W1C	1C W	1 to clear Write
W1S	1S W	1 to set Write
WP	P W	Requires privileged access Write
Reset or Default Value		
-n		Value after reset or the default value

22.11.1 ADRSTCR Register (Offset = 0h) [reset = 0h]

ADRSTCR is shown in [Figure 22-20](#) and described in [Table 22-8](#).

Return to [Summary Table](#).

Figure 22-20. ADRSTCR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							RESET
R-0h							R/WP-0h

Table 22-8. ADRSTCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	RESET	R/WP	0h	This bit is used to reset the ADC internal state machines and control and status registers. This reset state is held until this bit is cleared. Read in all modes, write in privileged mode. 0h = Module is released from the reset state. 1h = All the module's internal state machines and the control/status registers are reset.

22.11.2 ADOPMODECR Register (Offset = 4h) [reset = 00140000h]

ADOPMODECR is shown in [Figure 22-21](#) and described in [Table 22-9](#).

[Return to Summary Table.](#)

Figure 22-21. ADOPMODECR Register

31	30	29	28	27	26	25	24
10_12_BIT	RESERVED						COS
R/W-0h	R-0h						R/W-0h
23	22	21	20	19	18	17	16
RESERVED			CHN_TEST_EN				RAM_TEST_EN
R-0h			R/W-Ah				R/W-0h
15	14	13	12	11	10	9	8
RESERVED						POWER_DOWN	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
RESERVED			IDLE_PWRDN	RESERVED			ADC_EN
R-0h			R/W-0h	R-0h			R/W-0h

Table 22-9. ADOPMODECR Register Field Descriptions

Bit	Field	Type	Reset	Description
31	10_12_BIT	R/W	0h	This bit controls the resolution of the ADC core. It also affects the size of the conversion results stored in the results' RAM. Any operation mode: 0h = ADC core and digital logic are configured to be in 10-bit resolution. This is the default mode of operation. 1h = ADC core and digital logic are configured to be in 12-bit resolution.
30-25	RESERVED	R	0h	Reads return 0, writes have no effect.
24	COS	R/W	0h	This bit affects emulation operation only. It defines whether the ADC core clock (ADCLK) is immediately halted when the CPU enters debug mode or if it should continue operating normally. Note: If COS = 0 when the CPU enters the emulation mode, then the accuracy of the conversion results can be affected depending on how long the module stays in the emulation mode. Any operation mode: 0h = ADC module halts all ongoing conversions immediately after emulation mode is entered. 1h = ADC module continues all ongoing conversions as per the configurations of the three conversion groups.
23-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-17	CHN_TEST_EN	R/W	Ah	Enable the input channels' impedance measurement mode. This mode is reserved for use by TI. Any other value = Input impedance measurement mode is disabled. Any operation mode: 5h = Input impedance measurement mode is enabled. Ah = Input impedance measurement mode is disabled.
16	RAM_TEST_EN	R/W	0h	Enable the ADC Results' RAM Test Mode. Refer to ADC Results' RAM Test Mode subsection for more details. Any operation mode: 0h = ADC RAM Test Mode is disabled. The application cannot write to the ADC RAM by the CPU or the DMA. 1h = ADC RAM Test Mode is enabled. The application can directly write to the ADC RAM by the CPU or the DMA.

Table 22-9. ADOPMODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
15-9	RESERVED	R	0h	Reads return 0, writes have no effect.
8	POWER_DOWN	R/W	0h	ADC Power Down. This bit powers down only the ADC core, the digital logic in the sequencer stays active. To release the core from power down mode, this bit must be cleared. If a conversion is ongoing, the ADC module waits until the current conversion is completed before powering down the ADC core. Also refer to ADC Power-Up Delay Control Register (ADPWRUPDLYCTRL). Any operation mode: 0h = The state of the ADC core is controlled by the IDLE_PWRDN bit, or by a global power down mode entry. 1h = ADC core is in the power-down state.
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4	IDLE_PWRDN	R/W	0h	ADC Power Down When Idle. When this bit is set, the ADC module automatically powers down the ADC core whenever there are no conversions ongoing or pending. This is the enhanced power-down mode. Also refer to ADC Power-Up Delay Control Register (ADPWRUPDLYCTRL). Any operation mode: 0h = The ADC stays in the normal operating mode even if no conversions are ongoing or pending. The power down state is entered only by configuring the POWER_DOWN bit or via a global power down mode entry. 1h = Enhanced power down mode is enabled.
3-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADC_EN	R/W	0h	ADC Enable. This bit must be set to allow the ADC module to be configured to perform any conversions. Any operation mode: 0h = No ADC conversions occur. The input channel select registers: ADEVSEL, ADG1SEL, and ADG2SEL are held at their reset values. 1h = ADC can now be configured.

22.11.3 ADCLOCKCR Register (Offset = 8h) [reset = 0h]

ADCLOCKCR is shown in [Figure 22-22](#) and described in [Table 22-10](#).

Return to [Summary Table](#).

Figure 22-22. ADCLOCKCR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED																												PS			
R-0h																												R/W-0h			

Table 22-10. ADCLOCKCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	PS	R/W	0h	ADC Clock Prescaler. These bits define the prescaler value for the ADC core clock (ADCLK). The ADCLK is generated by dividing down the input bus clock (VCLK) to the ADC module from 0 to 1Fh. Note: The supported range for the ADC clock frequency is specified in the device datasheet. The ADC clock prescaler must be configured to meet this datasheet specification. Any operation mode: $t_{C(ADCLK)} = t_{C(VCLK)} \times (PS[4:0] + 1)$ where $t_{C(ADCLK)}$ is the period of the ADCLK and $t_{C(VCLK)}$ is the period of the VCLK.

22.11.4 ADCALCR Register (Offset = Ch) [reset = 0h]

ADCALCR is shown in [Figure 22-23](#) and described in [Table 22-11](#).

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Figure 22-23. ADCALCR Register

31	30	29	28	27	26	25	24
RESERVED							SELF_TEST
R-0h							R/W-0h
23	22	21	20	19	18	17	16
RESERVED							CAL_ST
R-0h							R/W1S-0h
15	14	13	12	11	10	9	8
RESERVED						BRIDGE_EN	HILO
R-0h						R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
RESERVED							CAL_EN
R-0h							R/W-0h

Table 22-11. ADCALCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-25	RESERVED	R	0h	Reads return 0, writes have no effect.
24	SELF_TEST	R/W	0h	ADC Self Test Enable. When this bit is Set, either AD _{REFHI} or AD _{REFLO} is connected through a resistor to the selected input channel. The desired con-version mode is configured in the group mode control registers. For more details on the ADC Self Test Mode, refer to ADC Self-Test Mode subsection. Any operation mode: 0h = ADC Self Test mode is disabled. 1h = ADC Self Test mode is enabled.
23-17	RESERVED	R	0h	Reads return 0, writes have no effect.
16	CAL_ST	R/W1S	0h	ADC Calibration Conversion Start. Setting the CAL_ST bit while the CAL_EN bit is set starts conversion of the selected reference voltage. The ADC module uses the sample time configured in the Event Group sample time configuration register (ADEVSAMP) for the calibration conversion. Any operation mode: 0h (W) = Writing 0 to this bit has no effect. 0h (R) = Calibration conversion has completed, or has not yet been started. 1h (W) = ADC module starts calibration conversion. 1h (R) = Calibration conversion is in the progress.
15-10	RESERVED	R	0h	Reads return 0, writes have no effect.
9	BRIDGE_EN	R/W	0h	Bridge Enable. When set with the HILO bit, BRIDGE_EN allows a reference voltage to be converted in calibration mode. Calibration Reference Voltages table defines the four different reference voltages that can be selected.
8	HILO	R/W	0h	ADC Self Test mode and Calibration Mode Reference Source Selection. In the ADC Self Test mode, this bit defines the test voltage to be combined through a resistor with the selected input pin voltage. Refer to ADC Self-Test Mode subsection for details on the ADC Self Test Mode. In the ADC Calibration Mode, this bit defines the reference source polarity. Refer to ADC Error Calibration Mode subsection for details on the ADC Calibration Mode. In the ADC module's normal operating mode, this bit has no effect.

Table 22-11. ADCALCR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	CAL_EN	R/W	0h	ADC Calibration Enable. When this bit is Set, the input channel multiplexor is disconnected and the calibration reference voltage is connected to the ADC core input. The calibration reference voltage is selected by the combination of the BRIDGE_EN and HILO. The actual conversion of this reference voltage starts when the CAL_ST bit is set. If the CAL_ST bit is already set when the CAL_EN bit is set, then the calibration conversion is immediately started. Refer to ADC Error Calibration Mode subsection for more details on the ADC calibration mode. Any operation mode: 0h = Calibration mode is disabled. 1h = Calibration mode is enabled.

22.11.5 ADEVMODECR Register (Offset = 10h) [reset = 0h]

ADEVMODECR is shown in [Figure 22-24](#) and described in [Table 22-12](#).

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The format of the ADEVMODECR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

Figure 22-24. ADEVMODECR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							No_Reset_on_ChSel
R-0h							R/W-0h
15	14	13	12	11	10	9	8
RESERVED						EV_DATA_FMT	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
RESERVED		EV_CHID	OVR_EV_RAM_IGN	RESERVED	EV_8BIT	EV_MODE	FRZ_EV
R-0h		R/W-0h	R/W-0h	R-0h	R/W-0h	R/W-0h	R/W-0h

Table 22-12. ADEVMODECR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reads return 0, writes have no effect.
16	No_Reset_on_ChSel	R/W	0h	No Event Group Results Memory Reset on New Channel Select. This bit determines whether the event group results' RAM is reset whenever a non-zero value is written to the event group channel select register. If the event group conversions are ongoing (active or frozen), then writing a non-zero value to the event group channel select register always resets the event group results RAM. Any operation mode: 0h = Event group results RAM is reset when a non-zero value is written to event group channel select register, even if event group conversions are completed. 1h = Event group results RAM is not reset when a non-zero value is written to event group channel select register, and event group conversions are completed.
15-10	RESERVED	R	0h	Reads return 0, writes have no effect.
9-8	EV_DATA_FMT	R/W	0h	Event Group Read Data Format. Note: This field is only applicable when the ADC module is configured to be a 12-bit ADC module. This field is reserved when the module is configured as a 10-bit ADC module. This field determines the format in which the conversion results are read out of the Event group results RAM when using the FIFO interface, that is, when reading from the ADEVBUFFER or ADEVEMUBUFFER locations. Any operation mode: 0h = Conversion results are read out in full 12-bit format. This is the default mode. 1h = Conversion results are read out in 10-bit format. Bits 11-2 of the 12-bit conversion result are returned as the 10-bit conversion result. 2h = Conversion results are read out in 8-bit format. Bits 11-4 of the 12-bit conversion result are returned as the 8-bit conversion result. 3h = Reserved. The full 12-bit conversion result is returned if programmed.

Table 22-12. ADEVMODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-6	RESERVED	R	0h	Reads return 0, writes have no effect.
5	EV_CHID	R/W	0h	Enable Channel Id for the Event Group conversion results to be read. This bit only affects the “read from FIFO” mode. The ADC always stores the channel id in the results RAM. Any 16-bit read performed in the “read from RAM” mode returns the 5-bit channel id along with the 10-bit conversion result. Any operation mode: 0h = Bits 14-10, the channel id field, of the data read from the Event Group results' FIFO is read as 00000b. 1h = Bits 14-10, the channel id field, of the data read from the Event Group results' FIFO contains the number of the ADC analog input to which the conversion result belongs.
4	OVR_EV_RAM_IGN	R/W	0h	This bit allows the ADC module to overwrite the contents of the Event Group results memory under an overrun condition. Any operation mode: 0h = The ADC cannot overwrite the contents of the Event Group results memory. When an overrun of this memory occurs, the software needs to read out all the contents of this memory before the ADC is able to write a new conversion result for the Event Group. 1h = When an overrun of the Event Group results memory occurs, the ADC proceeds to overwrite the contents with any new conversion results for the Event Group, starting with the first location in this memory.
3	RESERVED	R	0h	Reads return 0, writes have no effect.
2	EV_8BIT	R/W	0h	Event Group 8-bit result mode. Note: This bit is only applicable when the ADC module is configured to be a 10-bit ADC module. This bit is reserved when the module is configured as a 12-bit ADC module. This bit allows the Event Group conversion results to be read out in an 8-bit format. This bit only applies to the read from FIFO mode. The lower 2 bits of the 10-bit conversion result are discarded and the upper 8 bits are shifted right two places to form the 8-bit conversion result. Any operation mode: 0h = The Event Group conversion result is read out as a 10-bit value in the “read from Event Group FIFO” mode. 1h = The Event Group conversion result is read out as an 8-bit value in the “read from Event Group FIFO” mode.
1	EV_MODE	R/W	0h	Event Group Conversion Mode. This bit defines whether the input channels selected for conversion in the Event Group are converted only once per trigger, or are continuously converted. Any operation mode: 0h = The channels selected for conversion in the Event Group are converted only once when the selected event trigger condition occurs. 1h = The channels selected for conversion in the Event Group are converted continuously when the selected event trigger condition occurs.

Table 22-12. ADEVMODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	FRZ_EV	R/W	0h	Event Group Freeze Enable. This bit allows an Event Group conversion sequence to be frozen if a Group1 or a Group2 conversion is requested. The Event Group conversion is kept frozen while the Group1 or Group2 conversion is active, and continues from where it was frozen once the Group1 or Group2 conversions are completed. While the Event Group conversion is frozen, the EV_STOP status flag in the ADEVSR register indicates that the Event Group conversions have stopped. This bit gets cleared when the Event Group conversions resume. Any operation mode: 0h = Event Group conversions cannot be frozen. All the channels selected for conversion in the Event Group are converted before the ADC switches over to servicing any other conversion group. 1h = Event Group conversions are frozen whenever there is a request for conversion from Group1 or Group2.

22.11.6 ADG1MODECR Register (Offset = 14h) [reset = 0h]

ADG1MODECR is shown in [Figure 22-25](#) and described in [Table 22-13](#).

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The format of the ADG1MODECR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

Figure 22-25. ADG1MODECR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							No_Reset_on_ChSel
R-0h							R/W-0h
15	14	13	12	11	10	9	8
RESERVED						G1_DATA_FMT	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
RESERVED		G1_CHID	OVR_G1_RAM_IGN	G1_HW_TRIG	G1_8BIT	G1_MODE	FRZ_G1
R-0h		R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 22-13. ADG1MODECR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reads return 0, writes have no effect.
16	No_Reset_on_ChSel	R/W	0h	No Group1 Results Memory Reset on New Channel Select. This bit determines whether the group1 results' RAM is reset whenever a non-zero value is written to the group1 channel select register. If the group1 conversions are ongoing (active or frozen), then writing a nonzero value to the group1 channel select register always resets the group1 results RAM. Any operation mode: 0h = Group1 results RAM is reset when a non-zero value is written to group1 channel select register, even if group1 conversions are completed. 1h = Group1 results RAM is not reset when a non-zero value is written to group1 channel select register, and group1 conversions are completed.
15-10	RESERVED	R	0h	Reads return 0, writes have no effect.
9-8	G1_DATA_FMT	R/W	0h	Group1 Read Data Format. Note: This field is only applicable when the ADC module is configured to be a 12-bit ADC module. This field is reserved when the module is configured as a 10-bit ADC module. This field determines the format in which the conversion results are read out of the group1 results RAM when using the FIFO interface, that is, when reading from the ADG1BUFFER or ADG1EMUBUFFER locations. Any operation mode: 0h = Conversion results are read out in full 12-bit format. This is the default mode. 1h = Conversion results are read out in 10-bit format. Bits 11-2 of the 12-bit conversion result are returned as the 10-bit conversion result. 2h = Conversion results are read out in 8-bit format. Bits 11-4 of the 12-bit conversion result are returned as the 8-bit conversion result. 3h = Reserved. The full 12-bit conversion result is returned if programmed.

Table 22-13. ADG1MODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-6	RESERVED	R	0h	Reads return 0, writes have no effect.
5	G1_CHID	R/W	0h	Enable Channel Id for the Group1 conversion results to be read. This bit only affects the “read from FIFO” mode. The ADC always stores the channel id in the results RAM. Any 16-bit read performed in the “read from RAM” mode returns the 5-bit channel id along with the 10-bit conversion result. Any operation mode: 0h = Bits 14-10, the channel id field, of the data read from the Group1 results’ FIFO is read as 00000b. 1h = Bits 14-10, the channel id field, of the data read from the Group1 results’ FIFO contains the number of the ADC analog input to which the conversion result belongs.
4	OVR_G1_RAM_IGN	R/W	0h	This bit allows the ADC module to overwrite the contents of the Group1 results memory under an overrun condition. Any operation mode: 0h = The ADC cannot overwrite the contents of the Group1 results memory. When an overrun of this memory occurs, the software needs to read out all the contents of this memory before the ADC is able to write a new conversion result for the Group1. 1h = When an overrun of the Group1 results memory occurs, the ADC proceeds to overwrite the contents with any new conversion results for the Group1, starting with the first location in this memory.
3	G1_HW_TRIG	R/W	0h	Group1 Hardware Triggered. This bit allows the Group1 to be hardware triggered. The Group1 is software triggered by default. For more details on how to trigger a conversion group, refer to How to Start a Conversion subsection. Any operation mode: 0h = The Group1 is software-triggered. A Group1 conversion starts whenever the Group1 channel select register (ADG1SEL) is written with a non-zero value. 1h = The Group1 is hardware-triggered. A Group1 conversion starts whenever the Group1 channel select register has a non-zero value, and the specified hardware trigger occurs. The hardware trigger for the Group1 is specified in the Group1 Trigger Source register (ADG1SRC).
2	G1_8BIT	R/W	0h	Group1 8-bit result mode. Note: This bit is only applicable when the ADC module is configured to be a 10-bit ADC module. This bit is reserved when the module is configured as a 12-bit ADC module. This bit allows the Group1 conversion results to be read out in an 8-bit format. This bit only applies to the “read from FIFO” mode. The lower 2 bits of the 10-bit conversion result are discarded and the upper 8 bits are shifted right two places to form the 8-bit conversion result. Any operation mode: 0h = The Group1 conversion result is read out as a 10-bit value in the “read from Group1 FIFO” mode. 1h = The Group1 conversion result is read out as an 8-bit value in the “read from Group1 FIFO” mode.
1	G1_MODE	R/W	0h	Group1 Conversion Mode. This bit defines whether the input channels selected for conversion in the Group1 are converted only once, or are continuously converted. Any operation mode: 0h = The channels selected for conversion in the Group1 are converted only once. 1h = The channels selected for conversion in the Group1 are converted continuously.

Table 22-13. ADG1MODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	FRZ_G1	R/W	0h	Group1 Freeze Enable. This bit allows a Group1 conversion sequence to be frozen if an Event Group or a Group2 conversion is requested. The Group1 conversion is kept frozen while the Event Group or Group2 conversion is active, and continues from where it was frozen once the Event Group or Group2 conversions are completed. While the Group1 conversion is frozen, the G1_STOP status flag in the ADG1SR register indicates that the Group1 conversions have stopped. This bit gets cleared when the Group1 conversions resume. Any operation mode: 0h = Group1 conversions cannot be frozen. All the channels selected for conversion in the Group1 are converted before the ADC switches over to servicing any other conversion group. 1h = Group1 conversions are frozen whenever there is a request for conversion from Event Group or Group2.

22.11.7 ADG2MODECR Register (Offset = 18h) [reset = 0h]

ADG2MODECR is shown in [Figure 22-26](#) and described in [Table 22-14](#).

Return to [Summary Table](#).

The format of the ADG2MODECR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

Figure 22-26. ADG2MODECR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							No_Reset_on_ChSel
R-0h							R/W-0h
15	14	13	12	11	10	9	8
RESERVED						G2_DATA_FMT	
R-0h						R/W-0h	
7	6	5	4	3	2	1	0
RESERVED		G2_CHID	OVR_G2_RAM_IGN	G2_HW_TRIG	G2_8BIT	G2_MODE	FRZ_G2
R-0h		R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h

Table 22-14. ADG2MODECR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reads return 0, writes have no effect.
16	No_Reset_on_ChSel	R/W	0h	No Group2 Results Memory Reset on New Channel Select. This bit determines whether the group2 results' RAM is reset whenever a non-zero value is written to the group2 channel select register. If the group2 conversions are ongoing (active or frozen), then writing a nonzero value to the group2 channel select register always resets the group2 results RAM. Any operation mode: 0h = Group2 results RAM is reset when a non-zero value is written to group2 channel select register, even if group2 conversions are completed. 1h = Group2 results RAM is not reset when a non-zero value is written to group2 channel select register, and group2 conversions are completed.
15-10	RESERVED	R	0h	Reads return 0, writes have no effect.
9-8	G2_DATA_FMT	R/W	0h	Group2 Read Data Format. Note: This field is only applicable when the ADC module is configured to be a 12-bit ADC module. This field is reserved when the module is configured as a 10-bit ADC module. This field determines the format in which the conversion results are read out of the group1 results RAM when using the FIFO interface, that is, when reading from the ADG2BUFFER or ADG2EMUBUFFER locations. Any operation mode: 0h = Conversion results are read out in full 12-bit format. This is the default mode. 1h = Conversion results are read out in 10-bit format. Bits 11-2 of the 12-bit conversion result are returned as the 10-bit conversion result. 2h = Conversion results are read out in 8-bit format. Bits 11-4 of the 12-bit conversion result are returned as the 8-bit conversion result. 3h = Reserved. The full 12-bit conversion result is returned if programmed.

Table 22-14. ADG2MODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7-6	RESERVED	R	0h	Reads return 0, writes have no effect.
5	G2_CHID	R/W	0h	Enable Channel Id for the Group2 conversion results to be read. This bit only affects the “read from FIFO” mode. The ADC always stores the channel id in the results RAM. Any 16-bit read performed in the “read from RAM” mode returns the 5-bit channel id along with the 10-bit conversion result. Any operation mode: 0h = Bits 14-10, the channel id field, of the data read from the Group2 results’ FIFO is read as 00000b. 1h = Bits 14-10, the channel id field, of the data read from the Group2 results’ FIFO contains the number of the ADC analog input to which the conversion result belongs.
4	OVR_G2_RAM_IGN	R/W	0h	This bit allows the ADC module to overwrite the contents of the Group2 results memory under an overrun condition. Any operation mode: 0h = The ADC cannot overwrite the contents of the Group2 results memory. When an overrun of this memory occurs, the software needs to read out all the contents of this memory before the ADC is able to write a new conversion result for the Group2. 1h = When an overrun of the Group2 results memory occurs, the ADC proceeds to overwrite the contents with any new conversion results for the Group2, starting with the first location in this memory.
3	G2_HW_TRIG	R/W	0h	Group2 Hardware Triggered. This bit allows the Group2 to be hardware triggered. The Group2 is software triggered by default. For more details on how to trigger a conversion group, refer to How to Start a Conversion subsection. Any operation mode: 0h = The Group2 is software-triggered. A Group2 conversion starts whenever the Group2 channel select register (ADG2SEL) is written with a non-zero value. 1h = The Group2 is hardware-triggered. A Group2 conversion starts whenever the Group2 channel select register has a non-zero value, and the specified hardware trigger occurs. The hardware trigger for the Group2 is specified in the Group2 Trigger Source register (ADG2SRC).
2	G2_8BIT	R/W	0h	Group2 8-bit result mode. Note: This bit is only applicable when the ADC module is configured to be a 10-bit ADC module. This bit is reserved when the module is configured as a 12-bit ADC module. This bit allows the Group2 conversion results to be read out in an 8-bit format. This bit only applies to the “read from FIFO” mode. The lower 2 bits of the 10-bit conversion result are discarded and the upper 8 bits are shifted right two places to form the 8-bit conversion result. Any operation mode: 0h = The Group2 conversion result is read out as a 10-bit value in the “read from Group2 FIFO” mode. 1h = The Group2 conversion result is read out as an 8-bit value in the “read from Group2 FIFO” mode.
1	G2_MODE	R/W	0h	Group2 Conversion Mode. This bit defines whether the input channels selected for conversion in the Group2 are converted only once, or are continuously converted. Any operation mode: 0h = The channels selected for conversion in the Group2 are converted only once. 1h = The channels selected for conversion in the Group2 are converted continuously.

Table 22-14. ADG2MODECR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
0	FRZ_G2	R/W	0h	Group2 Freeze Enable. This bit allows a Group2 conversion sequence to be frozen if an Event Group or a Group1 conversion is requested. The Group2 conversion is kept frozen while the Event Group or Group1 conversion is active, and continues from where it was frozen once the Event Group or Group1 conversions are completed. While the Group2 conversion is frozen, the G2_STOP status flag in the ADG2SR register indicates that the Group2 conversions have stopped. This bit gets cleared when the Group2 conversions resume. Any operation mode: 0h = Group2 conversions cannot be frozen. All the channels selected for conversion in the Group2 are converted before the ADC switches over to servicing any other conversion group. 1h = Group2 conversions are frozen whenever there is a request for conversion from Event Group or Group1.

22.11.8 ADEVSRC Register (Offset = 1Ch) [reset = 0h]

ADEVSRC is shown in [Figure 22-27](#) and described in [Table 22-15](#).

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Figure 22-27. ADEVSRC Register

7	6	5	4	3	2	1	0
RESERVED			EV_EDG_BOTH	EV_EDG_SEL	EV_SRC		
R-0h			R/W-0h	R/W-0h	R/W-0h		

Table 22-15. ADEVSRC Register Field Descriptions

Bit	Field	Type	Reset	Description
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4	EV_EDG_BOTH	R/W	0h	EV Group Trigger Edge Polarity Select. This bit configures the event group to be triggered on both rising and falling edge detected on the selected trigger source. Any operation mode: 0h = The conversion is triggered only upon detecting an edge defined by the EV_EDGE_SEL bit. 1h = The conversion is triggered upon detecting either a rising or falling edge.
3	EV_EDG_SEL	R/W	0h	Event Group Trigger Edge Polarity Select. This bit determines the polarity of the transition on the selected source that triggers the Event Group conversion. Any operation mode: 0h = A high-to-low transition on the selected source triggers the Event Group conversion. 1h = A low-to-high transition on the selected source triggers the Event Group conversion.
2-0	EV_SRC	R/W	0h	Event Group Trigger Source. Any operation mode: 0h = The ADC module allows a trigger source to be selected for the Event Group from up to eight options. These options are device-specific and the device data manual must be referred to identify the actual trigger sources from 0 to 7h.

22.11.9 ADG1SRC Register (Offset = 20h) [reset = 0h]

ADG1SRC is shown in [Figure 22-28](#) and described in [Table 22-16](#).

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Figure 22-28. ADG1SRC Register

7	6	5	4	3	2	1	0
RESERVED			G1_EDG_BOTH	G1_EDG_SEL	G1_SRC		
R-0h			R/W-0h	R/W-0h	R/W-0h		

Table 22-16. ADG1SRC Register Field Descriptions

Bit	Field	Type	Reset	Description
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4	G1_EDG_BOTH	R/W	0h	Group1 Trigger Edge Polarity Select. This bit configures the group1 to be triggered on both rising and falling edge detected on the selected trigger source. Any operation mode: 0h = The conversion is triggered only upon detecting an edge defined by the G1_EDGE_SEL bit. 1h = The conversion is triggered upon detecting either a rising or falling edge.
3	G1_EDG_SEL	R/W	0h	Group1 Trigger Edge Polarity Select. This bit determines the polarity of the transition on the selected source that triggers the Group1 conversion. Any operation mode: 0h = A high-to-low transition on the selected source triggers the Group1 conversion. 1h = A low-to-high transition on the selected source triggers the Group1 conversion.
2-0	G1_SRC	R/W	0h	Group1 Trigger Source. Any operation mode: 0h = The ADC module allows a trigger source to be selected for the Group1 from up to eight options. These options are device-specific and the device data manual must be referred to identify the actual trigger sources from 0 to 7h.

22.11.10 ADG2SRC Register (Offset = 24h) [reset = 0h]

ADG2SRC is shown in [Figure 22-29](#) and described in [Table 22-17](#).

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Figure 22-29. ADG2SRC Register

7	6	5	4	3	2	1	0
RESERVED			G2_EDG_BOTH	G2_EDG_SEL	G2_SRC		
R-0h			R/W-0h	R/W-0h	R/W-0h		

Table 22-17. ADG2SRC Register Field Descriptions

Bit	Field	Type	Reset	Description
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4	G2_EDG_BOTH	R/W	0h	Group2 Trigger Edge Polarity Select. This bit configures the group2 to be triggered on both rising and falling edge detected on the selected trigger source. Any operation mode: 0h = The conversion is triggered only upon detecting an edge defined by the G2_EDGE_SEL bit. 1h = The conversion is triggered upon detecting either a rising or falling edge.
3	G2_EDG_SEL	R/W	0h	Group2 Trigger Edge Polarity Select. This bit determines the polarity of the transition on the selected source that triggers the Group2 conversion. Any operation mode: 0h = A high-to-low transition on the selected source triggers the Group2 conversion. 1h = A low-to-high transition on the selected source triggers the Group2 conversion.
2-0	G2_SRC	R/W	0h	Group2 Trigger Source. Any operation mode: 0h = The ADC module allows a trigger source to be selected for the Group2 from up to eight options. These options are device-specific and the device data manual must be referred to identify the actual trigger sources from 0 to 7h.

22.11.11 ADEVINTENA Register (Offset = 28h) [reset = 0h]

ADEVINTENA is shown in [Figure 22-30](#) and described in [Table 22-18](#).

Return to [Summary Table](#).

Figure 22-30. ADEVINTENA Register

7	6	5	4	3	2	1	0
RESERVED				EV_END_INT_EN	RESERVED	EV_OVR_INT_EN	EV_THR_INT_EN
R-0h				R/W-0h	R-0h	R/W-0h	R/W-0h

Table 22-18. ADEVINTENA Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	EV_END_INT_EN	R/W	0h	Event Group Conversion End Interrupt Enable. Refer to Group Conversion End Interrupt subsection for more details on the conversion end interrupts. Any operation mode: 0h = No interrupt is generated when conversion of all the channels selected for conversion in the Event Group is done. 1h = An Event Group conversion end interrupt is generated when conversion of all the channels selected for conversion in the Event Group is done.
2	RESERVED	R	0h	Reads return 0, writes have no effect.
1	EV_OVR_INT_EN	R/W	0h	Event Group Memory Overrun Interrupt Enable. A memory overrun occurs when the ADC tries to write a new conversion result to the Event Group results memory which is already full. For more details on the overrun interrupts, refer to Group Memory Overrun Interrupt subsection. Any operation mode: 0h = No interrupt is generated if an Event Group memory overrun occurs. 1h = An Event Group memory overrun interrupt is generated if an Event Group memory overrun condition occurs.
0	EV_THR_INT_EN	R/W	0h	Event Group Threshold Interrupt Enable. An Event Group threshold interrupt occurs when the programmed Event Group threshold counter counts down to 0. Refer to Group Memory Threshold Interrupt subsection for more details. Any operation mode: 0h = No interrupt is generated if the Event Group threshold counter reaches 0. 1h = An Event Group threshold interrupt is generated if the Event Group threshold counter reaches 0.

22.11.12 ADG1INTENA Register (Offset = 2Ch) [reset = 0h]

ADG1INTENA is shown in [Figure 22-31](#) and described in [Table 22-19](#).

Return to [Summary Table](#).

Figure 22-31. ADG1INTENA Register

7	6	5	4	3	2	1	0
RESERVED				G1_END_INT_EN	RESERVED	G1_OVR_INT_EN	G1_THR_INT_EN
R-0h				R/W-0h	R-0h	R/W-0h	R/W-0h

Table 22-19. ADG1INTENA Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	G1_END_INT_EN	R/W	0h	Group1 Conversion End Interrupt Enable. Refer to Group Conversion End Interrupt subsection for more details on the conversion end interrupts. Any operation mode: 0h = No interrupt is generated when conversion of all the channels selected for conversion in the Group1 is done. 1h = A Group1 conversion end interrupt is generated when conversion of all the channels selected for conversion in the Group1 is done.
2	RESERVED	R	0h	Reads return 0, writes have no effect.
1	G1_OVR_INT_EN	R/W	0h	Group1 Memory Overrun Interrupt Enable. A memory overrun occurs when the ADC tries to write a new conversion result to the Group1 results memory which is already full. For more details on the overrun interrupts, refer to Group Memory Overrun Interrupt subsection. Any operation mode: 0h = No interrupt is generated if a Group1 memory overrun occurs. 1h = A Group1 memory overrun interrupt is generated if a Group1 memory overrun condition occurs.
0	G1_THR_INT_EN	R/W	0h	Group1 Threshold Interrupt Enable. A Group1 threshold interrupt occurs when the programmed Group1 threshold counter counts down to 0. Refer to Group Memory Threshold Interrupt subsection for more details. Any operation mode: 0h = No interrupt is generated if the Group1 threshold counter reaches 0. 1h = A Group1 threshold interrupt is generated if the Group1 threshold counter reaches 0.

22.11.13 ADG2INTENA Register (Offset = 30h) [reset = 0h]

ADG2INTENA is shown in [Figure 22-32](#) and described in [Table 22-20](#).

Return to [Summary Table](#).

Figure 22-32. ADG2INTENA Register

7	6	5	4	3	2	1	0
RESERVED				G2_END_INT_EN	RESERVED	G2_OVR_INT_EN	G2_THR_INT_EN
R-0h				R/W-0h	R-0h	R/W-0h	R/W-0h

Table 22-20. ADG2INTENA Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	G2_END_INT_EN	R/W	0h	Group2 Conversion End Interrupt Enable. Refer to Group Conversion End Interrupt subsection for more details on the conversion end interrupts. Any operation mode: 0h = No interrupt is generated when conversion of all the channels selected for conversion in the Group2 is done. 1h = A Group2 conversion end interrupt is generated when conversion of all the channels selected for conversion in the Group2 is done.
2	RESERVED	R	0h	Reads return 0, writes have no effect.
1	G2_OVR_INT_EN	R/W	0h	Group2 Memory Overrun Interrupt Enable. A memory overrun occurs when the ADC tries to write a new conversion result to the Group2 results memory which is already full. For more details on the overrun interrupts, refer to Group Memory Overrun Interrupt subsection. Any operation mode: 0h = No interrupt is generated if a Group2 memory overrun occurs. 1h = A Group2 memory overrun interrupt is generated if a Group2 memory overrun condition occurs.
0	G2_THR_INT_EN	R/W	0h	Group2 Threshold Interrupt Enable. A Group2 threshold interrupt occurs when the programmed Group2 threshold counter counts down to 0. Refer to Group Memory Threshold Interrupt subsection for more details. Any operation mode: 0h = No interrupt is generated if the Group2 threshold counter reaches 0. 1h = A Group2 threshold interrupt is generated if the Group2 threshold counter reaches 0.

22.11.14 ADEVINTFLG Register (Offset = 34h) [reset = 4h]

ADEVINTFLG is shown in [Figure 22-33](#) and described in [Table 22-21](#).

Return to [Summary Table](#).

Figure 22-33. ADEVINTFLG Register

7	6	5	4	3	2	1	0
RESERVED				EV_END	EV_MEM_EMPTY	EV_MEM_OVERRUN	EV_THR_INT_FLG
R-0h				R/W1C-0h	R-1h	R-0h	R/W1C-0h

Table 22-21. ADEVINTFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	EV_END	R/W1C	0h	Event Group Conversion End. This bit is set only if the Event Group conversions are configured to be in the single-conversion mode. This bit is cleared by any one of the following ways: By writing a 1 to this bit. By writing a 1 to the Event Group status register bit 0 (EV_END). By reading one conversion result from the Event Group results' memory in the read from FIFO mode. By writing a new set of channels to the Event Group channel select register. Any operation mode read: 0h = All the channels selected for conversion in the Event Group have not yet been converted. 1h = All the channels selected for conversion in the Event Group have been converted. An Event Group conversion end interrupt is generated, if enabled, when this bit gets set.
2	EV_MEM_EMPTY	R	1h	Event Group Results Memory Empty. This is a read-only bit, writes have no effect. It is not a source of an interrupt from the ADC module. Any operation mode read: 0h = The Event Group results memory is not empty. 1h = The Event Group results memory is empty.
1	EV_MEM_OVERRUN	R	0h	Event Group Memory Overrun. This is a read-only bit, writes have no effect. Any operation mode read: 0h = Event Group results memory has not overrun. 1h = Event Group results memory has overrun.
0	EV_THR_INT_FLG	R/W1C	0h	Event Group Threshold Interrupt Flag. This bit is cleared by writing a 1, writing a 0 has no effect. Any operation mode read: 0h = The number of conversions completed for the Event Group is smaller than the threshold programmed in the Event Group interrupt threshold register. 1h = The number of conversions completed for the Event Group is equal to or greater than the threshold programmed in the Event Group interrupt threshold register.

22.11.15 ADG1INTFLG Register (Offset = 38h) [reset = 4h]

ADG1INTFLG is shown in [Figure 22-34](#) and described in [Table 22-22](#).

Return to [Summary Table](#).

Figure 22-34. ADG1INTFLG Register

7	6	5	4	3	2	1	0
RESERVED				G1_END	G1_MEM_EMPTY	G1_MEM_OVERRUN	G1_THR_INT_FLG
R-0h				R/W1C-0h	R-1h	R-0h	R/W1C-0h

Table 22-22. ADG1INTFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	G1_END	R/W1C	0h	Group1 Conversion End. This bit is set only if the Group1 conversions are configured to be in the single-conversion mode. This bit is cleared by any one of the following ways: By writing a 1 to this bit. By writing a 1 to the Group1 status register bit 0 (G1_END). By reading one conversion result from the Group1 results' memory in the read from FIFO mode. By writing a new set of channels to the Group1 channel select register. Any operation mode read: 0h = All the channels selected for conversion in the Group1 have not yet been converted. 1h = All the channels selected for conversion in the Group1 have been converted. A Group1 conversion end interrupt is generated, if enabled, when this bit gets set.
2	G1_MEM_EMPTY	R	1h	Group1 Results Memory Empty. This is a read-only bit, writes have no effect. It is not a source of an interrupt from the ADC module. Any operation mode read: 0h = The Group1 results memory is not empty. 1h = The Group1 results memory is empty.
1	G1_MEM_OVERRUN	R	0h	Group1 Memory Overrun. This is a read-only bit, writes have no effect. Any operation mode read: 0h = Group1 results memory has not overrun. 1h = Group1 results memory has overrun.
0	G1_THR_INT_FLG	R/W1C	0h	Group1 Threshold Interrupt Flag. This bit is cleared by writing a 1, writing a 0 has no effect. Any operation mode read: 0h = The number of conversions completed for the Group1 is smaller than the threshold programmed in the Group1 interrupt threshold register. 1h = The number of conversions completed for the Group1 is equal to or greater than the threshold programmed in the Group1 interrupt threshold register.

22.11.16 ADG2INTFLG Register (Offset = 3Ch) [reset = 4h]

ADG2INTFLG is shown in [Figure 22-35](#) and described in [Table 22-23](#).

Return to [Summary Table](#).

Figure 22-35. ADG2INTFLG Register

7	6	5	4	3	2	1	0
RESERVED				G2_END	G2_MEM_EMPTY	G2_MEM_OVERRUN	G2_THR_INT_FLG
R-0h				R/W1C-0h	R-1h	R-0h	R/W1C-0h

Table 22-23. ADG2INTFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	G2_END	R/W1C	0h	Group2 Conversion End. This bit is set only if the Group2 conversions are configured to be in the single-conversion mode. This bit is cleared by any one of the following ways: By writing a 1 to this bit. By writing a 1 to the Group2 status register bit 0 (G2_END). By reading one conversion result from the Group2 results' memory in the read from FIFO mode. By writing a new set of channels to the Group2 channel select register. Any operation mode read: 0h = All the channels selected for conversion in the Group2 have not yet been converted. 1h = All the channels selected for conversion in the Group2 have been converted. A Group2 conversion end interrupt is generated, if enabled, when this bit gets set.
2	G2_MEM_EMPTY	R	1h	Group2 Results Memory Empty. This is a read-only bit, writes have no effect. It is not a source of an interrupt from the ADC module. Any operation mode read: 0h = The Group2 results memory is not empty. 1h = The Group2 results memory is empty.
1	G2_MEM_OVERRUN	R	0h	Group2 Memory Overrun. This is a read-only bit, writes have no effect. Any operation mode read: 0h = Group2 results memory has not overrun. 1h = Group2 results memory has overrun.
0	G2_THR_INT_FLG	R/W1C	0h	Group2 Threshold Interrupt Flag. This bit is cleared by writing a 1, writing a 0 has no effect. Any operation mode read: 0h = The number of conversions completed for the Group2 is smaller than the threshold programmed in the Group2 interrupt threshold register. 1h = The number of conversions completed for the Group2 is equal to or greater than the threshold programmed in the Group2 interrupt threshold register.

22.11.17 ADEVTHRINTCR Register (Offset = 40h) [reset = 0h]

ADEVTHRINTCR is shown in [Figure 22-36](#) and described in [Table 22-24](#).

Return to [Summary Table](#).

Figure 22-36. ADEVTHRINTCR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Sign_Extension								EV_THR							
R-0h								R/W-0h							

Table 22-24. ADEVTHRINTCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0, writes have no effect.
15-9	Sign_Extension	R	0h	These bits always read the same as the bit 8 of this register.
8-0	EV_THR	R/W	0h	Event Group Threshold Counter. Before ADC conversions begin on the Event Group, this field is initialized to the number of conversion results that the Event Group memory should contain before interrupting the CPU. This counter decrements when the ADC module writes a new conversion result to the Event Group results' memory. The counter increments for each read of a conversion result from the Event Group results' memory in the "read from FIFO" mode. The threshold counter is not affected for a direct read from the Event Group results' memory. Also, a simultaneous ADC write and a CPU and DMA read from the Event Group FIFO leaves the threshold counter unchanged. In case of an Event Group Results' memory overrun condition, if new conversion results are not allowed to overwrite the existing memory contents, then the Event Group threshold counter is not decremented. Refer to Group Memory Threshold Interrupt subsection for more details on the threshold interrupts.

22.11.18 ADG1THRINTCR Register (Offset = 44h) [reset = 0h]

ADG1THRINTCR is shown in [Figure 22-37](#) and described in [Table 22-25](#).

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Figure 22-37. ADG1THRINTCR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Sign_Extension								G1_THR							
R-0h								R/W-0h							

Table 22-25. ADG1THRINTCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0, writes have no effect.
15-9	Sign_Extension	R	0h	These bits always read the same as the bit 8 of this register.
8-0	G1_THR	R/W	0h	Group1 Threshold Counter. Before ADC conversions begin on the Group1, this field is initialized to the number of conversion results that the Group1 memory should contain before interrupting the CPU. This counter decrements when the ADC module writes a new conversion result to the Group1 results' memory. The counter increments for each read of a conversion result from the Group1 results' memory in the "read from FIFO" mode. The threshold counter is not affected for a direct read from the group1 results' memory. Also, a simultaneous ADC write and a CPU and DMA read from the Group1 FIFO leaves the threshold counter unchanged. In case of an Group1 Results' memory overrun condition, if new conversion results are not allowed to overwrite the existing memory contents, then the Group1 threshold counter is not decremented. Refer to Group Memory Threshold Interrupt subsection for more details on the threshold interrupts.

22.11.19 ADG2THRINTCR Register (Offset = 48h) [reset = 0h]

ADG2THRINTCR is shown in [Figure 22-38](#) and described in [Table 22-26](#).

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Figure 22-38. ADG2THRINTCR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Sign_Extension								G2_THR							
R-0h								R/W-0h							

Table 22-26. ADG2THRINTCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0, writes have no effect.
15-9	Sign_Extension	R	0h	These bits always read the same as the bit 8 of this register.
8-0	G2_THR	R/W	0h	Group2 Threshold Counter. Before ADC conversions begin on the Group2, this field is initialized to the number of conversion results that the Group2 memory should contain before interrupting the CPU. This counter decrements when the ADC module writes a new conversion result to the Group2 results' memory. The counter increments for each read of a conversion result from the Group2 results' memory in the "read from FIFO" mode. The threshold counter is not affected for a direct read from the group2 results' memory. Also, a simultaneous ADC write and a CPU and DMA read from the Group2 FIFO leaves the threshold counter unchanged. In case of an Group2 Results' memory overrun condition, if new conversion results are not allowed to overwrite the existing memory contents, then the Group2 threshold counter is not decremented. Refer to Group Memory Threshold Interrupt subsection for more details on the threshold interrupts.

22.11.20 ADEVDMACR Register (Offset = 4Ch) [reset = 0h]

ADEVDMACR is shown in [Figure 22-39](#) and described in [Table 22-27](#).

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Figure 22-39. ADEVDMACR Register

31	30	29	28	27	26	25	24
RESERVED							EV_BLOCKS
R-0h							R/W-0h
23	22	21	20	19	18	17	16
EV_BLOCKS							
R/W-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				DMA_EV_END	EV_BLK_XFER	RESERVED	EV_DMA_EN
R-0h				R/W-0h	R/W-0h	R-0h	R/W-0h

Table 22-27. ADEVDMACR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-25	RESERVED	R	0h	Reads return 0, writes have no effect.
24-16	EV_BLOCKS	R/W	0h	Number of Event Group Result buffers to be transferred using DMA if the ADC module is configured to generate a DMA request. If the Event Group is configured to use the block transfer mode of the DMA module, then the ADC module generates a DMA request after the Event Group results' memory accumulates EV_BLOCKS number of conversion results. This feature is designed to be used in place of the threshold interrupt for the Event Group. As a result, the EV_THR field of the Event Group Interrupt Threshold Control Register and the EV_BLOCKS field of the Event Group DMA Control Register are the same. Any operation mode: 0h = No DMA transfer occurs even if EV_BLK_XFER is set to 1. 1h = One DMA request is generated if the EV_BLK_XFER is set to 1 and the specified number of Event Group conversion results have been accumulated from 1h to 1FFh.
15-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	DMA_EV_END	R/W	0h	Event Group Conversion End DMA Transfer Enable. If DMA_EV_END bit is set to 1, EV_DMA_EN bit is ignored and DMA requests are generated every time the DMA_EV_END flag in the event group status register is set. The DMA_EV_END bit must be set before enabling conversions for the event group. Any operation mode read: 0h = ADC module generates a DMA request for each write to the Event group results RAM if EV_DMA_EN is set. 1h = ADC module generates a DMA request when the ADC has completed the conversions for all channels selected for conversion in the event group.

Table 22-27. ADEVDMACR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
2	EV_BLK_XFER	R/W	0h	Event Group Block DMA Transfer Enable. If EV_BLK_XFER bit is set to 1, EV_DMA_EN bit is ignored and DMA requests are generated every time the Threshold Counter reaches 0 from a count value of 1. Any operation mode read: 0h = ADC module generates a DMA request for each write to the Event Group memory if EV_DMA_EN is set. 1h = ADC module generates a DMA request when the ADC has written EV_BLOCKS number of buffers into the Event Group memory.
1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	EV_DMA_EN	R/W	0h	Event Group DMA Transfer Enable. Any operation mode read: 0h = ADC module does not generate a DMA request when it writes the conversion result to the Event Group memory. 1h = ADC module generates a DMA transfer when the ADC has written to the Event Group memory. The EV_BLK_XFER bit must be cleared to 0 for this DMA request to be generated.

22.11.21 ADG1DMACR Register (Offset = 50h) [reset = 0h]

ADG1DMACR is shown in [Figure 22-40](#) and described in [Table 22-28](#).

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Figure 22-40. ADG1DMACR Register

31	30	29	28	27	26	25	24
RESERVED							G1_BLOCKS
R-0h							R/W-0h
23	22	21	20	19	18	17	16
G1_BLOCKS							
R/W-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				DMA_G1_END	G1_BLK_XFER	RESERVED	G1_DMA_EN
R-0h				R/W-0h	R/W-0h	R-0h	R/W-0h

Table 22-28. ADG1DMACR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-25	RESERVED	R	0h	Reads return 0, writes have no effect.
24-16	G1_BLOCKS	R/W	0h	Number of Group1 Result buffers to be transferred using DMA if the ADC module is configured to generate a DMA request. If the Group1 is configured to use the block transfer mode of the DMA module, then the ADC module generates a DMA request after the Group1 results' memory accumulates G1_BLOCKS number of conversion results. This feature is designed to be used in place of the threshold interrupt for the Group1. As a result, the G1_THR field of the Group1 Interrupt Threshold Control Register and the G1_BLOCKS field of the Group1 DMA Control Register are the same. Any operation mode: 0h = No DMA transfer occurs even if G1_BLK_XFER is set to 1. 1h = One DMA request is generated if the G1_BLK_XFER is set to 1 and the specified number of Group1 conversion results have been accumulated from 1h to 1FFh.
15-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	DMA_G1_END	R/W	0h	Group1 Conversion End DMA Transfer Enable. If DMA_G1_END bit is set to 1, G1_DMA_EN bit is ignored and DMA requests are generated every time the DMA_G1_END flag in the group 1 status register is set. The DMA_G1_END bit must be set before enabling conversions for the group 1. Any operation mode read: 0h = ADC module generates a DMA request for each write to the group1 results RAM if G1_DMA_EN is set. 1h = ADC module generates a DMA request when the ADC has completed the conversions for all channels selected for conversion in the group1.
2	G1_BLK_XFER	R/W	0h	Group1 Block DMA Transfer Enable. If G1_BLK_XFER bit is set to 1, G1_DMA_EN bit is ignored and DMA requests are generated every time the Threshold Counter reaches 0 from a count value of 1. Any operation mode read: 0h = ADC module generates a DMA request for each write to the Group1 memory if G1_DMA_EN is set. 1h = ADC module generates a DMA request when the ADC has written G1_BLOCKS number of buffers into the Group1 memory.

Table 22-28. ADG1DMACR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	G1_DMA_EN	R/W	0h	Group1 DMA Transfer Enable. Any operation mode read: 0h = ADC module does not generate a DMA request when it writes the conversion result to the Group1 memory. 1h = ADC module generates a DMA transfer when the ADC has written to the Group1 memory. The G1_BLK_XFER bit must be cleared to 0 for this DMA request to be generated.

22.11.22 ADG2DMACR Register (Offset = 54h) [reset = 0h]

ADG2DMACR is shown in [Figure 22-41](#) and described in [Table 22-29](#).

Return to [Summary Table](#).

Figure 22-41. ADG2DMACR Register

31	30	29	28	27	26	25	24
RESERVED							G2_BLOCKS
R-0h							R/W-0h
23	22	21	20	19	18	17	16
G2_BLOCKS							
R/W-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				DMA_G2_END	G2_BLK_XFER	RESERVED	G2_DMA_EN
R-0h				R/W-0h	R/W-0h	R-0h	R/W-0h

Table 22-29. ADG2DMACR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-25	RESERVED	R	0h	Reads return 0, writes have no effect.
24-16	G2_BLOCKS	R/W	0h	Number of Group2 Result buffers to be transferred using DMA if the ADC module is configured to generate a DMA request. If the Group2 is configured to use the block transfer mode of the DMA module, then the ADC module generates a DMA request after the Group2 results' memory accumulates G2_BLOCKS number of conversion results. This feature is designed to be used in place of the threshold interrupt for the Group2. As a result, the G2_THR field of the Group2 Interrupt Threshold Control Register and the G2_BLOCKS field of the Group2 DMA Control Register are the same. Any operation mode: 0h = No DMA transfer occurs even if G2_BLK_XFER is set to 1. 1h = One DMA request is generated if the G2_BLK_XFER is set to 1 and the specified number of Group2 conversion results have been accumulated from 1h to 1FFh.
15-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	DMA_G2_END	R/W	0h	Group2 Conversion End DMA Transfer Enable. If DMA_G2_END bit is set to 1, G2_DMA_EN bit is ignored and DMA requests are generated every time the DMA_G2_END flag in the group 2 status register is set. The DMA_G2_END bit must be set before enabling conversions for the group 2. Any operation mode read: 0h = ADC module generates a DMA request for each write to the group2 results RAM if G2_DMA_EN is set. 1h = ADC module generates a DMA request when the ADC has completed the conversions for all channels selected for conversion in the group2.
2	G2_BLK_XFER	R/W	0h	Group2 Block DMA Transfer Enable. If G2_BLK_XFER bit is set to 1, G2_DMA_EN bit is ignored and DMA requests are generated every time the Threshold Counter reaches 0 from a count value of 1. Any operation mode read: 0h = ADC module generates a DMA request for each write to the Group2 memory if G2_DMA_EN is set. 1h = ADC module generates a DMA request when the ADC has written G2_BLOCKS number of buffers into the Group2 memory.

Table 22-29. ADG2DMACR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	G2_DMA_EN	R/W	0h	Group2 DMA Transfer Enable. Any operation mode read: 0h = ADC module does not generate a DMA request when it writes the conversion result to the Group2 memory. 1h = ADC module generates a DMA transfer when the ADC has written to the Group2 memory. The G2_BLK_XFER bit must be cleared to 0 for this DMA request to be generated.

22.11.23 ADBNDCR Register (Offset = 58h) [reset = 0h]

ADBNDCR is shown in [Figure 22-42](#) and described in [Table 22-30](#).

Return to [Summary Table](#).

Refer to *How Results are Stored in the Results' Memory* subsection for further details on how the conversion results are stored in the ADC results' RAM.

Figure 22-42. ADBNDCR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED								BNDA							
R-0h								R/W-0h							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								BNDB							
R-0h								R/W-0h							

Table 22-30. ADBNDCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-25	RESERVED	R	0h	Reads return 0, writes have no effect.
24-16	BNDA	R/W	0h	Buffer Boundary A. These bits determine the memory available for the Event Group conversion results. The memory available is specified in terms of pairs of result buffers. A total of (2 x BNDA) buffers are available in the ADC results memory for storing Event Group conversion results from 0 to 1FFh. Any operation mode: 0h = Event Group conversions are not required. If Event Group conversions are performed with the BNDA value of 0, then the Event Group memory size defaults to 1024 words. For proper usage of the ADC results memory, configure the BNDA value to be non-zero and lower than the BNDB value.
15-9	RESERVED	R	0h	Reads return 0, writes have no effect.
8-0	BNDB	R/W	0h	Buffer Boundary B. These bits specify the number of buffers allocated for the Event Group plus the number of buffers allocated for the Group1. The number of buffer pairs allocated for storing Group1 conversion results is determined by subtracting BNDA from BNDB. As a result, BNDB must always be specified as greater than or equal to BNDA. A total of 2 x (BNDB - BNDA) buffers are available in the ADC results memory for storing Group1 conversion results from 0 to 1FFh. Any operation mode: 0h = Event Group as well as Group1 conversions are not required.

22.11.24 ADBNDEND Register (Offset = 5Ch) [reset = 0h]

ADBNDEND is shown in [Figure 22-43](#) and described in [Table 22-31](#).

Return to [Summary Table](#).

Figure 22-43. ADBNDEND Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							BUF_INIT_ACTIVE
R-0h							R-0h
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED					BNDEND		
R-0h					R/W-0h		

Table 22-31. ADBNDEND Register Field Descriptions

Bit	Field	Type	Reset	Description
31-17	RESERVED	R	0h	Reads return 0, writes have no effect.
16	BUF_INIT_ACTIVE	R	0h	ADC Results Memory Auto-initialization Status. Any operation mode: 0h = ADC Results Memory is currently not being initialized, and the ADC is available. If this bit is read as 0 after triggering an auto-initialization of the ADC results memory, then the ADC results memory has been completely initialized to 0. For devices requiring parity checking on the ADC results memory, the parity bit in the results memory is also initialized according to the parity polarity. The parity polarity as well as the auto-initialization process is controlled by the System module. Refer to the Architecture chapter for more details. 1h = ADC results memory is being initialized, and the ADC is not available for conversion.
15-3	RESERVED	R	0h	Reads return 0, writes have no effect.
2-0	BNDEND	R/W	0h	Buffer Boundary End. These bits specify the total number of memory buffers available for storing the ADC conversion results. These bits should be programmed to match the number of ADC conversion result buffers required to be used for the application. Any operation mode: 0h = 16 words available for storing ADC conversion results. 1h = 32 words available for storing ADC conversion results. 2h = 64 words available for storing ADC conversion results. This is the maximum configuration allowed since the device supports 64 buffers each for ADC1 as well as ADC2. 3h = Reserved 4h = Reserved 5h = Reserved 6h = Reserved 7h = Reserved

22.11.25 ADEVSAMP Register (Offset = 60h) [reset = 0h]

ADEVSAMP is shown in [Figure 22-44](#) and described in [Table 22-32](#).

Return to [Summary Table](#).

Figure 22-44. ADEVSAMP Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED											EV_ACQ																				
R-0h											R/W-0h																				

Table 22-32. ADEVSAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	EV_ACQ	R/W	0h	Event Group Acquisition Time. These bits define the sampling window (SW) for the Event Group conversions. $SW = EV_ACQ + 2$ in terms of ADCLK cycles. There are two factors that determine the minimum sampling window value required: First, the ADC module design requires that software is greater than or equal to 3 ADCLK cycles. Second, the ADC input impedance necessitates a certain minimum sampling time. This needs to be assured by configuring the EV_ACQ value properly considering the frequency of the ADCLK signal. Refer to the device datasheet to determine the minimum sampling time for this device.

22.11.26 ADG1SAMP Register (Offset = 64h) [reset = 0h]

ADG1SAMP is shown in [Figure 22-45](#) and described in [Table 22-33](#).

Return to [Summary Table](#).

Figure 22-45. ADG1SAMP Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED												G1_ACQ																			
R-0h												R/W-0h																			

Table 22-33. ADG1SAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	G1_ACQ	R/W	0h	Group1 Acquisition Time. These bits define the sampling window (SW) for the Group1 conversions. $SW = G1_ACQ + 2$ in terms of ADCLK cycles. There are two factors that determine the minimum sampling window value required: First, the ADC module design requires that software is greater than or equal to 3 ADCLK cycles. Second, the ADC input impedance necessitates a certain minimum sampling time. This needs to be assured by configuring the G1_ACQ value properly considering the frequency of the ADCLK signal. Refer to the device datasheet to determine the minimum sampling time for this device.

22.11.27 ADG2SAMP Register (Offset = 68h) [reset = 0h]

ADG2SAMP is shown in [Figure 22-46](#) and described in [Table 22-34](#).

Return to [Summary Table](#).

Figure 22-46. ADG2SAMP Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
RESERVED																				G2_ACQ																
R-0h																				R/W-0h																

Table 22-34. ADG2SAMP Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	G2_ACQ	R/W	0h	Group2 Acquisition Time. These bits define the sampling window (SW) for the Group2 conversions. $SW = G2_ACQ + 2$ in terms of ADCLK cycles. There are two factors that determine the minimum sampling window value required: First, the ADC module design requires that software is greater than or equal to 3 ADCLK cycles. Second, the ADC input impedance necessitates a certain minimum sampling time. This needs to be assured by configuring the G2_ACQ value properly considering the frequency of the ADCLK signal. Refer to the device datasheet to determine the minimum sampling time for this device.

22.11.28 ADEVSR Register (Offset = 6Ch) [reset = 8h]

ADEVSR is shown in [Figure 22-47](#) and described in [Table 22-35](#).

Return to [Summary Table](#).

Figure 22-47. ADEVSR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				EV_MEM_EMPTY	EV_BUSY	EV_STOP	EV_END
R-0h				R-1h	R-0h	R-0h	R/W-0h

Table 22-35. ADEVSR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	EV_MEM_EMPTY	R	1h	Event Group Results Memory Empty. This bit is effectively used only when the conversion results are read out of the Event Group results memory in the "read from FIFO" mode. Any operation mode read: 0h = The Event Group results memory has valid conversion results. 1h = The Event Group results memory is empty, or does not contain any unread conversion results.
2	EV_BUSY	R	0h	Event Group Conversion Busy. Any operation mode read: 0h = Event Group conversions are neither in progress nor frozen. 1h = Event Group conversions are either in progress, or are frozen for servicing some other group. This bit is always set when the Event Group is configured to be in the continuous conversion mode.
1	EV_STOP	R	0h	Event Group Conversion Stopped. Any operation mode read: 0h = Event Group conversions are not currently frozen. 1h = Event Group conversions are currently frozen.
0	EV_END	R/W	0h	Event Group Conversions Ended. This bit is cleared under the following conditions: By reading a conversion result from the Event Group results memory in the "read from FIFO" mode. By writing a new value to the Event Group channel select register ADEVSEL. By writing a 1 to this bit. By disabling the ADC module by clearing the ADC_EN bit in the ADC operating mode control register (ADOPMODECR). Any operation mode read: 0h = Event Group conversions have either not been started or have not yet completed since the last time this status bit was cleared. 1h = The conversion for all the channels selected in the Event Group has completed.

22.11.29 ADG1SR Register (Offset = 70h) [reset = 8h]

ADG1SR is shown in [Figure 22-48](#) and described in [Table 22-36](#).

Return to [Summary Table](#).

Figure 22-48. ADG1SR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				G1_MEM_EMPTY	G1_BUSY	G1_STOP	G1_END
R-0h				R-1h	R-0h	R-0h	R/W-0h

Table 22-36. ADG1SR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	G1_MEM_EMPTY	R	1h	Group1 Results Memory Empty. This bit is effectively used only when the conversion results are read out of the Group1 results memory in the "read from FIFO" mode. Any operation mode read: 0h = The Group1 results memory has valid conversion results. 1h = The Group1 results memory is empty, or does not contain any unread conversion results.
2	G1_BUSY	R	0h	Group1 Conversion Busy. Any operation mode read: 0h = Group1 conversions are neither in progress nor frozen. 1h = Group1 conversions are either in progress, or are frozen for servicing some other group. This bit is always set when the Group1 is configured to be in the continuous conversion mode.
1	G1_STOP	R	0h	Group1 Conversion Stopped. Any operation mode read: 0h = Group1 conversions are not currently frozen. 1h = Group1 conversions are currently frozen.
0	G1_END	R/W	0h	Group1 Conversions Ended. This bit is cleared under the following conditions: By reading a conversion result from the Group1 results memory in the "read from FIFO" mode. By writing a new value to the Group1 channel select register ADG1SEL. By writing a 1 to this bit. By disabling the ADC module by clearing the ADC_EN bit in the ADC operating mode control register (ADOPMODECR). Any operation mode read: 0h = Group1 conversions have either not been started or have not yet completed since the last time this status bit was cleared. 1h = The conversion for all the channels selected in the Group1 has completed.

22.11.30 ADG2SR Register (Offset = 74h) [reset = 8h]

ADG2SR is shown in [Figure 22-49](#) and described in [Table 22-37](#).

Return to [Summary Table](#).

Figure 22-49. ADG2SR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				G2_MEM_EMPTY	G2_BUSY	G2_STOP	G2_END
R-0h				R-1h	R-0h	R-0h	R/W-0h

Table 22-37. ADG2SR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3	G2_MEM_EMPTY	R	1h	Group2 Results Memory Empty. This bit is effectively used only when the conversion results are read out of the Group2 results memory in the "read from FIFO" mode. Any operation mode read: 0h = The Group2 results memory has valid conversion results. 1h = The Group2 results memory is empty, or does not contain any unread conversion results.
2	G2_BUSY	R	0h	Group2 Conversion Busy. Any operation mode read: 0h = Group2 conversions are neither in progress nor frozen. 1h = Group2 conversions are either in progress, or are frozen for servicing some other group. This bit is always set when the Group2 is configured to be in the continuous conversion mode.
1	G2_STOP	R	0h	Group2 Conversion Stopped. Any operation mode read: 0h = Group2 conversions are not currently frozen. 1h = Group2 conversions are currently frozen.
0	G2_END	R/W	0h	Group2 Conversions Ended. This bit is cleared under the following conditions: By reading a conversion result from the Group2 results memory in the "read from FIFO" mode. By writing a new value to the Group2 channel select register ADG2SEL. By writing a 1 to this bit. By disabling the ADC module by clearing the ADC_EN bit in the ADC operating mode control register (ADOPMODECR). Any operation mode read: 0h = Group2 conversions have either not been started or have not yet completed since the last time this status bit was cleared. 1h = The conversion for all the channels selected in the Group2 has completed.

22.11.31 ADEVSEL Register (Offset = 78h) [reset = 0h]

ADEVSEL is shown in [Figure 22-50](#) and described in [Table 22-38](#).

Return to [Summary Table](#).

Clearing ADEVSEL During a Conversion: Writing 0000h to ADEVSEL stops the Event Group conversions. This does not cause the ADC Event Group results Memory pointer or the Event Group Threshold Register to be reset.

Writing A Non-Zero Value To ADEVSEL During a Conversion: Writing a new value to ADEVSEL while a Channel in Event Group is being converted results in a new conversion sequence starting immediately with the highest priority channel in the new ADEVSEL selection. This also causes the ADC Event Group Results Memory pointer to be reset so that the memory allocated for storing the Event Group conversion results gets overwritten. Care should be taken to re-program the corresponding Interrupt Threshold Counter or DMA Threshold Counter again so that correct number of conversions happen before a Threshold interrupt or Block DMA request is generated.

ADC1 supports up to 24 channels and ADC2 supports up to 16 channels on the microcontroller.

Figure 22-50. ADEVSEL Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								EV_SEL																							
R-0h								R/W-0h																							

Table 22-38. ADEVSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-24	RESERVED	R	0h	Reads return 0, writes have no effect.
23-0	EV_SEL	R/W	0h	Event Group channels selected. Any non-zero value, the channels marked by the bit positions that are set to 1 are converted in ascending order when the Event Group is triggered. Any operation mode: 0h = No ADC input channel is selected for conversion in the Event Group.

22.11.32 ADG1SEL Register (Offset = 7Ch) [reset = 0h]

ADG1SEL is shown in [Figure 22-51](#) and described in [Table 22-39](#).

Return to [Summary Table](#).

Clearing ADG1SEL During a Conversion: Writing 0000h to ADG1SEL stops the Group1 conversions. This does not cause the ADC Group1 Results Memory pointer or the Group1 Threshold Register to be reset.

Writing A Non-Zero Value To ADG1SEL During a Conversion: Writing a new value to ADG1SEL while a Channel in Group1 is being converted results in a new conversion sequence starting immediately with the highest priority channel in the new ADG1SEL selection. This also causes the ADC Group1 Results Memory pointer to be reset so that the memory allocated for storing the Group1 conversion results gets overwritten. Care should be taken to re-program the corresponding Interrupt Threshold Counter or DMA Threshold Counter again so that correct number of conversions happen before a Threshold interrupt or Block DMA request is generated.

ADC1 supports up to 24 channels and ADC2 supports up to 16 channels on the microcontroller.

Figure 22-51. ADG1SEL Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								G1_SEL																							
R-0h								R/W-0h																							

Table 22-39. ADG1SEL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-24	RESERVED	R	0h	Reads return 0, writes have no effect.
23-0	G1_SEL	R/W	0h	Group1 channels selected. Any non-zero value, the channels marked by the bit positions that are set to 1 are converted in ascending order when the Group1 is triggered. Any operation mode: 0h = No ADC input channel is selected for conversion in the Group1.

22.11.33 ADG2SEL Register (Offset = 80h) [reset = 0h]

ADG2SEL is shown in [Figure 22-52](#) and described in [Table 22-40](#).

Return to [Summary Table](#).

Clearing ADG2SEL During a Conversion: Writing 0000h to ADG2SEL stops the Group2 conversions. This does not cause the ADC Group2 Results Memory pointer or the Group2 Threshold Register to be reset.

Writing A Non-Zero Value To ADG2SEL During a Conversion: Writing a new value to ADG2SEL while a Channel in Group2 is being converted results in a new conversion sequence starting immediately with the highest priority channel in the new ADG2SEL selection. This also causes the ADC Group2 Results Memory pointer to be reset so that the memory allocated for storing the Group2 conversion results gets overwritten. Care should be taken to re-program the corresponding Interrupt Threshold Counter or DMA Threshold Counter again so that correct number of conversions happen before a Threshold interrupt or Block DMA request is generated.

ADC1 supports up to 24 channels and ADC2 supports up to 16 channels on the microcontroller.

Figure 22-52. ADG2SEL Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								G2_SEL																							
R-0h								R/W-0h																							

Table 22-40. ADG2SEL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-24	RESERVED	R	0h	Reads return 0, writes have no effect.
23-0	G2_SEL	R/W	0h	Group2 channels selected. Any non-zero value, the channels marked by the bit positions that are set to 1 are converted in ascending order when the Group2 is triggered. Any operation mode: 0h = No ADC input channel is selected for conversion in the Group2.

22.11.34 ADCALR Register (Offset = 84h) [reset = 0h]

ADCALR is shown in [Figure 22-53](#) and described in [Table 22-41](#).

Return to [Summary Table](#).

The format of the ADCALR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

Figure 22-53. ADCALR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED												ADCALR																			
R-0h												R/W-0h																			

Table 22-41. ADCALR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	ADCALR	R/W	0h	ADC Calibration Result and Offset Error Correction Value. Note: The actual size of the ADCALR field is 12 bits or 10 bits depending on whether the ADC is configured to be in 12-bit or 10-bit resolution mode, respectively. Bits 11-10 are reserved when the module is configured as a 10-bit ADC module. The ADC module writes the results of the calibration conversions to this register. The application is required to use these conversion results and determine the ADC offset error. The application then computes the correction for the offset error and this correction value needs to be written back to the ADCALR register in the 2's complement form. During normal conversion (when calibration is disabled), the ADCALR register contents are automatically added to each digital output from the ADC core before it is stored in the ADC results memory. For more details on error calibration, refer to ADC Error Calibration Mode subsection. 0h = ADC Calibration Result and Offset Error Correction Value.

22.11.35 ADSTMSTATE Register (Offset = 88h) [reset = 0h]

ADSTMSTATE is shown in [Figure 22-54](#) and described in [Table 22-42](#).

Return to [Summary Table](#).

Figure 22-54. ADSTMSTATE Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED												SM_STATE			
R-0h												R-0h			

Table 22-42. ADSTMSTATE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3-0	SM_STATE	R	0h	ADC State Machine Current State. These bits reflect the current state of the state machine and are reserved for use by TI for debug purposes.

22.11.36 ADLASTCONV Register (Offset = 8Ch) [reset = X]

ADLASTCONV is shown in [Figure 22-55](#) and described in [Table 22-43](#).

Return to [Summary Table](#).

Figure 22-55. ADLASTCONV Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								LAST_CONV																							
R-0h								R-X																							

Table 22-43. ADLASTCONV Register Field Descriptions

Bit	Field	Type	Reset	Description
31-24	RESERVED	R	0h	Reads return 0, writes have no effect.
23-0	LAST_CONV	R	X	ADC Input Channel's Last Converted Value. This register indicates whether the last converted value for a particular input channel was lower or higher than the mid-point of the reference voltage. In other words, this register acts as a digital input register and is read by the application to determine the digital level at the input pins. This data is only valid for an input channel if it has been converted at least once. Any operation mode read for each bit of this register: 0h = A level lower than the midpoint reference voltage was measured at the last conversion for this channel. 1h = A level higher than or equal to the midpoint reference voltage was measured at the last conversion for this channel.

22.11.37 ADEVBUFFER_0 to ADEVBUFFER_7 Register (Offset = 90h to ACh) [reset = 8000000h]

ADEVBUFFER_0 to ADEVBUFFER_7 is shown in [Figure 22-56](#) and [Figure 22-57](#), and described in [Table 22-44](#).

Return to [Summary Table](#).

There are 8 ADC Event Group Results' FIFO Registers (ADEVBUFFER) at offsets 90h to ACh. The format of the data read from the ADEVBUFFER locations is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

Figure 22-56. ADEVBUFFER_0 to ADEVBUFFER_7 Register (12-bit)

31	30	29	28	27	26	25	24
EV_EMPTY	RESERVED						
R-1h	R-0h						
23	22	21	20	19	18	17	16
RESERVED				EV_CHID			
R-0h				R-0h			
15	14	13	12	11	10	9	8
RESERVED				EV_DR			
R-0h				R-0h			
7	6	5	4	3	2	1	0
EV_DR							
R-0h							

Figure 22-57. ADEVBUFFER_0 to ADEVBUFFER_7 Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
EV_EMPTY	EV_CHID				EV_DR		
R-1h	R-0h				R-0h		
7	6	5	4	3	2	1	0
EV_DR							
R-0h							

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Table 22-44. ADEVBUFFER_0 to ADEVBUFFER_7 Register Field Descriptions

Bit	Field	Type	Reset	Description
31	EV_EMPTY	R	1h	Event Group FIFO Empty. This bit is applicable only when the "read from FIFO" mode is used for reading the Event Group conversion results. Any operation mode read: 0h = The data in the EV_DR field of this buffer is valid. 1h = The data in the EV_DR field of this buffer is not valid and there are no valid data in the Event Group results memory.
30-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-16	EV_CHID	R	0h	Event Group Channel Id. These bits are also applicable only when the "read from FIFO" mode is used for reading the Event Group conversion results. The conversion result in the EV_DR field of this buffer is from the ADC input channel number denoted by the EV_CHID field from 1h to 1Fh. Any operation mode read: 0h = The conversion result in the EV_DR field of this buffer is from the ADC input channel 0, or the channel id mode is disabled in the Event Group mode control register (ADEVMODECR).
15-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	EV_DR	R	0h	Event Group Digital Conversion Result. The Event Group results' FIFO location is aliased eight times, so that any word-aligned read from the address range 90h to AFh results in one conversion result to be read from the Event Group results' memory. This allows the ARM LDMIA instruction to read out up to 8 conversion results from the Event Group results' memory with just one instruction.

22.11.38 ADG1BUFFER_0 to ADG1BUFFER_7 Register (Offset = B0h to CCh) [reset = X]

ADG1BUFFER_0 to ADG1BUFFER_7 is shown in [Figure 22-58](#) and [Figure 22-59](#), and described in [Table 22-45](#).

Return to [Summary Table](#).

There are 8 ADC Group1 Results FIFO Registers (ADG1BUFFER) at offsets B0h to CCh. The format of the data read from the ADG1BUFFER locations is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

The reset value of ADG1BUFFER comes from RAM and is random after reset.

Figure 22-58. ADG1BUFFER_0 to ADG1BUFFER_7 Register (12-bit)

31	30	29	28	27	26	25	24
G1_EMPTY	RESERVED						
R-1h	R-0h						
23	22	21	20	19	18	17	16
RESERVED				G1_CHID			
R-0h				R-X			
15	14	13	12	11	10	9	8
RESERVED				G1_DR			
R-0h				R-X			
7	6	5	4	3	2	1	0
G1_DR							
R-X							

Figure 22-59. ADG1BUFFER_0 to ADG1BUFFER_7 Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
G1_EMPTY	G1_CHID				G1_DR		
R-1h	R-X				R-X		
7	6	5	4	3	2	1	0
G1_DR							
R-X							

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Table 22-45. ADG1BUFFER_0 to ADG1BUFFER_7 Register Field Descriptions

Bit	Field	Type	Reset	Description
31	G1_EMPTY	R	1h	Group1 FIFO Empty. This bit is applicable only when the "read from FIFO" mode is used for reading the Group1 conversion results. Any operation mode read: 0h = The data in the G1_DR field of this buffer is valid. 1h = The data in the G1_DR field of this buffer is not valid and there are no valid data in the Group1 results memory.
30-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-16	G1_CHID	R	X	Group1 Channel Id. These bits are also applicable only when the "read from FIFO" mode is used for reading the Group1 conversion results. The conversion result in the G1_DR field of this buffer is from the ADC input channel number denoted by the G1_CHID field from 1h to 1Fh. Any operation mode read: 0h = The conversion result in the G1_DR field of this buffer is from the ADC input channel 0, or the channel id mode is disabled in the Group1 mode control register (ADG1MODECR).
15-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	G1_DR	R	X	Group1 Digital Conversion Result. The Group1 results' FIFO location is aliased eight times, so that any word-aligned read from the address range B0h to CFh results in one conversion result to be read from the Group1 results' memory. This allows the ARM LDMIA instruction to read out up to 8 conversion results from the Group1 results' memory with just one instruction.

22.11.39 ADG2BUFFER_0 to ADG2BUFFER_7 Register (Offset = D0h to ECh) [reset = X]

ADG2BUFFER_0 to ADG2BUFFER_7 is shown in [Figure 22-60](#) and [Figure 22-61](#), and described in [Table 22-46](#).

Return to [Summary Table](#).

There are 8 ADC Group2 Results FIFO Registers (ADG2BUFFER) at offsets D0 to ECh. The format of the data read from the ADG2BUFFER locations is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

The reset value of ADG2BUFFER comes from RAM and is random after reset.

Figure 22-60. ADG2BUFFER_0 to ADG2BUFFER_7 Register (12-bit)

31	30	29	28	27	26	25	24
G2_EMPTY	RESERVED						
R-1h				R-0h			
23	22	21	20	19	18	17	16
RESERVED				G2_CHID			
R-0h				R-X			
15	14	13	12	11	10	9	8
RESERVED				G2_DR			
R-0h				R-X			
7	6	5	4	3	2	1	0
G2_DR							
R-X							

Figure 22-61. ADG2BUFFER_0 to ADG2BUFFER_7 Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
G2_EMPTY	G2_CHID				G2_DR		
R-1h	R-X				R-X		
7	6	5	4	3	2	1	0
G2_DR							
R-X							

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Table 22-46. ADG2BUFFER_0 to ADG2BUFFER_7 Register Field Descriptions

Bit	Field	Type	Reset	Description
31	G2_EMPTY	R	1h	Group2 FIFO Empty. This bit is applicable only when the "read from FIFO" mode is used for reading the Group2 conversion results. Any operation mode read: 0h = The data in the G2_DR field of this buffer is valid. 1h = The data in the G2_DR field of this buffer is not valid and there are no valid data in the Group2 results memory.
30-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-16	G2_CHID	R	X	Group2 Channel Id. These bits are also applicable only when the "read from FIFO" mode is used for reading the Group2 conversion results. The conversion result in the G2_DR field of this buffer is from the ADC input channel number denoted by the G2_CHID field from 1h to 1Fh. Any operation mode read: 0h = The conversion result in the G2_DR field of this buffer is from the ADC input channel 0, or the channel id mode is disabled in the Group2 mode control register (ADG2MODECR).
15-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	G2_DR	R	X	Group2 Digital Conversion Result. The Group2 results' FIFO location is aliased eight times, so that any word-aligned read from the address range D0h to EFh results in one conversion result to be read from the Group2 results' memory. This allows the ARM LDMIA instruction to read out up to 8 conversion results from the Group2 results' memory with just one instruction.

22.11.40 ADEVEMUBUFFER Register (Offset = F0h) [reset = X]

ADEVEMUBUFFER is shown in [Figure 22-62](#) and [Figure 22-63](#), and described in [Table 22-47](#).

Return to [Summary Table](#).

The format of the data read from the ADEVEMUBUFFER locations is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

A read from this location also gives out one conversion result from the Event Group results' memory along with the EV_EMPTY status bit and the optional channel id. However, this read does not affect any of the status flags in the Event Group interrupt flag register or the Event Group status register. This register is useful for debuggers.

The reset value of ADEVEMUBUFFER comes from RAM and is random after reset.

Figure 22-62. ADEVEMUBUFFER Register (12-bit)

31	30	29	28	27	26	25	24
EV_EMPTY	RESERVED						
R-1h	R-0h						
23	22	21	20	19	18	17	16
RESERVED				EV_CHID			
R-0h				R-X			
15	14	13	12	11	10	9	8
RESERVED				EV_DR			
R-0h				R-X			
7	6	5	4	3	2	1	0
EV_DR							
R-X							

Figure 22-63. ADEVEMUBUFFER Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
EV_EMPTY	EV_CHID				EV_DR		
R-1h	R-X				R-X		
7	6	5	4	3	2	1	0
EV_DR							
R-X							

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Table 22-47. ADEVEMUBUFFER Register Field Descriptions

Bit	Field	Type	Reset	Description
31	EV_EMPTY	R	1h	Event Group FIFO Empty. This bit is applicable only when the "read from FIFO" mode is used for reading the Event Group conversion results. Any operation mode read: 0h = The data in the EV_DR field of this buffer is valid. 1h = The data in the EV_DR field of this buffer is not valid and there are no valid data in the Event Group results memory.
30-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-16	EV_CHID	R	X	Event Group Channel Id. These bits are also applicable only when the "read from FIFO" mode is used for reading the Event Group conversion results. The conversion result in the EV_DR field of this buffer is from the ADC input channel number denoted by the EV_CHID field from 1h to 1Fh. Any operation mode read: 0h = The conversion result in the EV_DR field of this buffer is from the ADC input channel 0, or the channel id mode is disabled in the Event Group mode control register (ADEVMODECR).
15-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	EV_DR	R	X	Event Group Digital Conversion Result. The Event Group results' FIFO location is aliased eight times, so that any word-aligned read from the address range 90h to AFh results in one conversion result to be read from the Event Group results' memory. This allows the ARM LDmia instruction to read out up to 8 conversion results from the Event Group results' memory with just one instruction.

22.11.41 ADG1EMUBUFFER Register (Offset = F4h) [reset = X]

ADG1EMUBUFFER is shown in [Figure 22-64](#) and [Figure 22-65](#), and described in [Table 22-48](#).

Return to [Summary Table](#).

The format of the data read from the ADG1EMUBUFFER locations is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

A read from this location also gives out one conversion result from the Group1 results' memory along with the G1_EMPTY status bit and the optional channel id. However, this read does not affect any of the status flags in the Group1 interrupt flag register or the Group1 status register. This register is useful for debuggers.

The reset value of ADG1EMUBUFFER comes from RAM and is random after reset.

Figure 22-64. ADG1EMUBUFFER Register (12-bit)

31	30	29	28	27	26	25	24
G1_EMPTY	RESERVED						
R-1h	R-0h						
23	22	21	20	19	18	17	16
RESERVED				G1_CHID			
R-0h				R-X			
15	14	13	12	11	10	9	8
RESERVED				G1_DR			
R-0h				R-X			
7	6	5	4	3	2	1	0
G1_DR							
R-X							

Figure 22-65. ADG1EMUBUFFER Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
G1_EMPTY	G1_CHID				G1_DR		
R-1h	R-X				R-X		
7	6	5	4	3	2	1	0
G1_DR							
R-X							

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Table 22-48. ADG1EMUBUFFER Register Field Descriptions

Bit	Field	Type	Reset	Description
31	G1_EMPTY	R	1h	Group1 FIFO Empty. This bit is applicable only when the "read from FIFO" mode is used for reading the Group1 conversion results. Any operation mode read: 0h = The data in the G1_DR field of this buffer is valid. 1h = The data in the G1_DR field of this buffer is not valid and there are no valid data in the Group1 results memory.
30-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-16	G1_CHID	R	X	Group1 Channel Id. These bits are also applicable only when the "read from FIFO" mode is used for reading the Group1 conversion results. The conversion result in the G1_DR field of this buffer is from the ADC input channel number denoted by the G1_CHID field from 1h to 1Fh. Any operation mode read: 0h = The conversion result in the G1_DR field of this buffer is from the ADC input channel 0, or the channel id mode is disabled in the Group1 mode control register (ADG1MODECR).
15-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	G1_DR	R	X	Group1 Digital Conversion Result. The Group1 results' FIFO location is aliased eight times, so that any word-aligned read from the address range B0h to CFh results in one conversion result to be read from the Group1 results' memory. This allows the ARM LDMIA instruction to read out up to 8 conversion results from the Group1 results' memory with just one instruction.

22.11.42 ADG2EMUBUFFER Register (Offset = F8h) [reset = X]

ADG2EMUBUFFER is shown in [Figure 22-66](#) and [Figure 22-67](#), and described in [Table 22-49](#).

Return to [Summary Table](#).

The format of the data read from the ADG2EMUBUFFER locations is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module.

A read from this location also gives out one conversion result from the Group2 results' memory along with the G2_EMPTY status bit and the optional channel id. However, this read does not affect any of the status flags in the Group2 interrupt flag register or the Group2 status register. This register is useful for debuggers.

The reset value of ADG2EMUBUFFER comes from RAM and is random after reset.

Figure 22-66. ADG2EMUBUFFER Register (12-bit)

31	30	29	28	27	26	25	24
G2_EMPTY	RESERVED						
R-1h	R-0h						
23	22	21	20	19	18	17	16
RESERVED				G2_CHID			
R-0h				R-X			
15	14	13	12	11	10	9	8
RESERVED				G2_DR			
R-0h				R-X			
7	6	5	4	3	2	1	0
G2_DR							
R-X							

Figure 22-67. ADG2EMUBUFFER Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
G2_EMPTY	G2_CHID				G2_DR		
R-1h	R-X				R-X		
7	6	5	4	3	2	1	0
G2_DR							
R-X							

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Table 22-49. ADG2EMUBUFFER Register Field Descriptions

Bit	Field	Type	Reset	Description
31	G2_EMPTY	R	1h	Group2 FIFO Empty. This bit is applicable only when the "read from FIFO" mode is used for reading the Group2 conversion results. Any operation mode read: 0h = The data in the G2_DR field of this buffer is valid. 1h = The data in the G2_DR field of this buffer is not valid and there are no valid data in the Group2 results memory.
30-21	RESERVED	R	0h	Reads return 0, writes have no effect.
20-16	G2_CHID	R	X	Group2 Channel Id. These bits are also applicable only when the "read from FIFO" mode is used for reading the Group2 conversion results. The conversion result in the G2_DR field of this buffer is from the ADC input channel number denoted by the G2_CHID field from 1h to 1Fh. Any operation mode read: 0h = The conversion result in the G2_DR field of this buffer is from the ADC input channel 0, or the channel id mode is disabled in the Group2 mode control register (ADG2MODECR).
15-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	G2_DR	R	X	Group2 Digital Conversion Result. The Group2 results' FIFO location is aliased eight times, so that any word-aligned read from the address range D0h to EFh results in one conversion result to be read from the Group2 results' memory. This allows the ARM LDMIA instruction to read out up to 8 conversion results from the Group2 results' memory with just one instruction.

22.11.43 ADEVTDIR Register (Offset = FCh) [reset = 0h]

ADEVTDIR is shown in [Figure 22-68](#) and described in [Table 22-50](#).

Return to [Summary Table](#).

Figure 22-68. ADEVTDIR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_DIR
R-0h							R/W-0h

Table 22-50. ADEVTDIR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_DIR	R/W	0h	ADEVT Pin Direction. Any operation mode: 0h = ADEVT is an input pin, the output buffer is disabled. 1h = ADEVT is an output pin, the output buffer is enabled.

22.11.44 ADEVTOUT Register (Offset = 100h) [reset = X]

ADEVTOUT is shown in [Figure 22-69](#) and described in [Table 22-51](#).

Return to [Summary Table](#).

The reset value of ADEVTOUT is unknown.

Figure 22-69. ADEVTOUT Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_OUT
R-0h							R/W-X

Table 22-51. ADEVTOUT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_OUT	R/W	X	ADEVT Pin Output Value. This bit determines the logic level to be output to the ADEVT pin when the pin is configured to be an output pin. Any operation mode: 0h = Output logic LOW on the ADEVT pin. 1h = Output logic HIGH on the ADEVT pin.

22.11.45 ADEVTIN Register (Offset = 104h) [reset = X]

ADEVTIN is shown in [Figure 22-70](#) and described in [Table 22-52](#).

Return to [Summary Table](#).

The reset value of ADEVTIN is unknown.

Figure 22-70. ADEVTIN Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_IN
R-0h							R-X

Table 22-52. ADEVTIN Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_IN	R	X	ADEVT Pin Input Value. This is a read-only bit that reflects the logic level on the ADEVT pin. Any operation mode read: 0h = Logic LOW present on the ADEVT pin. 1h = Logic HIGH present on the ADEVT pin.

22.11.46 ADEVTSET Register (Offset = 108h) [reset = X]

ADEVTSET is shown in [Figure 22-71](#) and described in [Table 22-53](#).

Return to [Summary Table](#).

The reset value of ADEVTSET is unknown.

Figure 22-71. ADEVTSET Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_SET
R-0h							R/W-X

Table 22-53. ADEVTSET Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_SET	R/W	X	ADEVT Pin Set. This bit drives the output of the ADEVT pin high. A read from this bit always returns the current state of the ADEVT pin. Any operation mode: 0h = Output value on the ADEVT pin is unchanged. 1h = Output logic HIGH on the ADEVT pin, if the pin is configured to be an output pin.

22.11.47 ADEVTCLR Register (Offset = 10Ch) [reset = X]

ADEVTCLR is shown in [Figure 22-72](#) and described in [Table 22-54](#).

Return to [Summary Table](#).

The reset value of ADEVTCLR is unknown.

Figure 22-72. ADEVTCLR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_CLR
R-0h							R/W-X

Table 22-54. ADEVTCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_CLR	R/W	X	ADEVT Pin Clear. A read from this bit always returns the current state of the ADEVT pin. Any operation mode: 0h = Output value on the ADEVT pin is unchanged. 1h = Output logic LOW on the ADEVT pin, if the pin is configured to be an output pin.

22.11.48 ADEVTPDR Register (Offset = 110h) [reset = 0h]

ADEVTPDR is shown in [Figure 22-73](#) and described in [Table 22-55](#).

Return to [Summary Table](#).

Figure 22-73. ADEVTPDR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_PDR
R-0h							R/W-0h

Table 22-55. ADEVTPDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_PDR	R/W	0h	ADEVT Pin Open Drain Enable. This bit enables the open-drain capability for the ADEVT pin if it is configured to be an output and a logic HIGH is being driven on to the pin. Any operation mode: 0h = Output value on the ADEVT pin is logic HIGH. 1h = The ADEVT pin is tristated.

22.11.49 ADEVTPDIS Register (Offset = 114h) [reset = 0h]

ADEVTPDIS is shown in [Figure 22-74](#) and described in [Table 22-56](#).

Return to [Summary Table](#).

Figure 22-74. ADEVTPDIS Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_PD
R-0h							DIS R/W-0h

Table 22-56. ADEVTPDIS Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_PD	R/W	0h	ADEVT Pin Pull Control Disable. This bit enables or disables the pull control on the ADEVT pin if it is configured to be an input pin. Any operation mode: 0h = Pull on ADEVT pin is enabled. 1h = Pull on ADEVT pin is disabled.

22.11.50 ADEVT_PSEL Register (Offset = 118h) [reset = 0h]

ADEVT_PSEL is shown in [Figure 22-75](#) and described in [Table 22-57](#).

Return to [Summary Table](#).

Figure 22-75. ADEVT_PSEL Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							ADEVT_PSEL
R-0h							R/W-0h

Table 22-57. ADEVT_PSEL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	ADEVT_PSEL	R/W	0h	ADEVT Pin Pull Control Select. This bit selects a pull-down or pull-up on the ADEVT pin if it is configured to be an input pin. Any operation mode: 0h = Pull down is selected on ADEVT pin. 1h = Pull up is selected on ADEVT pin.

22.11.51 ADEVSAMPDISEN Register (Offset = 11Ch) [reset = 0h]

ADEVSAMPDISEN is shown in [Figure 22-76](#) and described in [Table 22-58](#).

Return to [Summary Table](#).

Figure 22-76. ADEVSAMPDISEN Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
EV_SAMP_DIS_CYC							
R/W-0h							
7	6	5	4	3	2	1	0
RESERVED							EV_SAMP_DIS_EN
R-0h							R/W-0h

Table 22-58. ADEVSAMPDISEN Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0, writes have no effect.
15-8	EV_SAMP_DIS_CYC	R/W	0h	Event Group sample cap discharge cycles. These bits specify the duration in terms of ADCLK cycles for which the ADC internal sampling capacitor is allowed to discharge before sampling the input channel voltage.
7-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	EV_SAMP_DIS_EN	R/W	0h	Event Group sample cap discharge enable. Any operation mode: 0h = Event Group sample cap discharge mode is disabled. 1h = Event Group sample cap discharge mode is enabled. The ADC internal sampling capacitor is connected to the V_{refLo} reference voltage for a duration specified by the EV_SAMP_DIS_CYC field. After this discharge time has expired, the selected ADC input channel is sampled and converted normally based on the Event Group settings.

22.11.52 ADG1SAMPDISEN Register (Offset = 120h) [reset = 0h]

ADG1SAMPDISEN is shown in [Figure 22-77](#) and described in [Table 22-59](#).

Return to [Summary Table](#).

Figure 22-77. ADG1SAMPDISEN Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
G1_SAMP_DIS_CYC							
R/W-0h							
7	6	5	4	3	2	1	0
RESERVED							G1_SAMP_DIS_EN
R-0h							R/W-0h

Table 22-59. ADG1SAMPDISEN Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0, writes have no effect.
15-8	G1_SAMP_DIS_CYC	R/W	0h	Group1 sample cap discharge cycles. These bits specify the duration in terms of ADCLK cycles for which the ADC internal sampling capacitor is allowed to discharge before sampling the input channel voltage.
7-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	G1_SAMP_DIS_EN	R/W	0h	Group1 sample cap discharge enable. Any operation mode: 0h = Group1 sample cap discharge mode is disabled. 1h = Group1 sample cap discharge mode is enabled. The ADC internal sampling capacitor is connected to the V_{refLo} reference voltage for a duration specified by the G1_SAMP_DIS_CYC field. After this discharge time has expired, the selected ADC input channel is sampled and converted normally based on the Group1 settings.

22.11.53 ADG2SAMPDISEN Register (Offset = 124h) [reset = 0h]

ADG2SAMPDISEN is shown in [Figure 22-78](#) and described in [Table 22-60](#).

Return to [Summary Table](#).

Figure 22-78. ADG2SAMPDISEN Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
G2_SAMP_DIS_CYC							
R/W-0h							
7	6	5	4	3	2	1	0
RESERVED							G2_SAMP_DIS_EN
R-0h							R/W-0h

Table 22-60. ADG2SAMPDISEN Register Field Descriptions

Bit	Field	Type	Reset	Description
31-16	RESERVED	R	0h	Reads return 0, writes have no effect.
15-8	G2_SAMP_DIS_CYC	R/W	0h	Group2 sample cap discharge cycles. These bits specify the duration in terms of ADCLK cycles for which the ADC internal sampling capacitor is allowed to discharge before sampling the input channel voltage.
7-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	G2_SAMP_DIS_EN	R/W	0h	Group2 sample cap discharge enable. Any operation mode: 0h = Group2 sample cap discharge mode is disabled. 1h = Group2 sample cap discharge mode is enabled. The ADC internal sampling capacitor is connected to the V_{refLo} reference voltage for a duration specified by the G2_SAMP_DIS_CYC field. After this discharge time has expired, the selected ADC input channel is sampled and converted normally based on the Group2 settings.

22.11.54 ADMAGINT1CR Register (Offset = 128h) [reset = 0h]

ADMAGINT1CR is shown in [Figure 22-79](#) and [Figure 22-80](#), and described in [Table 22-61](#).

Return to [Summary Table](#).

The format of the ADMAGINT1CR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module. The ADC module supports up to three magnitude compare interrupts.

Figure 22-79. ADMAGINT1CR Register (12-bit)

31	30	29	28	27	26	25	24
RESERVED				MAG_THR1			
R-0h				R/W-0h			
23	22	21	20	19	18	17	16
MAG_THR1							
R/W-0h							
15	14	13	12	11	10	9	8
CHN_THR_COMP1	CMP_GE_LT1	RESERVED	COMP_CHID1				
R/W-0h	R/W-0h	R-0h	R/W-0h				
7	6	5	4	3	2	1	0
RESERVED				MAG_CHID1			
R-0h				R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Figure 22-80. ADMAGINT1CR Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED				MAG_THR1			
R-0h				R/W-0h			
23	22	21	20	19	18	17	16
MAG_THR1							
R/W-0h							
15	14	13	12	11	10	9	8
CHN_THR_COMP1	CMP_GE_LT1	RESERVED	COMP_CHID1				
R/W-0h	R/W-0h	R-0h	R/W-0h				
7	6	5	4	3	2	1	0
RESERVED				MAG_CHID1			
R-0h				R/W-0h			

Table 22-61. ADMAGINT1CR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-28	RESERVED	R	0h	Reads return 0, writes have no effect.
27-16	MAG_THR1	R/W	0h	These bits specify the 12-bit or 10-bit compare value that the ADC uses for the comparison with the MAG_CHID1 channel's conversion result.
15	CHN_THR_COMP1	R/W	0h	Channel OR Threshold comparison. Both the MAG_CHID1 and the COMP_CHID1 channel must have been converted at least once for the ADC to perform the comparison. Any operation mode: 0h = The ADC module compares the MAG_CHID1 channel's conversion result with the fixed threshold value specified by the MAG_THR1 field. 1h = The ADC module compares the MAG_CHID1 channel's conversion result with the last conversion result for the COMP_CHID1 channel.
14	CMP_GE_LT1	R/W	0h	Greater than or equal to OR Less than comparison operator. Any operation mode: 0h = The ADC module checks if the conversion result is lower than the reference value (fixed threshold or COMP_CHID1 conversion result). 1h = The ADC module checks if the conversion result is greater than or equal to the reference value (fixed threshold or COMP_CHID1 conversion result).
13	RESERVED	R	0h	Reads return 0, writes have no effect.
12-8	COMP_CHID1	R/W	0h	These bits specify the channel number from 0 to 31 whose last conversion result is compared with the MAG_CHID1 channel's conversion result.
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	MAG_CHID1	R/W	0h	These bits specify the channel number from 0 to 31 for which the conversion result needs to be monitored by the ADC.

22.11.55 ADMAGINT1MASK Register (Offset = 12Ch) [reset = 0h]

ADMAGINT1MASK is shown in [Figure 22-81](#) and described in [Table 22-62](#).

Return to [Summary Table](#).

The format of the ADMAGINT1MASK is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module. The ADC module supports up to three magnitude compare interrupts.

Figure 22-81. ADMAGINT1MASK Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED				MAG_INT1_MASK											
R-0h				R/W-0h											

Table 22-62. ADMAGINT1MASK Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	MAG_INT1_MASK	R/W	0h	These bits specify the mask for the comparison in order to generate the magnitude compare interrupt #1. Note: The actual size of the MAG_INT1_MASK field is 12 bits or 10 bits depending on whether the ADC is configured to be in 12-bit or 10-bit resolution mode, respectively. Bits 11-10 are reserved when the module is configured as a 10-bit ADC module. Any operation mode: 0h = The ADC module does not mask the corresponding bit for the comparison. 1h = The ADC module masks the corresponding bit for the comparison.

22.11.56 ADMAGINT2CR Register (Offset = 130h) [reset = 0h]

ADMAGINT2CR is shown in [Figure 22-82](#) and [Figure 22-83](#), and described in [Table 22-63](#).

Return to [Summary Table](#).

The format of the ADMAGINT2CR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module. The ADC module supports up to three magnitude compare interrupts.

Figure 22-82. ADMAGINT2CR Register (12-bit)

31	30	29	28	27	26	25	24
RESERVED						MAG_THR2	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
MAG_THR2							
R/W-0h							
15	14	13	12	11	10	9	8
CHN_THR_COMP2	CMP_GE_LT2	RESERVED	COMP_CHID2				
R/W-0h	R/W-0h	R-0h	R/W-0h				
7	6	5	4	3	2	1	0
RESERVED				MAG_CHID2			
R-0h				R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Figure 22-83. ADMAGINT2CR Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED						MAG_THR2	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
MAG_THR2							
R/W-0h							
15	14	13	12	11	10	9	8
CHN_THR_COMP2	CMP_GE_LT2	RESERVED	COMP_CHID2				
R/W-0h	R/W-0h	R-0h	R/W-0h				
7	6	5	4	3	2	1	0
RESERVED				MAG_CHID2			
R-0h				R/W-0h			

Table 22-63. ADMAGINT2CR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-28	RESERVED	R	0h	Reads return 0, writes have no effect.
27-16	MAG_THR2	R/W	0h	These bits specify the 12-bit or 10-bit compare value that the ADC uses for the comparison with the MAG_CHID2 channel's conversion result.
15	CHN_THR_COMP2	R/W	0h	Channel OR Threshold comparison. Both the MAG_CHID2 and the COMP_CHID2 channel must have been converted at least once for the ADC to perform the comparison. Any operation mode: 0h = The ADC module compares the MAG_CHID2 channel's conversion result with the fixed threshold value specified by the MAG_THR2 field. 1h = The ADC module compares the MAG_CHID2 channel's conversion result with the last conversion result for the COMP_CHID2 channel.
14	CMP_GE_LT2	R/W	0h	Greater than or equal to OR Less than comparison operator. Any operation mode: 0h = The ADC module checks if the conversion result is lower than the reference value (fixed threshold or COMP_CHID2 conversion result). 1h = The ADC module checks if the conversion result is greater than or equal to the reference value (fixed threshold or COMP_CHID2 conversion result).
13	RESERVED	R	0h	Reads return 0, writes have no effect.
12-8	COMP_CHID2	R/W	0h	These bits specify the channel number from 0 to 31 whose last conversion result is compared with the MAG_CHID2 channel's conversion result.
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	MAG_CHID2	R/W	0h	These bits specify the channel number from 0 to 31 for which the conversion result needs to be monitored by the ADC.

22.11.57 ADMAGINT2MASK Register (Offset = 134h) [reset = 0h]

ADMAGINT2MASK is shown in [Figure 22-84](#) and described in [Table 22-64](#).

Return to [Summary Table](#).

The format of the ADMAGINT2MASK is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module. The ADC module supports up to three magnitude compare interrupts.

Figure 22-84. ADMAGINT2MASK Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED				MAG_INT2_MASK											
R-0h				R/W-0h											

Table 22-64. ADMAGINT2MASK Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	MAG_INT2_MASK	R/W	0h	These bits specify the mask for the comparison in order to generate the magnitude compare interrupt #2. Note: The actual size of the MAG_INT2_MASK field is 12 bits or 10 bits depending on whether the ADC is configured to be in 12-bit or 10-bit resolution mode, respectively. Bits 11-10 are reserved when the module is configured as a 10-bit ADC module. Any operation mode: 0h = The ADC module does not mask the corresponding bit for the comparison. 1h = The ADC module masks the corresponding bit for the comparison.

22.11.58 ADMAGINT3CR Register (Offset = 138h) [reset = 0h]

ADMAGINT3CR is shown in [Figure 22-85](#) and [Figure 22-86](#), and described in [Table 22-65](#).

Return to [Summary Table](#).

The format of the ADMAGINT3CR is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module. The ADC module supports up to three magnitude compare interrupts.

Figure 22-85. ADMAGINT3CR Register (12-bit)

31	30	29	28	27	26	25	24
RESERVED						MAG_THR3	
R-0h						R/W-0h	
23	22	21	20	19	18	17	16
MAG_THR3							
R/W-0h							
15	14	13	12	11	10	9	8
CHN_THR_COMP3	CMP_GE_LT3	RESERVED	COMP_CHID3				
R/W-0h	R/W-0h	R-0h	R/W-0h				
7	6	5	4	3	2	1	0
RESERVED				MAG_CHID3			
R-0h				R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; W1toCl = Write 1 to clear bit; -n = value after reset

Figure 22-86. ADMAGINT3CR Register (10-bit)

31	30	29	28	27	26	25	24
RESERVED				MAG_THR3			
R-0h				R/W-0h			
23	22	21	20	19	18	17	16
MAG_THR3							
R/W-0h							
15	14	13	12	11	10	9	8
CHN_THR_COMP3	CMP_GE_LT3	RESERVED	COMP_CHID3				
R/W-0h	R/W-0h	R-0h	R/W-0h				
7	6	5	4	3	2	1	0
RESERVED				MAG_CHID3			
R-0h				R/W-0h			

Table 22-65. ADMAGINT3CR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-28	RESERVED	R	0h	Reads return 0, writes have no effect.
27-16	MAG_THR3	R/W	0h	These bits specify the 12-bit or 10-bit compare value that the ADC uses for the comparison with the MAG_CHID3 channel's conversion result.
15	CHN_THR_COMP3	R/W	0h	Channel OR Threshold comparison. Both the MAG_CHID3 and the COMP_CHID3 channel must have been converted at least once for the ADC to perform the comparison. Any operation mode: 0h = The ADC module compares the MAG_CHID3 channel's conversion result with the fixed threshold value specified by the MAG_THR3 field. 1h = The ADC module compares the MAG_CHID3 channel's conversion result with the last conversion result for the COMP_CHID3 channel.
14	CMP_GE_LT3	R/W	0h	Greater than or equal to OR Less than comparison operator. Any operation mode: 0h = The ADC module checks if the conversion result is lower than the reference value (fixed threshold or COMP_CHID3 conversion result). 1h = The ADC module checks if the conversion result is greater than or equal to the reference value (fixed threshold or COMP_CHID3 conversion result).
13	RESERVED	R	0h	Reads return 0, writes have no effect.
12-8	COMP_CHID3	R/W	0h	These bits specify the channel number from 0 to 31 whose last conversion result is compared with the MAG_CHID3 channel's conversion result.
7-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	MAG_CHID3	R/W	0h	These bits specify the channel number from 0 to 31 for which the conversion result needs to be monitored by the ADC.

22.11.59 ADMAGINT3MASK Register (Offset = 13Ch) [reset = 0h]

ADMAGINT3MASK is shown in [Figure 22-87](#) and described in [Table 22-66](#).

Return to [Summary Table](#).

The format of the ADMAGINT3MASK is different based on whether the ADC module is configured to be a 12-bit or a 10-bit ADC module. The ADC module supports up to three magnitude compare interrupts.

Figure 22-87. ADMAGINT3MASK Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED				MAG_INT3_MASK											
R-0h				R/W-0h											

Table 22-66. ADMAGINT3MASK Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-0	MAG_INT3_MASK	R/W	0h	These bits specify the mask for the comparison in order to generate the magnitude compare interrupt #3. Note: The actual size of the MAG_INT3_MASK field is 12 bits or 10 bits depending on whether the ADC is configured to be in 12-bit or 10-bit resolution mode, respectively. Bits 11-10 are reserved when the module is configured as a 10-bit ADC module. Any operation mode: 0h = The ADC module does not mask the corresponding bit for the comparison. 1h = The ADC module masks the corresponding bit for the comparison.

22.11.60 ADMAGINTENASET Register (Offset = 158h) [reset = 0h]

ADMAGINTENASET is shown in [Figure 22-88](#) and described in [Table 22-67](#).

Return to [Summary Table](#).

Figure 22-88. ADMAGINTENASET Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED					MAG_INT_ENA_SET		
R-0h					R/W-0h		

Table 22-67. ADMAGINTENASET Register Field Descriptions

Bit	Field	Type	Reset	Description
31-3	RESERVED	R	0h	Reads return 0, writes have no effect.
2-0	MAG_INT_ENA_SET	R/W	0h	Each of these three bits, when set, enable the corresponding magnitude compare interrupt. Any operation mode for each bit: 0h = The enable status of the corresponding magnitude compare interrupt is left unchanged. 1h = The corresponding magnitude compare interrupt is enabled.

22.11.61 ADMAGINTENACLR Register (Offset = 15Ch) [reset = 0h]

ADMAGINTENACLR is shown in [Figure 22-89](#) and described in [Table 22-68](#).

Return to [Summary Table](#).

Figure 22-89. ADMAGINTENACLR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED					MAG_INT_ENA_CLR		
R-0h					R/W-0h		

Table 22-68. ADMAGINTENACLR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-3	RESERVED	R	0h	Reads return 0, writes have no effect.
2-0	MAG_INT_ENA_CLR	R/W	0h	Each of these three bits, when set, enable the corresponding magnitude compare interrupt. Any operation mode for each bit: 0h = The enable status of the corresponding magnitude compare interrupt is left unchanged. 1h = The corresponding magnitude compare interrupt is disabled.

22.11.62 ADMAGINTFLG Register (Offset = 160h) [reset = 0h]

ADMAGINTFLG is shown in [Figure 22-90](#) and described in [Table 22-69](#).

Return to [Summary Table](#).

Figure 22-90. ADMAGINTFLG Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED					MAG_INT_FLG		
R-0h					R/W-0h		

Table 22-69. ADMAGINTFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
31-3	RESERVED	R	0h	Reads return 0, writes have no effect.
2-0	MAG_INT_FLG	R/W	0h	Magnitude Compare Interrupt Flags. These bits are polled by the application to determine if the magnitude compares have been evaluated as true. When a magnitude compare interrupt flag is set, the corresponding magnitude compare interrupt is generated if enabled. Any operation mode for each bit: 0h (W) = The corresponding flag is left unchanged. 0h (R) = The condition for the corresponding magnitude threshold interrupt was false. 1h (W) = The corresponding flag is cleared. The flag is also cleared by reading from the magnitude compare interrupt offset register. 1h (R) = The condition for the corresponding magnitude threshold interrupt was true.

22.11.63 ADMAGINTOFF Register (Offset = 164h) [reset = 0h]

ADMAGINTOFF is shown in [Figure 22-91](#) and described in [Table 22-70](#).

Return to [Summary Table](#).

Figure 22-91. ADMAGINTOFF Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED												MAG_INT_OFF			
R-0h												RC-0h			

Table 22-70. ADMAGINTOFF Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3-0	MAG_INT_OFF	RC	0h	Magnitude Compare Interrupt Offset. This field indexes the currently highest-priority magnitude compare interrupt. Interrupt 1 has the highest priority and interrupt 3 has the lowest priority among the magnitude compare interrupts. Writes to these bits have no effect. A read from this register clears this register as well as the corresponding magnitude compare interrupt flag in the ADMAGINTFLG register. However, a read from this register in emulation mode does not affect this register or the interrupt status flags. Any operation mode read: 0h = No magnitude compare interrupt is pending. 1h = Magnitude compare interrupt # 1 is pending. 2h = Magnitude compare interrupt # 2 is pending. 3h = Magnitude compare interrupt # 3 is pending. 4h = Reserved 5h = Reserved 6h = Reserved 7h = Reserved

22.11.64 ADEVFIFORESETCR Register (Offset = 168h) [reset = 0h]

ADEVFIFORESETCR is shown in [Figure 22-92](#) and described in [Table 22-71](#).

Return to [Summary Table](#).

Figure 22-92. ADEVFIFORESETCR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							EV_FIFO_RESET
R-0h							R/W-0h

Table 22-71. ADEVFIFORESETCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	EV_FIFO_RESET	R/W	0h	ADC Event Group FIFO Reset. The application sets this bit in case of an overrun condition. This allows the ADC module to overwrite the contents of the Event Group results memory starting from the first location. The EV_FIFO_RESET bit only has the desired effect when the Event Group results memory is in an overrun condition. It must be used when the data already available in the results memory is discarded. If the application needs the Event Group memory to always be overwritten with the latest available conversion results, then the OVR_EV_RAM_IGN bit in the Event Group operating mode control register (ADEVMODECR) needs to be set to 1. 1h = The ADC module resets its internal Event Group results memory pointers. Then this bit automatically gets cleared, so that the ADC module allows the Event Group results memory to be overwritten only once each time this bit is set to 1. As a result, the EV_FIFO_RESET bit is always read as a 0.

22.11.65 ADG1FIFORESETCR Register (Offset = 16Ch) [reset = 0h]

ADG1FIFORESETCR is shown in [Figure 22-93](#) and described in [Table 22-72](#).

Return to [Summary Table](#).

Figure 22-93. ADG1FIFORESETCR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							G1_FIFO_RESET
R-0h							R/W-0h

Table 22-72. ADG1FIFORESETCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	G1_FIFO_RESET	R/W	0h	ADC Group1 FIFO Reset. The application sets this bit in case of an overrun condition. This allows the ADC module to overwrite the contents of the Group1 results memory starting from the first location from 0 to 1. The G1_FIFO_RESET bit only has the desired effect when the Group1 results memory is in an overrun condition. It must be used when the data already available in the results memory is discarded. If the application needs the Group1 memory to always be overwritten with the latest available conversion results, then the OVR_G1_RAM_IGN bit in the Group1 operating mode control register (ADG1MODECR) needs to be set to 1. 1h = The ADC module resets its internal Group1 results memory pointers. Then this bit automatically gets cleared, so that the ADC module allows the Group1 results memory to be overwritten only once each time this bit is set to 1. As a result, the G1_FIFO_RESET bit is always read as a 0.

22.11.66 ADG2FIFORESETCR Register (Offset = 170h) [reset = 0h]

ADG2FIFORESETCR is shown in [Figure 22-94](#) and described in [Table 22-73](#).

Return to [Summary Table](#).

Figure 22-94. ADG2FIFORESETCR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							G2_FIFO_RESET
R-0h							R/W-0h

Table 22-73. ADG2FIFORESETCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-1	RESERVED	R	0h	Reads return 0, writes have no effect.
0	G2_FIFO_RESET	R/W	0h	ADC Group2 FIFO Reset. The application sets this bit in case of an overrun condition. This allows the ADC module to overwrite the contents of the Group2 results memory starting from the first location from 0 to 1. The G2_FIFO_RESET bit only has the desired effect when the Group2 results memory is in an overrun condition. It must be used when the data already available in the results memory is discarded. If the application needs the Group2 memory to always be overwritten with the latest available conversion results, then the OVR_G2_RAM_IGN bit in the Group2 operating mode control register (ADG2MODECR) needs to be set to 1. 1h = The ADC module resets its internal Group2 results memory pointers. Then this bit automatically gets cleared, so that the ADC module allows the Group2 results memory to be overwritten only once each time this bit is set to 1. As a result, the G2_FIFO_RESET bit is always read as a 0.

22.11.67 ADEVRAMWRADDR Register (Offset = 174h) [reset = 0h]

ADEVRAMWRADDR is shown in [Figure 22-95](#) and described in [Table 22-74](#).

Return to [Summary Table](#).

Figure 22-95. ADEVRAMWRADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								EV_RAM_ADDR							
R-0h								R/W-0h							

Table 22-74. ADEVRAMWRADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-9	RESERVED	R	0h	Reads return 0, writes have no effect.
8-0	EV_RAM_ADDR	R/W	0h	Event Group results memory write pointer. This field shows the address of the location where the next Event Group conversion result is stored. This is specified in terms of the buffer number. The application reads this register to determine the number of valid Event Group conversion results available until that time.

22.11.68 ADG1RAMWRADDR Register (Offset = 178h) [reset = 0h]

ADG1RAMWRADDR is shown in [Figure 22-96](#) and described in [Table 22-75](#).

Return to [Summary Table](#).

Figure 22-96. ADG1RAMWRADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								G1_RAM_ADDR							
R-0h								R/W-0h							

Table 22-75. ADG1RAMWRADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-9	RESERVED	R	0h	Reads return 0, writes have no effect.
8-0	G1_RAM_ADDR	R/W	0h	Group1 results memory write pointer. This field shows the address of the location where the next Group1 conversion result is stored. This is specified in terms of the buffer number. The application reads this register to determine the number of valid Group1 conversion results available until that time.

22.11.69 ADG2RAMWRADDR Register (Offset = 17Ch) [reset = 0h]

ADG2RAMWRADDR is shown in [Figure 22-97](#) and described in [Table 22-76](#).

Return to [Summary Table](#).

Figure 22-97. ADG2RAMWRADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED								G2_RAM_ADDR							
R-0h								R/W-0h							

Table 22-76. ADG2RAMWRADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-9	RESERVED	R	0h	Reads return 0, writes have no effect.
8-0	G2_RAM_ADDR	R/W	0h	Group2 results memory write pointer. This field shows the address of the location where the next Group2 conversion result is stored. This is specified in terms of the buffer number. The application reads this register to determine the number of valid Group2 conversion results available until that time.

22.11.70 ADPARCR Register (Offset = 180h) [reset = 5h]

ADPARCR is shown in [Figure 22-98](#) and described in [Table 22-77](#).

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Figure 22-98. ADPARCR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							TEST
R-0h							R/WP-0h
7	6	5	4	3	2	1	0
RESERVED				PARITY_ENA			
R-0h				R/WP-5h			

Table 22-77. ADPARCR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-9	RESERVED	R	0h	Reads return 0, writes have no effect.
8	TEST	R/WP	0h	This bit maps the parity bits into the ADC results' RAM frame so that the application accesses them. Any operation mode read, privileged mode write: 0h = The parity bits are not memory-mapped. 1h = The parity bits are memory mapped.
7-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3-0	PARITY_ENA	R/WP	5h	Enable parity checking. These bits enable or disable the parity check on read operations and the parity calculation on write operations to the ADC results memory. If parity checking is enabled and a parity error is detected, the ADC module sends a parity error signal to the System module. Any other value other than 5h, parity check is enabled. Any operation mode read, privileged mode write: 5h = Parity check is disabled.

22.11.71 ADPARADDR Register (Offset = 184h) [reset = X]

ADPARADDR is shown in [Figure 22-99](#) and described in [Table 22-78](#).

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Figure 22-99. ADPARADDR Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED				ERROR_ADDRESS			
R-0h				R-X			
7	6	5	4	3	2	1	0
ERROR_ADDRESS						RESERVED	
R-X						R-0h	

Table 22-78. ADPARADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-12	RESERVED	R	0h	Reads return 0, writes have no effect.
11-2	ERROR_ADDRESS	R	X	These bits hold the address of the first parity error generated in the ADC results' RAM. Reading bits [11:0] provides the 32-bit aligned address. This error address is frozen from being updated until it is read by the application. In emulation mode, this address is maintained frozen even when read.
1-0	RESERVED	R	0h	Reads return 0, writes have no effect.

22.11.72 ADPWRUPDLYCTRL Register (Offset = 188h) [reset = 0h]

ADPWRUPDLYCTRL is shown in [Figure 22-100](#) and described in [Table 22-79](#).

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Figure 22-100. ADPWRUPDLYCTRL Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
RESERVED																						PWRUP_DLY															
R-0h																						R/W-0h															

Table 22-79. ADPWRUPDLYCTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-10	RESERVED	R	0h	Reads return 0, writes have no effect.
9-0	PWRUP_DLY	R/W	0h	This register defines the number of VCLK cycles that the ADC state machine has to wait after releasing the ADC core from power down before starting a new conversion. Refer to ADC Power-Down Mode subsection for more details.

22.11.73 ADEVCHNSELMODECTRL Register (Offset = 190h) [reset = 5h]

ADEVCHNSELMODECTRL is shown in [Figure 22-101](#) and described in [Table 22-80](#).

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Figure 22-101. ADEVCHNSELMODECTRL Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				EV_ENH_CHNSEL_MODE_ENABLE			
R-0h				R/W-5h			

Table 22-80. ADEVCHNSELMODECTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3-0	EV_ENH_CHNSEL_MODE_ENABLE	R/W	5h	Enable enhanced channel selection mode for Event group. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. Writing any value other than 5h or Ah to this field has no effect on the selected channel selection mode for the Event group, and the ADC module continues to use the channel selection mode that was previously programmed channel selection mode. 5h (R) = Indicates that the enhanced channel selection mode for Event group is not enabled. The default sequential channel selection mode is used for Event group conversions. 5h (W) = Disables the enhanced channel selection mode for Event group and enables the sequential channel selection mode. Ah (W) = Enables the enhanced channel selection mode for Event group. Ah (R) = Indicates that the enhanced channel selection mode for Event group is enabled.

22.11.74 ADG1CHNSELMODECTRL Register (Offset = 194h) [reset = 5h]

ADG1CHNSELMODECTRL is shown in [Figure 22-102](#) and described in [Table 22-81](#).

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Figure 22-102. ADG1CHNSELMODECTRL Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				G1_ENH_CHNSEL_MODE_ENABLE			
R-0h				R/W-5h			

Table 22-81. ADG1CHNSELMODECTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3-0	G1_ENH_CHNSEL_MODE_ENABLE	R/W	5h	Enable enhanced channel selection mode for Group1. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. Writing any value other than 5h or Ah to this field has no effect on the selected channel selection mode for the Group1, and the ADC module continues to use the channel selection mode that was previously programmed channel selection mode. 5h (R) = Indicates that the enhanced channel selection mode for Group1 is not enabled. The default sequential channel selection mode is used for Group1 conversions. 5h (W) = Disables the enhanced channel selection mode for Group1 and enables the sequential channel selection mode. Ah (W) = Enables the enhanced channel selection mode for Group1. Ah (R) = Indicates that the enhanced channel selection mode for Group1 is enabled.

22.11.75 ADG2CHNSELMODECTRL Register (Offset = 198h) [reset = 5h]

ADG2CHNSELMODECTRL is shown in [Figure 22-103](#) and described in [Table 22-82](#).

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Figure 22-103. ADG2CHNSELMODECTRL Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				G2_ENH_CHNSEL_MODE_ENABLE			
R-0h				R/W-5h			

Table 22-82. ADG2CHNSELMODECTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-4	RESERVED	R	0h	Reads return 0, writes have no effect.
3-0	G2_ENH_CHNSEL_MODE_ENABLE	R/W	5h	Enable enhanced channel selection mode for Group2. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. Writing any value other than 5h or Ah to this field has no effect on the selected channel selection mode for the Group2, and the ADC module continues to use the channel selection mode that was previously programmed channel selection mode. 5h (R) = Indicates that the enhanced channel selection mode for Group2 is not enabled. The default sequential channel selection mode is used for Group2 conversions. 5h (W) = Disables the enhanced channel selection mode for Group2 and enables the sequential channel selection mode. Ah (W) = Enables the enhanced channel selection mode for Group2. Ah (R) = Indicates that the enhanced channel selection mode for Group2 is enabled.

22.11.76 ADEVCURRCOUNT Register (Offset = 19Ch) [reset = 0h]

ADEVCURRCOUNT is shown in [Figure 22-104](#) and described in [Table 22-83](#).

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Figure 22-104. ADEVCURRCOUNT Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				EV_CURRENT_COUNT			
R-0h				R/W-0h			

Table 22-83. ADEVCURRCOUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	EV_CURRENT_COUNT	R/W	0h	The CURRENT_COUNT value for the Event group conversions when enhanced channel selection mode is enabled. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. A read from the ADEVCURRCOUNT register returns the value of the current index into the Event group's look-up table. This register resets to 0 on any of the following conditions: - A peripheral reset occurs. - An ADC software reset occurs via the ADC Reset Control Register (ADRSTCR). - EV_CURRENT_COUNT becomes equal to EV_MAX_COUNT. - Application write 0s to ADEVCURRCOUNT register. - Event group's result RAM is reset.

22.11.77 ADEVMAXCOUNT Register (Offset = 1A0h) [reset = 0h]

ADEVMAXCOUNT is shown in [Figure 22-105](#) and described in [Table 22-84](#).

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Figure 22-105. ADEVMAXCOUNT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED											EV_MAX_COUNT				
R-0h											R/W-0h				

Table 22-84. ADEVMAXCOUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	EV_MAX_COUNT	R/W	0h	The MAX_COUNT value for the Event group conversions when enhanced channel selection mode is enabled. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. It is recommended to clear the Event group's Current Count register (ADEVCURRCOUNT) whenever the EV_MAX_COUNT is changed.

22.11.78 ADG1CURRCOUNT Register (Offset = 1A4h) [reset = 0h]

ADG1CURRCOUNT is shown in [Figure 22-106](#) and described in [Table 22-85](#).

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Figure 22-106. ADG1CURRCOUNT Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				G1_CURRENT_COUNT			
R-0h				R/W-0h			

Table 22-85. ADG1CURRCOUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	G1_CURRENT_COUNT	R/W	0h	The CURRENT_COUNT value for the Group1 conversions when enhanced channel selection mode is enabled. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. A read from the ADG1CURRCOUNT register returns the value of the current index into the Group1's look-up table. This register resets to 0 on any of the following conditions: - A peripheral reset occurs. - An ADC software reset occurs via the ADC Reset Control Register (ADRSTCR). - G1_CURRENT_COUNT becomes equal to G1_MAX_COUNT. - Application write 0s to ADG1CURRCOUNT register. - Group1's result RAM is reset.

22.11.79 ADG1MAXCOUNT Register (Offset = 1A8h) [reset = 0h]

ADG1MAXCOUNT is shown in [Figure 22-107](#) and described in [Table 22-86](#).

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Figure 22-107. ADG1MAXCOUNT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED											G1_MAX_COUNT				
R-0h											R/W-0h				

Table 22-86. ADG1MAXCOUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	G1_MAX_COUNT	R/W	0h	The MAX_COUNT value for the Group1 conversions when enhanced channel selection mode is enabled. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. It is recommended to clear the Group1's Current Count register (ADG1CURRCOUNT) whenever the G1_MAX_COUNT is changed.

22.11.80 ADG2CURRCOUNT Register (Offset = 1ACh) [reset = 0h]

ADG2CURRCOUNT is shown in [Figure 22-108](#) and described in [Table 22-87](#).

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Figure 22-108. ADG2CURRCOUNT Register

31	30	29	28	27	26	25	24
RESERVED							
R-0h							
23	22	21	20	19	18	17	16
RESERVED							
R-0h							
15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				G2_CURRENT_COUNT			
R-0h				R/W-0h			

Table 22-87. ADG2CURRCOUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	G2_CURRENT_COUNT	R/W	0h	The CURRENT_COUNT value for the Group2 conversions when enhanced channel selection mode is enabled. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. A read from the ADG2CURRCOUNT register returns the value of the current index into the Group2's look-up table. This register resets to 0 on any of the following conditions: - A peripheral reset occurs. - An ADC software reset occurs via the ADC Reset Control Register (ADRSTCR). - G2_CURRENT_COUNT becomes equal to G2_MAX_COUNT. - Application write 0s to ADG2CURRCOUNT register. - Group2's result RAM is reset.

22.11.81 ADG2MAXCOUNT Register (Offset = 1B0h) [reset = 0h]

ADG2MAXCOUNT is shown in [Figure 22-109](#) and described in [Table 22-88](#).

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Figure 22-109. ADG2MAXCOUNT Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RESERVED															
R-0h															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED											G2_MAX_COUNT				
R-0h											R/W-0h				

Table 22-88. ADG2MAXCOUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
31-5	RESERVED	R	0h	Reads return 0, writes have no effect.
4-0	G2_MAX_COUNT	R/W	0h	The MAX_COUNT value for the Group2 conversions when enhanced channel selection mode is enabled. Refer to Enhanced Channel Selection Mode subsection for a description of the enhanced channel selection mode. It is recommended to clear the Group2's Current Count register (ADG2CURRCOUNT) whenever the G2_MAX_COUNT is changed.